



2026

Formula Hybrid + Electric

Rules

October 23, 2025



Formula Hybrid + Electric Rules Committee

Formula Hybrid + Electric gratefully acknowledges the contributions of the following people, who have donated countless hours and immeasurable effort into making the Formula Hybrid + Electric competition a stellar, multidisciplinary learning experience, while keeping the competition as safe as possible.

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RULES CHANGES FOR 2026

The following is a list of noteworthy changes from the 2025 Formula Hybrid + Electric rules. It is not complete and is not binding. If there are any differences between this summary and the official rules, the rules will prevail. Therefore, it is the responsibility of the competitors to read the published Rules thoroughly.

Rule Number	Section	Change
A9.4	Administrative Regulations	Added new rule
D7.13.6	Dynamic Events	To adjust the rules to the long-time practice
T3.8.1	Technical – SES Submission	SES must reflect the vehicle being entered
T3.8.8	Technical – SES Submission	Year update
T3.12.4	Main Hoop Bracing	Increased visibility requirements
T13.1	Technical – Car Number	Increased number size
T14.1	Technical - Helmet	Removes noncompliant helmet examples

Electrical Rules Updates

EV1.2.2	Electrical Systems Overview	Added to comply with another ruleset
EV1.2.3	Electrical Systems Overview	Increases Safety
EV2.8.3	Accumulator – Isolation Relays	Clarifies suitable sealants
EV2.3.5	Energy Storage	Clarifies verbiage for technical inspection
EV2.8.3	Isolation Relays	Removes ambiguity
EV2.8.5	Isolation Relays	Removes voltage spike example
EV2.10.2	Pre-charge Discharge Circuits	Time based pre-charge should not be allowed
EV2.10.3	Pre-charge Discharge Circuits	Creates more robust pre-charge circuitry designs
EV2.10.7	Pre-charge Discharge Circuits	Fusing the pre-charge circuit should be allowed.
EV2.11.2	Accumulator Management System	Increases Safety
EV4.1.7	Grounded Low Voltage System	GLV batteries must disable their output when fully discharged
EV5.3.1	Insulation, Spacing, and Segregation	Tightened creepage/clearance tolerances
EV5.5.3	PCB Isolation	Grammatical Clarification
EV6.1.6	Fusing Typo	
EV8.1	Grounding	Adopted SAE rule

EV8.1.3 Grounding	Deleted redundancy
EV7.8.1 Grounding	Specifies use of four terminal ohm meter
EV9.1 TSSI Adopted SAE rule	
EV10.1.3 IMD Test	Removed constraints on IMD test
EV12.2.2 Charging	Added additional grounding tolerances
Figure 37 Grounded Low Voltage System	Added row for Electric BRB

Advice from Our Rules Committee

Formula Hybrid + Electric's Rules Committee welcomes you to the most challenging of the SAE Collegiate Design Series competitions. Many of us are former Formula Hybrid + Electric competitors who are now professionals in the engineering industry. **We have two goals: to have a safe competition and to see every team on the track.**

Top 10 Tips for Building a Formula Hybrid + Electric Racecar and Passing Tech Inspection

Start work early. Everything takes longer than you expect.

Consider starting from an existing FSAE chassis and concentrating on the powertrain.

Read all rules carefully. If you don't understand something, ask us for clarification.

Use the Project Management techniques and tools. They will make life easier—and will help you as engineers. Most importantly, they will help you arrive at the competition with a *finished* car.

Take advantage of our Mentor Program. These experts will help you solve issues—and will be valuable career contacts.

Your car should be running on its own power at least a month prior to competition.

Make brake testing an early priority.

Take advantage of the extra day of electrical tech inspection on Sunday. That will give you extra time if you need to make modifications. This is also a good time to have your documentation reviewed.

Watch out for the rules tagged with the "attention" symbol. These rules have a history of tripping up teams.



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PART A - ADMINISTRATIVE REGULATIONS

ARTICLE A1 FORMULA HYBRID + ELECTRIC OVERVIEW AND COMPETITION

A1.1 Formula Hybrid + Electric Competition Objective

- A1.1.1 The Formula Hybrid + Electric competition challenges teams of university undergraduate and graduate students to conceive, design, fabricate, develop and compete with small, formula-style, hybrid-powered and electric cars.
- A1.1.2 The Formula Hybrid + Electric competition is intended as an educational program requiring students to work across disciplinary boundaries, such as those of electrical and mechanical engineering.
- A1.1.3 To give teams the maximum design flexibility and the freedom to express their creativity and imagination there are very few restrictions on the overall vehicle design apart from the requirement for a mechanical/electrical hybrid or electric-only drivetrain.
- A1.1.4 Teams typically spend eight to twelve months designing, building, testing and preparing their vehicles before a competition. The competitions themselves give teams the chance to demonstrate and prove both their creativity and their engineering skills in comparison to teams from other universities around the world.

A1.2 Energy Limits

- A1.2.1 Competitiveness and high efficiency designs are encouraged through limits on accumulator capacities and the amount of energy that a team has available to complete the endurance event.
- A1.2.2 The accumulator capacities and endurance energy allocation will be reviewed by the Formula Hybrid + Electric rules committee each year, and posted as early in the season as possible.

Hybrid (and Hybrid In Progress)	
Endurance Energy Allocation	31.25 MJ
Maximum Accumulator Capacity	4,449 Wh
Electric	
Maximum Accumulator Capacity	5,400 Wh

Table 1 – 2024 Energy and Accumulator Limits

- A1.2.3 Vehicles may run with accumulator capacities greater than the Table 1 value if equipped with an FH+E approved energy meter. Any endurance laps that reach or exceed the Table 1 maximum energy value will not be counted. See also **D7.5.1**- Energy for Endurance Event.
- A1.2.4 Accumulator capacities are calculated as:

Energy (Wh) = Number of Cells x Vnom x Cell Capacity (Inom in AH) x 80%

Segment Energy (**Table 8**) is calculated as

Energy (MJ) = Number of Cells x Vnom x Cell Capacity (Inom in AH) x 0.0036.

See [Appendix A](#) for energy calculations for capacitors.

A1.3 Vehicle Design Objectives

For the purpose of this competition, the students are to assume that a manufacturing firm has engaged them to design, fabricate and demonstrate a prototype hybrid-electric or all electric vehicle for evaluation as a production item. The intended market is the nonprofessional weekend autocross competitor. Therefore, the car must balance exceptional performance with fuel efficiency. Performance will be evaluated in terms of its acceleration, braking, and handling qualities. Fuel efficiency will be evaluated during the 44 km endurance event. The car must be easy to maintain and reliable. It should accommodate drivers whose stature varies from a 5th percentile female to a 95th percentile male. In addition, the car's marketability is enhanced by other factors such as aesthetics, comfort and use of common parts. The manufacturing firm is planning to produce four (4) cars per day for a limited production run. The challenge to the design team is to develop a prototype car that best meets these goals and intents. Each design will be compared and judged with other competing designs to determine the best overall car.

A1.4 Good Engineering Practices

- A1.4.1 Vehicles entered into Formula Hybrid + Electric competitions are expected to be designed and fabricated in accordance with good engineering practices.

Note in particular, that the high-voltage electrical systems in a Formula Hybrid + Electric car present health and safety risks unique to a hybrid/electric vehicle, and that carelessness or poor engineering can result in serious injury or death.

- A1.4.2 The organizers have produced several advisory publications that are available on the Formula Hybrid + Electric website. It is expected that all team members will familiarize themselves with these publications, and will apply the information in them appropriately.

A1.5 Judging Categories

The cars are judged in a series of static and dynamic events including: technical inspections, project management skills, engineering design, solo performance trials, and high-performance track endurance. These events are scored to determine how well the car performs.

Static Events		
	Project Management	150
	Engineering Design	200
	Virtual Racing Challenge ¹	0
Dynamic Events		
	Acceleration	100
	Autocross	200
	Endurance	350
Total Points		1000

Table 2 - Event Points

A1.5.1 A team's final score will equal the sum of their event scores plus or minus penalty and/or bonus points.

Note: If a team's penalty points exceed the sum of their event scores, their final score will be Zero (0). i.e. negative final scores will not be given.

ARTICLE A2 FORMULA HYBRID + ELECTRIC VEHICLE CATEGORIES

A2.1 Hybrid

A2.1.1 A Hybrid vehicle is defined as a vehicle using a propulsion system which comprises both a 4-stroke Internal Combustion Engine (ICE) and electrical storage (accumulator) with electric motor drive.

A2.1.2 A hybrid drive system may deploy the ICE and electric motor(s) in any configuration, including series and/or parallel. Coupling through the road surface is permitted.

A2.1.3 To qualify as a hybrid, vehicles must have a drive system utilizing one or more electric motors with a minimum continuous power rating of 2.5 kW (sum total of all motors) and one or more I.C engines with a minimum (sum total) power rating of 2.5 kW.

A2.2 Electric

An Electric vehicle is defined as a vehicle wherein the accumulator is charged from an external electrical source (and/or through regenerative braking) and propelled by electric drive only.

There is no minimum power requirement for electric-only drive motors.

A2.3 Hybrid in Progress (HIP)

A Hybrid-in-Progress is a not-yet-completed hybrid vehicle, which includes both an internal combustion engine and electric motor.

Note: The purpose of the HIP category is to give teams that are on a 2-year development/build cycle an opportunity to enter and compete their vehicle alongside the regular entries. The HIP

¹ Virtual Racing Challenge will be awarded trophies without points going towards the Event Points

is regarded, for all scoring purposes, as a hybrid, but will run the dynamic events on electric power only.

- A2.3.1 To qualify as an HIP, a vehicle must have been designed, and intended for completion as a hybrid vehicle.
- A2.3.2 Teams planning to enter a vehicle in the HIP category will initially register as a Hybrid. To change to the HIP category, the team must submit a request to the organizers in writing before the start of the design event.

Note: The advantages of entering as an HIP are:

- (a) Receive a full technical inspection of the vehicle and electrical drive systems.
- (b) Participate in all the competition events. (Provided tech inspection is passed).
- (c) Receive feedback from the design judges.

Note: Teams can maximize the benefits of an HIP entry by including the full-hybrid designs in their document submissions and design event presentations, as well as including the full multi-year program in their Project Management materials.

- (d) When the vehicle is completed and entered as a hybrid, in a subsequent competition, it is considered an all-new vehicle, and not a second-year entry.

A2.4 Static Events Only (SEO)

- A2.4.1 SEO is a category that may only be declared after arrival at the competition. All teams must initially register as either Hybrid/HIP or Electric.
- A2.4.2 A team may declare themselves as SEO and participate in the design² and other static events even if the vehicle is in an unfinished state.
 - (a) An SEO vehicle may not participate in any of the dynamic events.
 - (b) An SEO vehicle may continue the technical inspection process, but will be given a lower priority than the non-SEO teams.
- A2.4.3 An SEO declaration must be submitted in writing to the organizers before the scheduled start of the design events.
- A2.4.4 A vehicle that declares SEO in one year, will not be penalized by the design judges as a multi-year vehicle the following year per [A7.5](#).

A2.5 Electric vs. Hybrid Vehicles

- A2.5.1 The Electric and Hybrid categories are separate. Although they compete in the same events, and may be on the endurance course at the same time, they are scored separately and receive separate awards.
- A2.5.2 The event scoring formulas will maintain separate baselines (T_{max} , T_{min}) for Hybrid and Electric categories.

Note: Electric vehicles, because they are not carrying the extra weight of engines and generating systems, may demonstrate higher performances in some of the dynamic events.

² Provided the team has met the document submission requirements of [A9.3\(c\)](#)

Design scores should not be compared, as the engineering challenge between the two classes is different and scored accordingly.

ARTICLE A3 THE FORMULA HYBRID + ELECTRIC COMPETITION

A3.1 Open Registration

The Formula Hybrid + Electric Competition has an open registration policy and will accept registrations by student teams representing universities in any country.

A3.2 Official Announcements and Competition Information

A3.2.1 Teams should read any newsletters published by SAE or Formula Hybrid + Electric and to be familiar with all official announcements concerning the competition and rules interpretations released by the Formula Hybrid + Electric Rules Committee.

A3.2.2 Formula Hybrid + Electric posts announcements to the “Announcements” page of the Formula Hybrid + Electric website at <https://www.formula-hybrid.org/announcements>

A3.3 Official Language

The official language of the Formula Hybrid + Electric competition is English.

ARTICLE A4 FORMULA HYBRID + ELECTRIC RULES AND ORGANIZER AUTHORITY

A4.1 Rules Authority

A4.1.1 The Formula Hybrid + Electric Rules are the responsibility of the Formula Hybrid + Electric Rules Committee and are issued under the authority of the SAE Collegiate Design Series Committee. Official announcements from the Formula Hybrid + Electric Rules Committee are to be considered part of, and have the same validity as, these rules.

A4.1.2 Ambiguities or questions concerning the meaning or intent of these rules will be resolved by the Formula Hybrid + Electric Rules Committee, SAE or by the individual competition organizers as appropriate. (See [ARTICLE A12](#))

A4.2 Rules Validity

The Formula Hybrid + Electric Rules posted on the Formula Hybrid + Electric website and dated for the calendar year of the competition are the rules in effect for the competition. Rule sets dated for other years are invalid.

A4.3 Rules Compliance

A4.3.1 By entering a Formula Hybrid + Electric competition the team, members of the team as individuals, faculty advisors and other personnel of the entering university agree to comply with, and be bound by, these rules and all rule interpretations or procedures issued or announced by SAE, the Formula Hybrid + Electric Rules Committee or the organizers.

A4.3.2 Any rules or regulations pertaining to the use of the competition site by teams or individuals and which are posted, announced and/or otherwise publicly available are incorporated into these rules by reference. As examples, all event site waiver requirements, speed limits, parking and facility use rules apply to Formula Hybrid + Electric participants.

A4.3.3 All team members, faculty advisors and other university representatives are required to cooperate with, and follow all instructions from, competition organizers, officials and judges.

A4.4 Understanding the Rules

Teams, team members as individuals and faculty advisors, are responsible for reading and understanding the rules in effect for the competition in which they are participating.

A4.5 Participating in the Competition

Teams, team members as individuals, faculty advisors and other representatives of a registered university who are present on-site at a competition are considered to be “participating in the competition” from the time they arrive at the event site until they depart the site at the conclusion of the competition or earlier by withdrawing.

A4.6 Violations of Intent

A4.6.1 The violation of intent of a rule will be considered a violation of the rule itself.

A4.6.2 Questions about the intent or meaning of a rule may be addressed to the Formula Hybrid + Electric Rules Committee or by the individual competition organizers as appropriate.

A4.7 Right to Impound

SAE and other competition organizing bodies reserve the right to impound any onsite registered vehicles at any time during a competition for inspection and examination by the organizers, officials and technical inspectors. The organizers may also impound any equipment deemed hazardous by the technical inspectors.

A4.8 Restriction on Vehicle Use

Teams are cautioned that the vehicles designed in compliance with these Formula Hybrid + Electric Rules are intended for competition operation only at the official Formula Hybrid + Electric competitions.

ARTICLE A5 RULES FORMAT AND USE

A5.1.1 Definition of Terms

Must - designates a requirement

Must NOT - designates a prohibition or restriction

Should - gives an expectation

May - gives permission, not a requirement and not a recommendation

A5.1.2 Capitalized Terms

Items or areas which have specific definitions or are covered by specific rules are capitalized.

For example, “Rules Questions” or “Primary Structure”.

A5.1.3 Headings

The article, section and paragraph headings in these rules are provided only to facilitate reading: they do not affect the paragraph contents.

A5.1.4 Applicability

Unless otherwise designated, all rules apply to all vehicles at all times.

A5.1.5 Figures and Illustrations

Figures and illustrations give clarification or guidance, but are rules only when referred to in the text of a rule.

A5.1.6 Change Identification

Any summary of changed rules and/or changed portions marked in the rules themselves are provided for courtesy, and may or may not include all changes.

A5.2 General Authority

SAE and the competition organizing bodies reserve the right to revise the schedule of any competition and/or interpret or modify the competition rules at any time and in any manner that is, in their sole judgment, required for the efficient operation of the event or the Formula Hybrid + Electric series as a whole.

A5.3 SAE Technical Standards Access

- A5.3.1** A cooperative program of SAE International University Programs and Technical Standards Board is making some of SAE's Technical Standards available to teams registered for any SAE International hosted competition at no cost. A list of accessible standards can be found online www.fsaeonline.com and accessed under the team's registration online www.sae.org.

INDIVIDUAL PARTICIPATION REQUIREMENTS

A5.4 Eligibility Limits

Eligibility is limited to undergraduate and graduate students to ensure that this is an engineering design competition.

A5.5 Student Status

- A5.5.1** Team members must be enrolled as degree seeking undergraduate or graduate students in the college or university of the team with which they are participating. Team members who have graduated during the twelve (12) month period prior to the competition remain eligible to participate.
- A5.5.2** Teams which are formed with members from two or more Universities are treated as a single team. A student at any University making up the team may compete at any event where the team participates. The multiple Universities are in effect treated as one University with two campuses and all eligibility requirements are enforced.

A5.6 Society Membership

- A5.6.1** Team members must be members of at least one of the following societies:

- (a) SAE
- (b) IEEE
- (c) SAE Australasia
- (d) SAE Brazil
- (e) ATA
- (f) IMechE
- (g) VDI

A5.6.2 Proof of membership, such as membership card, is required at the competition. Students who are members of one of the societies listed above are not required to join any of the other societies in order to participate in the Formula Hybrid + Electric competition.

A5.6.3 Students can join

SAE at: <http://www.sae.org/students>

IEEE at <https://www.ieee.org/membership/join/index.html>

Note: SAE membership is required to complete the on-line vehicle registration process, so at least one team member must be a member of SAE.

A5.7 Age

Team members must be at least eighteen (18) years of age.

A5.8 Driver's License

Team members who will drive a competition vehicle at any time during a competition must hold a valid, government issued driver's license. All drivers will be required to submit copies of their driver's licenses prior to the competition. Additional instructions will be sent to teams 6 weeks before the competition.

A5.9 Driver Restrictions

Drivers who have driven for a professional racing team in a national or international series at any time may not drive in any competition event. A "professional racing team" is defined as a team that provides racing cars and enables drivers to compete in national or international racing series and employs full time staff in order to achieve this.

A5.10 Liability Waiver

All on-site participants, including students, faculty, volunteers and guests, are required to sign a liability waiver upon registering on-site.

A5.11 Medical Insurance

Individual medical insurance coverage is required and is the sole responsibility of the participant.

ARTICLE A6 INDIVIDUAL REGISTRATION REQUIREMENTS

A6.1 SAE Student Members

A6.1.1 If your qualifying professional society membership is with the SAE, you should link yourself to your respective school, and complete the following information on the SAE website:

- (a) Medical insurance (provider, policy/ID number, telephone number)
- (b) Driver's license (state/country, ID number)
- (c) Emergency contact data (point of contact (parent/guardian, spouse), relationship, and phone number)

A6.1.2 To do this you will need to go to "Registration" page under the specific event the team is registered and then click on the "Register Your Team / Update Team Information" link. At this

point, if you are properly affiliated to the school/college/university, a link will appear with your team name to select. Once you have selected the link, the registration page will appear. Selecting the “Add New Member” button will allow individuals to include themselves with the rest of the team. This can also be completed by team captain and faculty advisor for all team members.

- A6.1.3 All students, both domestic and international, must affiliate themselves online or submit the International Student Registration form by March 1, 2026. For additional assistance, please contact collegiatecompetitions@sae.org

A6.2 Onsite Registration Requirement

- A6.2.1 Onsite registration is required of all team members and faculty advisors
- A6.2.2 Registration must be completed and the credentials and/or other identification issued by the organizers properly worn before the car can be unloaded, uncrated or worked upon in any manner.
- A6.2.3 The following is required at registration:
- (a) Government issued driver’s license or passport and
 - (b) Medical insurance card or documentation
 - (c) Proof of professional society membership (such as card or member number)
- A6.2.4 All international student participants (or unaffiliated faculty advisors) who are not SAE members are required to complete the International Student Registration form for the entire team found under “Competition Resources” on the event specific webpage. Upon completion, email the form to collegiatecompetitions@sae.org
- A6.2.5 All students, both domestic and international, must affiliate themselves online or submit the International Student Registration form prior to the date shown in the Action Deadlines on the Formula Hybrid + Electric website. For additional assistance, please contact collegiatecompetitions@sae.org

NOTE: When your team is registering for a competition, only the student or faculty advisor completing the registration needs to be linked to the school. All other students and faculty can affiliate themselves after registration has been completed.

A6.3 Team Advisor

- A6.3.1 Each team must have a Professional Advisor appointed by the university. The Advisor, or an official university representative, must accompany the team to the competition and will be considered by competition officials to be the official university representative. If the appointed advisor cannot attend the competition, the team must have an alternate college or university approved adult attend. 
- A6.3.2 Advisors are expected to review their team’s Structural Equivalency, Impact Attenuator data and both ESFs prior to submission. Advisors are not required to certify the accuracy of these documents, but should perform a “sanity check” and look for omissions.
- A6.3.3 Advisors may advise their teams on general engineering and engineering project management theory, but must not design any part of the vehicle nor directly participate in the development of any documentation or presentation. Additionally, Advisors may neither fabricate nor assemble any components nor assist in the preparation, maintenance, testing or operation of the vehicle.

In Brief –Advisors must not design, build or repair any part of the car.

A6.4 Rules and Safety Officer (RSO)

A6.4.1 Each team must appoint a person to be the “Rules and Safety Officer (RSO)”.

A6.4.2 The RSO must:

- (a) Be present at the entire Formula Hybrid + Electric event.
- (b) Be responsible for understanding the Formula Hybrid + Electric rules prior to the competition and ensuring that competing vehicles comply with all those rules requirements.
- (c) System Documentation – Have vehicle designs, plans, schematics and supporting documents available for review by the officials as needed.
- (d) Component Documentation – Have manufacturer’s documentation and information available on all components of the electrical system.
- (e) Be responsible for team safety while at the event. This includes issues such as:
 - (i) Use of safety glasses and other safety equipment
 - (ii) Control of shock hazards such as charging equipment and accessible high voltage sources
 - (iii) Control of fire hazards such as fuel, sources of ignition (grinding, welding etc.)
 - (iv) Safe working practices (lock-out/tag-out, clean work area, use of jack stands etc.)
- (f) Be the point of contact between the team and Formula Hybrid + Electric organizers should rules or safety issues arise.

A6.4.3 If the RSO is also a driver in a dynamic event, a backup RSO must be appointed who will take responsibility for sections **A6.4.2(e)** and **A6.4.2(f)** (above) while the primary RSO is in the vehicle.

A6.4.4 Preferably, the RSO should be the team's faculty advisor or a member of the university's professional staff, but the position may be held by a student member of the team.

A6.4.5 Contact information for the primary and backup RSOs (Name, Cell Phone number, etc.) must be provided to the organizers during registration.

ARTICLE A7 VEHICLE ELIGIBILITY

A7.1 Student Developed Vehicle

Vehicles entered into Formula SAE competitions must be conceived, designed, fabricated and maintained by the student team members without direct involvement from professional engineers, automotive engineers, racers, machinists or related professionals.

A7.2 Information Sources

The student team may use any literature or knowledge related to car design and information from professionals or from academics as long as the information is given as a discussion of alternatives with their pros and cons.

A7.3 Professional Assistance

Professionals may not make design decisions or drawings and the Faculty Advisor may be required to sign a statement of compliance with this restriction.

A7.4 Student Fabrication

It is the intent of the SAE Collegiate Design Series competitions to provide direct hands-on experience to the students. Therefore, students should perform all fabrication tasks whenever possible.

A7.5 Vehicles Entered for Multiple Years

Formula Hybrid + Electric does not require that teams design and build a new vehicle from scratch each year.

A7.5.1 Teams may enter the same vehicle used in previous competitions, provided it complies with all the Formula Hybrid + Electric rules in effect at the competition in which it is entered.

A7.5.2 Rules waivers issued to vehicles are valid for only one year. It is assumed that teams will address the issues requiring a waiver before entering the vehicle in a subsequent year.

NOTE 1: Design judges will look more favorably on vehicles that have clearly documented upgrades and improvements since last entered in a Formula Hybrid + Electric competition.

NOTE 2: Because the Formula Hybrid + Electric competition emphasizes high-efficiency drive systems, engineered improvements to the drive train and control systems will be weighted more heavily by the design judges than updates to the chassis, suspension etc.

A7.6 Entries per University

Universities may enter up to two vehicles per competition. Note that there will be a registration wait period imposed for a second vehicle.

ARTICLE A8 REGISTRATION

Registration for the Formula Hybrid + Electric competition must be completed on-line. Online registration must be done by either (a) an SAE member or (b) the official faculty advisor connected with the registering university and recorded as such in the SAE record system.

Note: It typically takes at least 1 working day between the time you complete an online SAE membership application and our system recognizes you as eligible to register your team.

A8.1 Registration Cap

The Formula Hybrid + Electric competition is capped at 35 entries. Registrations received after the entry cap is reached will be placed on a waiting list. If no slots become available prior to the competition, the entry fee will be refunded.

Registration Fees

A8.1.1 Registration fees must be paid to the organizer by the deadline specified on the Formula Hybrid + Electric website.

A8.1.2 Registration fees are not refundable.

A8.2 Withdrawals

Registered teams that find that they will not be able to attend the Formula Hybrid + Electric competition are requested to officially withdraw by notifying the organizers at the following address not later than one (1) week before the event: formulahybridcomp@gmail.com

A8.3 United States Visas

A8.3.1 Teams requiring visas to enter to the United States are advised to apply at least sixty (60) days prior to the competition. Although many visa applications go through without an unreasonable delay, occasionally teams have had difficulties and, in several instances, visas were not issued before the competition.

Don't wait – apply early for your visa.

Note: After your team has registered for the Formula Hybrid + Electric competition, the organizers can provide an acknowledgement your registration. We do not issue letters of invitation or participation certificates.

A8.3.2 Neither SAE staff nor any competition organizers are permitted to give advice on visas, customs regulations or vehicle shipping regulations concerning the United States or any other country.

ARTICLE A9 VEHICLE DOCUMENTS, DEADLINES AND PENALTIES

A9.1 Required Documents

The following documents supporting each vehicle must be submitted by the action deadlines posted on the Formula Hybrid + Electric website or otherwise published by the organizers.

Document	Section
Project Management Plan	S2.3
Electrical System Form (ESF-1)	EV13.1
Change Management Report	S2.4
Structural Equivalency Spreadsheet (SES)	T3.8
Impact Attenuator Data (IA)	T3.21
Program Submission	ARTICLE A10(f)
Site Pre-Registration	ARTICLE A10(l)
Design Report	S3.2.1
Design Specification Sheet	S3.2.2
Electrical System Form (ESF-2)	EV13.2

Table 3 - Required Documents

A9.2 Document Submission Policies

Note: Volunteer examiners and judges evaluate all the required submissions and it is essential that they have enough time to complete their work. There are no exceptions to the document submission deadlines and late submissions will incur penalties.

- A9.2.1 Document submission penalties or bonus points are factored into a team's final score. They do not alter the individual event scores.
- A9.2.2 All documents must be submitted using the following document title, unless otherwise noted in individual document requirements: **Car Number_University_Document_Competition Year** i.e. **000_Oxford University_SES_2026**
Teams are responsible for submitting properly labeled documents and will accrue applicable penalty points if documents with corrected formatting are late.
- A9.2.3 Teams must submit the required documents online at <https://formulahybridupload.supportsystem.com/>
- A9.2.4 Document submission deadlines are listed in **GMT (Greenwich Mean Time)** unless otherwise explicitly stated. This means that deadlines are typically **7:00 PM Eastern Standard Time**. [Please use this website for time conversions](https://time.is/GMT) or <https://time.is/GMT>.
- A9.2.5 The time and date that the document is uploaded is recorded in GMT (Greenwich Mean Time) and constitutes the official record for deadline compliance.
- A9.2.6 The official time and date of document receipt will be posted on the Formula Hybrid + Electric website.
Teams are responsible for ensuring the accuracy of these postings and must notify the organizers within three days of their submission to report a discrepancy. After three days the submission time and date will become final.

A9.3 Late submissions

Documents received after their deadlines will be penalized as follows:

- (a) **Structural Equivalency Spreadsheet (SES)** – The penalty for late SES submission is 10 points per day and is capped at negative fifty (-50) points. However, teams are advised that the SES's are evaluated in the order in which they are received and that late submissions will be reviewed last. Late SES approval could delay the completion of your vehicle. We strongly recommend you submit your SES as early as possible.
- (b) **Impact Attenuator Report (IA)** - The penalty for late IA submissions is 10 points per day and is capped at negative fifty (-50) points.
- (c) **Design Reports** – The Design Report and Design Spec Sheet collectively constitute the "Design Documents".
 - (i) Design Report 10 points/day up to 100 points
 - (ii) Design Spec Sheet 5 points/day up to 50 points

NOTE: The total penalty for late arrival of any design documents will not exceed 100 points.

- (d) **Program Submissions** – There are no penalties for late program submissions. However, late submissions will not be included in the race program, which can be an important tool for future team fund raising.
- (e) **Project Management Project Plan** - Late submission or failure to submit the Project Plan will be penalized at five (5) points per day and is capped at negative fifty-five (-55) points.

- (f) **Change Management Report** - Late submission or failure to submit the interim report will be penalized at five (5) points per day and is capped at negative forty (-40) points.
- (g) **ESF** - The penalty for late ESF submissions is 10 points per day and is capped at negative twenty-five points (-25) per ESF or fifty (-50) points total.
- (h) **Site Pre-Registration** – The penalty for late submission of Site Pre-Registration will be five (5) points. Teams may submit additional team member information up to one week prior to the competition, and all drivers must submit a photo of their driver’s license prior to April 26th on the “Document Upload Page”.

A9.4 Early submissions

In some cases, documents submitted before their deadline can earn a team bonus points as follows:

- (a) **Structural Equivalency Spreadsheet (SES)**
 - (i) Approved documents that were submitted 30 days or more before the SES deadline will receive 20 bonus points.
 - (ii) Approved documents that were submitted between 29 and 15 days before the SES deadline will receive 10 bonus points.
- (b) **Electrical System Forms (ESF-1 and ESF-2)**
 - (i) Approved documents that were submitted 30 days or more before each ESF deadline will receive 10 bonus points.
 - (ii) Approved documents that were submitted between 29 and 15 days before each ESF deadline will receive 5 bonus points.
 - (iii) A complete ESF-2 is required before a team may compete in the event. Submissions after the ESF deadline may result in penalties.

Note 1: The qualifying dates for bonus points will be listed on the Formula Hybrid + Electric website.

Note 2: The number of bonus points will be based on the submission date of the document, not on the approval date. Documents submitted early that are not approved will not qualify for bonus points.

Note 3: Some bonus point deadlines may occur before the closing date of registration. If a team has not previously registered for a Formula Hybrid + Electric competition, they may not be listed on the documents submission page. (A9.2.3) In that case, a team should submit the document via email to formulahybridcomp@gmail.com

ARTICLE A10 FORMS AND DOCUMENTS

The following forms and documents are available on the Formula Hybrid + Electric website:

- (a) 2026 Formula Hybrid + Electric Rules (This Document)
- (b) Structural Equivalency Spreadsheet (SES)
- (c) Impact Attenuator (IA) Data Sheet
- (d) Electrical Systems Form (ESF-1) template
- (e) Electrical Systems Form (ESF-2) template
- (f) Program Information Sheet (Team information for the Event Program)
- (g) Mechanical Inspection Sheet (For reference)
- (h) Electrical Inspection Sheet (For reference)
- (i) Design Specification Sheet
- (j) Design Event Judging Form (For reference)
- (k) Project Management Judging Form (For reference)
- (l) Site pre-registration form

Note: Formula Hybrid + Electric strives to provide student engineering teams with timely and useful information to assist in the design and construction of their vehicles. Check the Formula Hybrid + Electric website often for new or updated advisory publications.

ARTICLE A11 PROTESTS

A11.1 Protests - General

It is recognized that thousands of hours of work have gone into fielding a vehicle and that teams are entitled to all the points they can earn. We also recognize that there can be differences in the interpretation of rules, the application of penalties and the understanding of procedures. The officials and SAE staff will make every effort to fully review all questions and resolve problems and discrepancies quickly and equitably

A11.2 Preliminary Review – Required

If a team has a question about scoring, judging, policies, or any official action it must be brought to the organizer's or SAE staff's attention for an informal preliminary review before a protest can be filed.

A11.3 Cause for Protest

A team may protest any rule interpretation, score, or official action (unless specifically excluded from protest) which they feel has caused some actual, non-trivial, harm to their team, or has had substantive effect on their score. Teams may not protest rule interpretations or actions that have not caused them any substantive damage.

A11.4 Protest Format and Forfeit

All protests must be filed in writing and presented to the organizer or SAE staff by the team captain. In order to have a protest considered, a team must post a twenty-five (25) point protest bond which will be forfeited if their protest is rejected.

A11.5 Protest Period

Protests concerning any aspect of the competition must be filed within one-half hour (30 minutes) of the posting of the scores of the event to which the protest relates.

A11.6 Decision

The decision of the competition protest committee regarding any protest is final.

ARTICLE A12 QUESTIONS ABOUT THE FORMULA HYBRID + ELECTRIC RULES

A12.1 Question Publication

By submitting a question to the Formula Hybrid + Electric Rules Committee or the competition's organizing body you and your team agree that both your question and the official answer can be reproduced and distributed by SAE or Formula Hybrid + Electric, in both complete and edited versions, in any medium or format anywhere in the world.

A12.2 Question Types

- A12.2.1 The Committee will answer questions that are not already answered in the rules or FAQs or that require new or novel rule interpretations. The Committee will not respond to questions that are already answered in the rules. For example, if a rule specifies a minimum dimension for a part the Committee will not answer questions asking if a smaller dimension can be used.

A12.3 Frequently Asked Questions

- A12.3.1 Before submitting a question, check the Frequently Asked Questions section of the Formula Hybrid + Electric website.

A12.4 Question Submission

Questions must be submitted on the Formula Hybrid + Electric Support page:
<https://formulahybridelectric.supportsystem.com/>

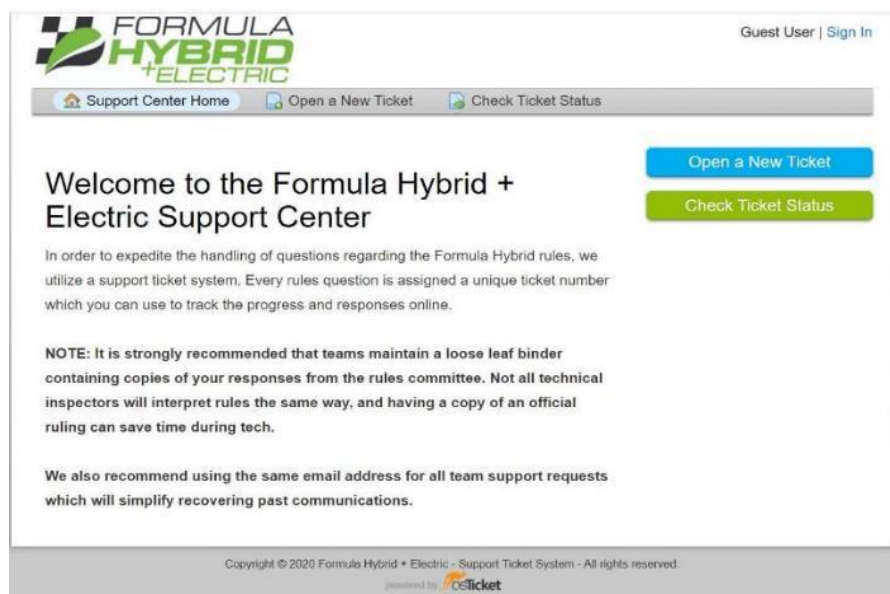


Figure 1 - Formula Hybrid + Electric Support Page

A12.5 Question Format

The following information is required:

- (a) Submitter's Name
- (b) Submitter's Email
- (c) Topic (Select from the pull-down menu)
- (d) University Name (Registered teams will find their University name in a pull-down list)

You may type your question into the "Message" box, or upload a document.

You will receive a confirmation email with a link to enable you to check on your question's status.

A12.6 Response Time

Please allow a minimum of 3 days for a response. The Rules Committee will respond as quickly as possible, however, responses to questions presenting new issues, or of unusual complexity, may take more than one week.

Please do not resend questions.

PART T - GENERAL TECHNICAL REQUIREMENTS

ARTICLE T1 VEHICLE REQUIREMENTS AND RESTRICTIONS ***

T1.1 Technical Inspection

- T1.1.1 The following requirements and restrictions will be enforced through technical inspection. Noncompliance must be corrected and the car re-inspected before the car is allowed to operate under power.

T1.2 Modifications and Repairs

- T1.2.1 Once the vehicle has been presented for judging in the Design Events, or submitted for Technical Inspection, and until the vehicle is approved to compete in the dynamic events, i.e. all the inspection stickers are awarded, the only modifications permitted to the vehicle are those directed by the Inspector(s) and noted on the Inspection Form.
- T1.2.2 Once the vehicle is approved to compete in the dynamic events, the ONLY modifications permitted to the vehicle are:
- (a) Adjustment of belts, chains and clutches
 - (b) Adjustment of brake bias
 - (c) Adjustment of the driver restraint system, head restraint, seat and pedal assembly
 - (d) Substitution of the head restraint or seat inserts for different drivers
 - (e) Adjustment to engine operating parameters, e.g. fuel mixture and ignition timing and any software calibration changes
 - (f) Adjustment of mirrors
 - (g) Adjustment of the suspension where no part substitution is required, (except that springs, sway bars and shims may be changed)
 - (h) Adjustment of tire pressure
 - (i) Adjustment of wing angle (but not the location)
 - (j) Replenishment of fluids
 - (k) Replacement of worn tires or brake pads The replacement tires and/or brake pads must be identical in material, composition and size to those presented and approved at Technical Inspection.
 - (l) The changing of wheels and tires for “wet” or “damp” conditions as allowed in [D3.1](#) of the Formula Hybrid + Electric Rules.
 - (m) Recharging of Grounded Low Voltage (GLV) supplies.
 - (n) Recharging of Accumulators. (See [EV12.2](#))
 - (o) Adjustment of motor controller operating parameters.
 - (p) Changes to vehicle’s software and operating parameters.
- T1.2.3 The vehicle must maintain all required specifications, e.g. ride height, suspension travel, braking capacity, sound level and wing location throughout the competition.
- T1.2.4 Once the vehicle is approved for competition, any damage to the vehicle that requires repair, e.g. crash damage, electrical or mechanical damage will void the Inspection Approval. Upon

the completion of the repair and before re-entering into any dynamic competition, the vehicle **MUST** be re-submitted to Technical Inspection for re-approval.

ARTICLE T2 GENERAL DESIGN REQUIREMENTS

T2.1 Vehicle Configuration

- T2.1.1 The vehicle must be open-wheeled and open-cockpit (a formula style body) with four (4) wheels that are not in a straight line.
- T2.1.2 Definition of "Open Wheel" – Open Wheel vehicles must satisfy all of the following criteria:
- (a) The top 180 degrees of the wheels/tires must be unobstructed when viewed from vertically above the wheel.
 - (b) The wheels/tires must be unobstructed when viewed from the side.
 - (c) No part of the vehicle may enter a keep-out-zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm behind the outer diameter of the front and rear tires in side view elevation of the vehicle with the tires steered straight ahead. This keep-out zone will extend laterally from the outside plane of the wheel/tire to the inboard plane of the wheel/tire. See [Figure 2](#) below.

Note: The dry tires will be used for all inspections.

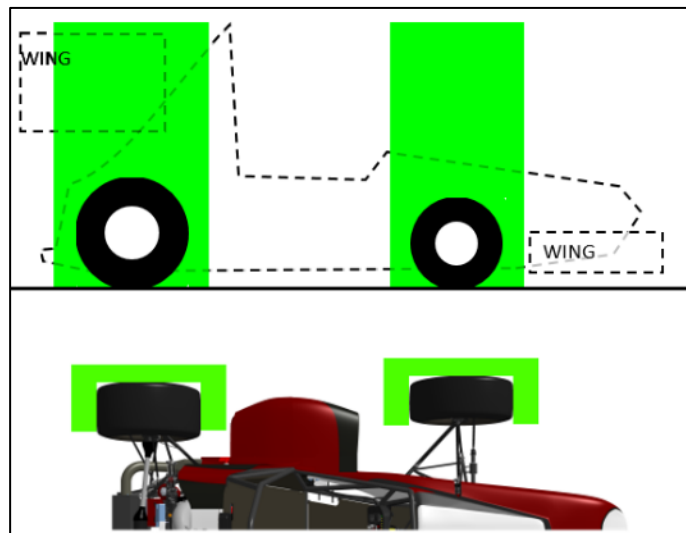


Figure 2 - Open Wheel Definition

T2.2 Bodywork

There must be no openings through the bodywork into the driver compartment from the front of the vehicle back to the roll bar main hoop or firewall other than that required for the cockpit opening. Minimal openings around the front suspension components are allowed.

T2.3 Wheelbase

The car must have a wheelbase of at least 1524 mm. The wheelbase is measured from the center of ground contact of the front and rear tires with the wheels pointed straight ahead.

T2.4 Vehicle Track

The smaller track of the vehicle (front or rear) must be no less than 75% of the larger track.

T2.5 Visible Access

All items on the Inspection Form must be clearly visible to the technical inspectors without using instruments such as endoscopes or mirrors. Visible access can be provided by removing body panels or by providing removable access panels.

ARTICLE T3 DRIVER'S CELL

T3.1 General Requirements

T3.1.1 Among other requirements, the vehicle's structure must include two roll hoops that are braced, a front bulkhead with support system and Impact Attenuator, and side impact structures.

Note: Many teams will be retrofitting Formula SAE cars for Formula Hybrid + Electric. In most cases these vehicles will be considerably heavier than what the original frame and suspension was designed to carry. It is important to analyze the structure of the car and to strengthen it as required to ensure that it will handle the additional stresses.

The technical inspectors will also be paying close attention to the mounting of accumulator systems. These can be very heavy and must be adequately fastened to the main structure of the vehicle.

T3.2 Definitions

The following definitions apply throughout the Rules document:

- (a) Main Hoop - A roll bar located alongside or just behind the driver's torso.
- (b) Front Hoop - A roll bar located above the driver's legs, in proximity to the steering wheel.
- (c) Roll Hoops – Both the Front Hoop and the Main Hoop are classified as "Roll Hoops"
- (d) Roll Hoop Bracing Supports – The structure from the lower end of the Roll Hoop Bracing back to the Roll Hoop(s).
- (e) Frame Member - A minimum representative single piece of uncut, continuous tubing.
- (f) Frame - The "Frame" is the fabricated structural assembly that supports all functional vehicle systems. This assembly may be a single welded structure, multiple welded structures or a combination of composite and welded structures.
- (g) Primary Structure – The Primary Structure is comprised of the following Frame components:
 - (i) Main Hoop
 - (ii) Front Hoop
 - (iii) Roll Hoop Braces and Supports
 - (iv) Side Impact Structure
 - (v) Front Bulkhead

- (vi) Front Bulkhead Support System
- (vii) All Frame Members, guides and supports that transfer load from the Driver's Restraint System into items (i) through (vi).
- (h) Major Structure of the Frame – The portion of the Frame that lies within the envelope defined by the Primary Structure. The upper portion of the Main Hoop and the Main Hoop Bracing are not included in defining this envelope.
- (i) Front Bulkhead – A planar structure that defines the forward plane of the Major Structure of the Frame and functions to provide protection for the driver's feet.
- (j) Impact Attenuator – A deformable, energy absorbing device located forward of the Front Bulkhead.
- (k) Side Impact Zone – The area of the side of the car extending from the top of the floor to 350 mm above the ground and from the Front Hoop back to the Main Hoop.
- (l) Node-to-node triangulation – An arrangement of frame members projected onto a plane, where a co-planar load applied in any direction, at any node, results in only tensile or compressive forces in the frame members. This is also what is meant by “properly triangulated”.

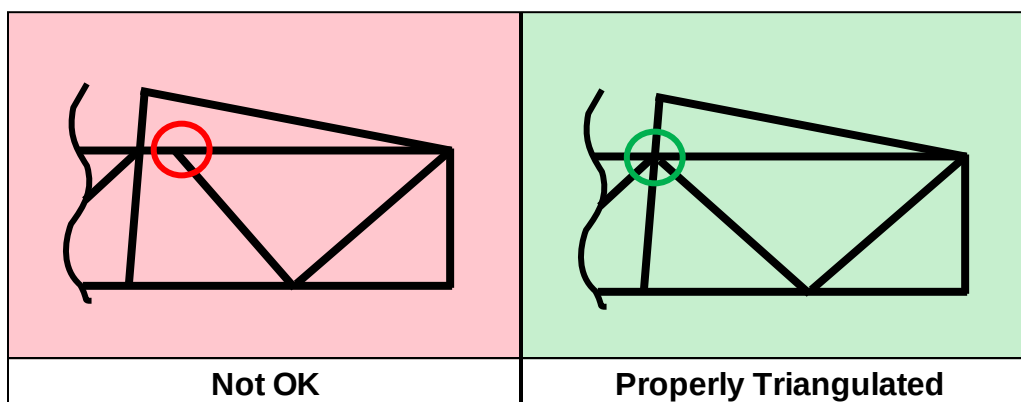


Figure 3 - Triangulation

T3.3 Minimum Material Requirements

T3.3.1 Baseline Steel Material

The Primary Structure of the car must be constructed of:

Either: Round, mild or alloy, steel tubing (minimum 0.1% carbon) of the minimum dimensions specified in [Table 4](#).

Or: Approved alternatives per Rules [T3.3](#), [T3.3.2](#), [T3.5](#) and [T3.6](#).

ITEM or APPLICATION	OUTSIDE DIMENSION x WALL THICKNESS	
	Inch	Metric
Main & Front Hoops, Shoulder Harness Mounting Bar	Round: 1.0" x 0.095"	Round: 25.0 mm x 2.50 mm
Side Impact Structure	Round: 1.0" x 0.065"	Round: 25.0 mm x 1.75 mm
Roll Hoop Bracing	Square: 1.0" x 1.0" x 0.049"	Square: 25.0 mm x 25.0 mm x 1.25 mm
Front Bulkhead		Square: 26.0 mm x 26.0 mm x 1.2 mm
Driver's Restraint Harness Attachment (except for Shoulder Harness Mounting Bar)		
Main Hoop Bracing Supports	Round: 1.0" x 0.049"	Round: 25.0 mm x 1.5 mm
Front Bulkhead Supports		Round: 26.0 mm x 1.2 mm
Protection of Tractive System Components		

Table 4 - Baseline Steel

Note 1: The use of alloy steel does not allow the wall thickness to be thinner than that used for mild steel.

Note 2: For a specific application using tubing of the specified outside diameter but with greater wall thickness, **or** of the specified wall thickness and a greater outside diameter, **or** replacing round tubing with square tubing of the same or larger size to those listed above, are NOT rules deviations requiring approval.

Note 3: Except for inspection holes, any holes drilled in any regulated tubing require the submission of an SES.

Note 4: Baseline steel properties used for calculations to be submitted in an SES may not be lower than the following:

Bending and buckling strength calculations:

Young's Modulus (E) =	200 GPa (29,000 ksi)
Yield Strength (S_y) =	305 MPa (44.2 ksi)
Ultimate Strength (S_u) =	365 MPa (52.9 ksi)

Welded monocoque attachment points or welded tube joint calculations:

Yield Strength (S_y) =	180 MPa (26 ksi)
Ultimate Strength (S_u) =	300 MPa (43.5 ksi)

T3.3.2 When a cutout, or a hole greater in diameter than 3/16 inch (4 mm), is made in a regulated tube, e.g. to mount the safety harness or suspension and steering components, in order to regain the baseline, cold rolled strength of the original tubing, the tubing must be reinforced by the use of

a welded insert or other reinforcement. The welded strength figures given above must be used for the additional material. And the details, including dimensioned drawings, must be included in the SES.

T3.4 Alternative Tubing and Material - General

- T3.4.1 Alternative tubing geometry and/or materials may be used except that the Main Roll Hoop and Main Roll Hoop Bracing must be made from steel, i.e. the use of aluminum or titanium tubing or composites for these components is prohibited.
- T3.4.2 Titanium or magnesium on which welding has been utilized may not be used for any part of the Primary Structure. This includes the attachment of brackets to the tubing or the attachment of the tubing to other components.
- T3.4.3 If a team chooses to use alternative tubing and/or materials they must still submit a “Structural Equivalency Spreadsheet” per Rule **T3.8**. The teams must submit calculations for the material they have chosen, demonstrating equivalence to the minimum requirements found in Section **T3.3.1** for yield and ultimate strengths in bending, buckling and tension, for buckling modulus and for energy dissipation.

Note: The Buckling Modulus is defined as EI , where, E = modulus of Elasticity, and I = area moment of inertia about the weakest axis.

- T3.4.4 To be considered as a structural tube in the SES Submission (**T3.8**) tubing cannot have an outside dimension less than 25 mm or a wall thickness less than that listed in **T3.5** or **T3.6**.
- T3.4.5 If a bent tube is used anywhere in the primary structure, other than the front and main roll hoops, an additional tube must be attached to support it. The attachment point must be the position along the tube where it deviates farthest from a straight line connecting both ends. The support tube must have the same diameter and thickness as the bent tube. The support tube must terminate at a node of the chassis.
- T3.4.6 Any chassis design that is a hybrid of the baseline and monocoque rules, must meet all relevant rules requirements, e.g. a sandwich panel side impact structure in a tube frame chassis must meet the requirements of rules **T3.27**, **T3.28**, **T3.29**, **T3.30** and **T3.33**.

Note: It is allowable for the properties of tubes and laminates to be combined to prove equivalence. E.g. in a side-impact structure consisting of one tube as per **T3.3** and a laminate panel, the panel only needs to be equivalent to two side-impact tubes.

T3.5 Alternative Steel Tubing

Minimum Wall Thickness Allowed:

MATERIAL & APPLICATION	MINIMUM WALL THICKNESS
Front and Main Roll Hoops Shoulder Harness Mounting Bar	2.0 mm
Roll Hoop Bracing Roll Hoop Bracing Supports Side Impact Structure Front Bulkhead Front Bulkhead Support Driver's Harness Attachment (Except for Shoulder Harness Mounting Bar - above) Protection of accumulators Protection of TSV components	1.2 mm

Table 5 - Steel Tubing Minimum Wall Thicknesses

Note 1: All steel is treated equally - there is no allowance for alloy steel tubing, e.g. SAE 4130, to have a thinner wall thickness than that used with mild steel.

Note 2: To maintain EI with a thinner wall thickness than specified in **T3.3.1**, the outside diameter MUST be increased.

Note 3: To maintain the equivalent yield and ultimate tensile strength the same cross-sectional area of steel as the baseline tubing specified in **T3.3.1** must be maintained.

T3.6 Aluminum Tubing Requirements

T3.6.1 Minimum Wall Thickness of Aluminum Tubing is 3.0 mm

T3.6.2 The equivalent yield strength must be considered in the “as-welded” condition, (Reference: *WELDING ALUMINUM* (latest Edition) by the Aluminum Association, or *THE WELDING HANDBOOK*, Volume 4, 9th Ed., by The American Welding Society), unless the team demonstrates and shows proof that the frame has been properly solution heat treated and artificially aged.

T3.6.3 Should aluminum tubing be solution heat-treated and age hardened to increase its strength after welding; the team must supply sufficient documentation as to how the process was performed. This includes, but is not limited to, the heat-treating facility used, the process applied, and the fixturing used.

T3.7 Composite Materials

T3.7.1 If any composite or other material is used, the team must present documentation of material type, e.g. purchase receipt, shipping document or letter of donation, and of the material properties. Details of the composite lay-up technique as well as the structural material used (cloth type, weight, and resin type, number of layers, core material, and skin material if metal) must also be submitted. The team must submit calculations demonstrating equivalence of their composite structure to one of similar geometry made to the minimum requirements found in Section **T3.3.1**. Equivalency calculations must be submitted for energy dissipation, yield and ultimate strengths in bending, buckling, and tension. Submit the completed “Structural Equivalency Spreadsheet” per Section **T3.8**

Note: Some composite materials present unique electrical shock hazards, and may require additional engineering and fabrication effort to minimize those hazards. See: [ARTICLE EV8](#).

T3.7.2 Composite materials are not allowed for the Main Hoop or the Front Hoop.

T3.8 Structural Documentation – SES Submission

All equivalency calculations must prove equivalency relative to steel grade SAE/AISI 1010.

T3.8.1 All teams must submit a Structural Equivalency Spreadsheet (SES) that represents the vehicle entered in the 2026 FH+E competition even if they are not planning to use alternative materials or tubing sizes to those specified in [T3.3.1](#) Baseline Steel Materials.

T3.8.2 The use of alternative materials or tubing sizes to those specified in [T3.3.1](#) “Baseline Steel Material,” is allowed, provided they have been judged by a technical review to have equal or superior properties to those specified in [T3.3.1](#).

T3.8.3 Approval of alternative material or tubing sizes will be based upon the engineering judgment and experience of the chief technical inspector or their appointee.

T3.8.4 The technical review is initiated by completing the “Structural Equivalency Spreadsheet” (SES) which can be downloaded from the Formula Hybrid + Electric website.

T3.8.5 Structural Equivalency Spreadsheet – Submission

SEs must be submitted via the Formula Hybrid + Electric Document Upload page. See Section [A9.2](#).

Do Not Resubmit SES’s unless instructed to do so.

T3.8.6 Vehicles completed under an approved SES must be fabricated in accordance with the materials and processes described in the SES.

T3.8.7 Teams must bring a copy of the approved SES with them to Technical Inspection.

Comment - The resubmission of an SES that was written and submitted for a competition in a previous year is strongly discouraged. Each team is expected to perform their own tests and to submit SEs based on their original work. Understanding the engineering that justifies the equivalency is essential to discussing your work with the officials.

T3.8.8 An approved SES for a Formula SAE 2025 or 2026 competition may be submitted in place of the Formula Hybrid + Electric specific SES required by [T3.8.4](#).

T3.9 Main and Front Roll Hoops – General Requirements

T3.9.1 The driver’s head and hands must not contact the ground in any rollover attitude.

T3.9.2 The Frame must include both a Main Hoop and a Front Hoop as shown in [Figures from Top: 4](#).

T3.9.3 When seated normally and restrained by the Driver’s Restraint System, the helmet of a 95th percentile male (anthropometrical data; See [Table 6](#) and [Figure 5](#)) and all of the team’s drivers must:

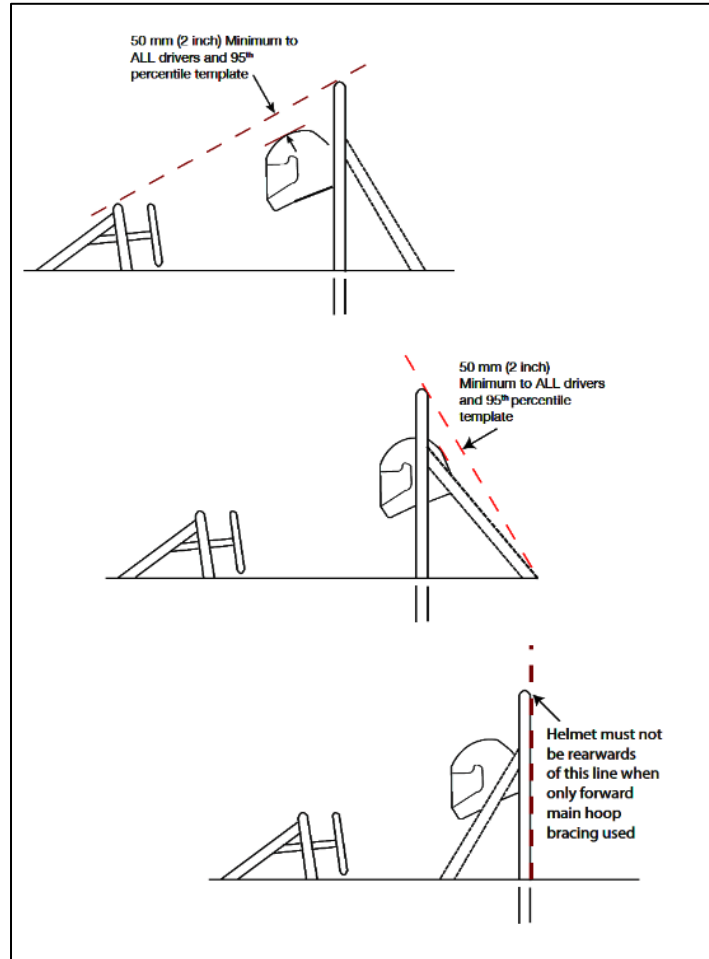
(a) Be a minimum of 50.8 mm from the straight line drawn from the top of the main hoop to the top of the front hoop. ([Figures from Top: 4a](#))

- (b) Be a minimum of 50.8 mm from the straight line drawn from the top of the main hoop to the lower end of the main hoop bracing if the bracing extends rearwards. (**Figures from Top: 4b**)
- (c) Be no further rearwards than the rear surface of the main hoop if the main hoop bracing extends forwards. (**Figures from Top: 4c**)

A two-dimensional template used to represent the 95th percentile male is made to the following dimensions:

- A circle of diameter 200 mm will represent the hips and buttocks.
- A circle of diameter 200 mm will represent the shoulder/cervical region.
- A circle of diameter 300 mm will represent the head (with helmet).
- A straight line measuring 490 mm will connect the centers of the two 200 mm circles.
- A straight line measuring 280 mm will connect the centers of the upper 200 mm circle and the 300 mm head circle.

Table 6 - 95th Percentile Male Template Dimensions



Figures from Top: 4a, 4b, and 4c- Roll Hoops and Helmet Clearance

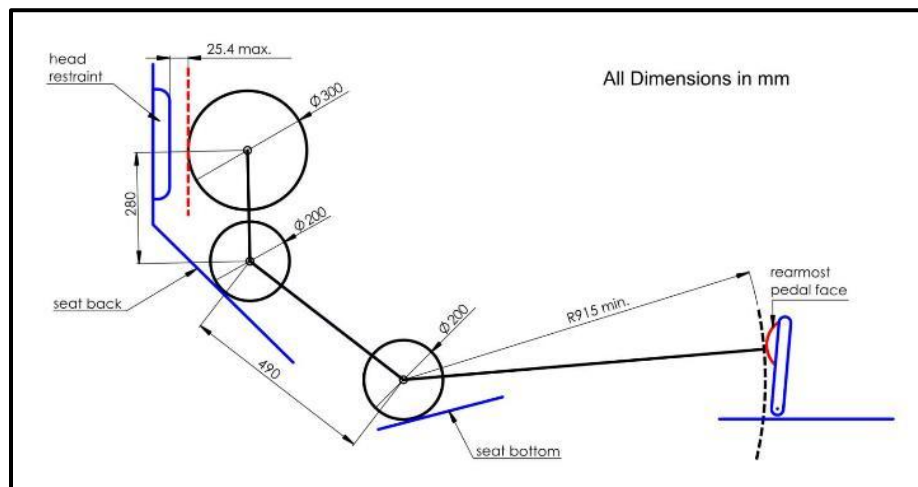


Figure 5 - Percy -- 95th Percentile Male with Helmet

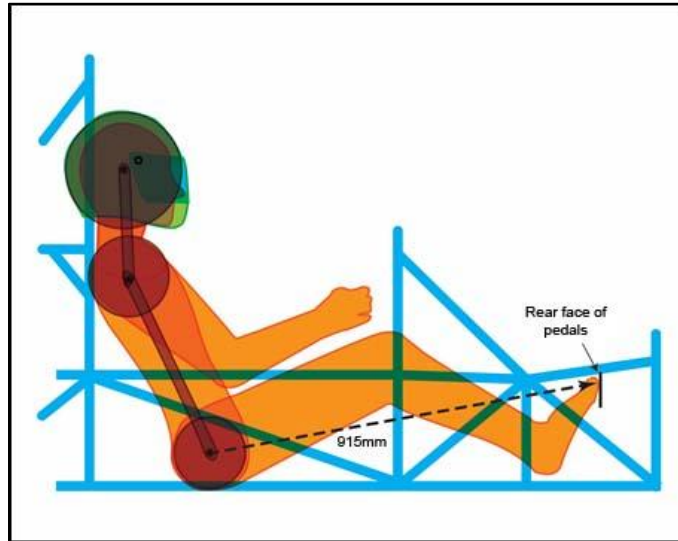


Figure 6 – 95th Percentile Template Positioning

- T3.9.4 The 95th percentile male template (Percy) will be positioned as follows: (See [Figure 6](#))
- (a) The seat will be adjusted to the rearmost position,
 - (b) The pedals will be placed in the most forward position.
 - (c) The bottom 200 mm circle will be placed on the seat bottom such that the distance between the center of this circle and the rearmost face of the pedals is no less than 915 mm.
 - (d) The middle 200 mm circle, representing the shoulders, will be positioned on the seat back.
 - (e) The upper 300 mm circle will be positioned no more than 25.4 mm away from the head restraint (i.e. where the driver's helmet would normally be located while driving).



IMPORTANT: If the requirements of [T3.9.3](#) are not met with the 95th percentile male template, the car will not receive a Technical Inspection Sticker and will not be allowed to compete in the dynamic events.

- T3.9.5 Drivers who do not meet the helmet clearance requirements of [T3.9.3](#) will not be allowed to drive in the competition.
- T3.9.6 The minimum radius of any bend, measured at the tube centerline, must be at least three times the tube outside diameter. Bends must be smooth and continuous with no evidence of crimping or wall failure.
- T3.9.7 The Main Hoop and Front Hoop must be securely integrated into the Primary Structure using gussets and/or tube triangulation.
- T3.10 Main Hoop**
- T3.10.1 The Main Hoop must be constructed of a single piece of uncut, continuous, closed section steel tubing per Rule [T3.3.1](#)

- T3.10.2 The use of aluminum alloys, titanium alloys or composite materials for the Main Hoop is prohibited.
- T3.10.3 The Main Hoop must extend from the lowest Frame Member on one side of the Frame, up, over and down the lowest Frame Member on the other side of the Frame.
- T3.10.4 In the side view of the vehicle, the portion of the Main Roll Hoop that lies above its attachment point to the Major Structure of the Frame must be within ten degrees (10°) of the vertical.
- T3.10.5 In the side view of the vehicle, any bends in the Main Roll Hoop above its attachment point to the Major Structure of the Frame must be braced to a node of the Main Hoop Bracing Support structure with tubing meeting the requirements of Roll Hoop Bracing as per Rule [T3.3.1](#)
- T3.10.6 In the front view of the vehicle, the vertical members of the Main Hoop must be at least 380 mm apart (inside dimension) at the location where the Main Hoop is attached to the Major Structure of the Frame.

T3.11 Front Hoop

- T3.11.1 The Front Hoop must be constructed of closed section metal tubing per Rule [T3.3.1](#).
- T3.11.2 The Front Hoop must extend from the lowest Frame Member on one side of the Frame, up, over and down to the lowest Frame Member on the other side of the Frame.
- T3.11.3 With proper gusseting and/or triangulation, it is permissible to fabricate the Front Hoop from more than one piece of tubing.
- T3.11.4 The top-most surface of the Front Hoop must be no lower than the top of the steering wheel in any angular position.
- T3.11.5 The Front Hoop must be no more than 250 mm forward of the steering wheel. This distance must be measured horizontally, on the vehicle centerline, from the rear surface of the Front Hoop to the forward most surface of the steering wheel rim with the steering in the straight-ahead position.
- T3.11.6 In side view, no part of the Front Hoop can be inclined at more than twenty degrees (20°) from the vertical.

T3.12 Main Hoop Bracing

- T3.12.1 Main Hoop braces must be constructed of closed section steel tubing per Rule [T3.3.1](#).
- T3.12.2 The Main Hoop must be supported by two braces extending in the forward or rearward direction on both the left and right sides of the Main Hoop.
- T3.12.3 In the side view of the Frame, the Main Hoop and the Main Hoop braces must not lie on the same side of the vertical line through the top of the Main Hoop, i.e. if the Main Hoop leans forward, the braces must be forward of the Main Hoop, and if the Main Hoop leans rearward, the braces must be rearward of the Main Hoop.
- T3.12.4 The Main Hoop braces must be attached as near as possible to the top of the Main Hoop but not more than 160 mm below the top-most surface of the Main Hoop. The included angle formed by the Main Hoop and the Main Hoop braces must be at least thirty degrees (30°) in side view. See: [Figure 7](#)

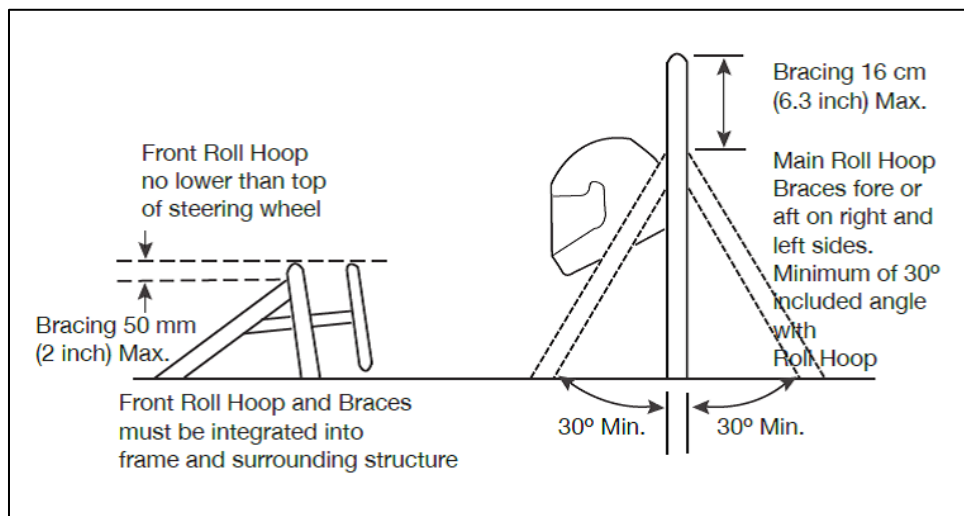


Figure 7 - Main and Front Hoop Bracing

- T3.12.5 The Main Hoop braces must be straight, i.e. without any bends.
- T3.12.6 The attachment of the Main Hoop braces must be capable of transmitting all loads from the Main Hoop into the Major Structure of the Frame without failing. From the lower end of the braces there must be a properly triangulated structure back to the lowest part of the Main Hoop and the node at which the upper side impact tube meets the Main Hoop. This structure must meet the minimum requirements for Main Hoop Bracing Supports (see Rule [T3.3](#)) or an SES approved alternative. Bracing loads must not be fed solely into the engine, transmission or differential, or through suspension components.
- T3.12.7 If any item which is outside the envelope of the Primary Structure is attached to the Main Hoop braces, then additional bracing must be added to prevent bending loads in the braces in any rollover attitude.
- T3.13 Front Hoop Bracing**
- T3.13.1 Front Hoop braces must be constructed of material per Rule [T3.3.1](#).
- T3.13.2 The Front Hoop must be supported by two braces extending in the forward direction, one on the left side and one on the right side of the Front Hoop.
- T3.13.3 The Front Hoop braces must be constructed such that they protect the driver's legs and should extend to the structure in front of the driver's feet.
- T3.13.4 The Front Hoop braces must be attached as near as possible to the top of the Front Hoop but not more than 50.8 mm below the top-most surface of the Front Hoop. See: [Figure 7](#)
- T3.13.5 If the Front Hoop leans rearwards by more than ten degrees (10°) from the vertical, it must be supported by additional bracing to the rear. This bracing must be constructed of material per Rule [T3.3.1](#).

T3.14 Other Bracing Requirements

Where the braces are not welded to steel Frame Members, the braces must be securely attached to the Frame using 8 mm Metric Grade 8.8 (5/16 in SAE Grade 5), or stronger, bolts. Mounting plates welded to the Roll Hoop braces must be at least 2.0 mm thick steel.

T3.15 Other Side Tube Requirements

If there is a Roll Hoop brace or other frame tube alongside the driver, at the height of the neck of any of the team's drivers, a metal tube or piece of sheet metal must be firmly attached to the Frame to prevent the drivers' shoulders from passing under the roll hoop brace or frame tube, and his/her neck contacting this brace or tube.

T3.16 Mechanically Attached Roll Hoop Bracing

T3.16.1 Roll Hoop bracing may be mechanically attached.

T3.16.2 Any non-permanent joint at either end must be either a double-lug joint as shown in [Figure 8](#) and [Figure 9](#) or a sleeved butt joint as shown in [Figure 10](#).

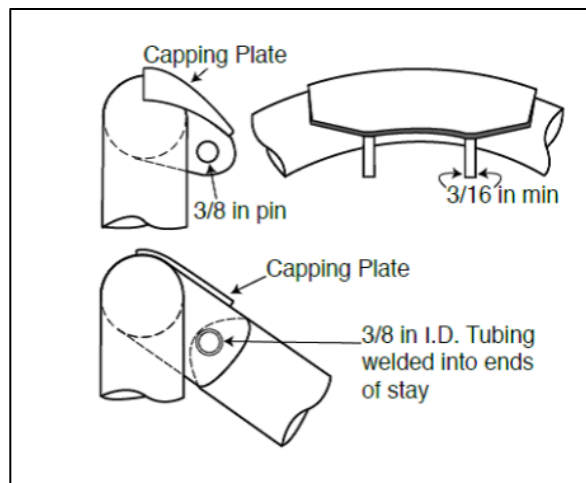


Figure 8 – Double-Lug Joint

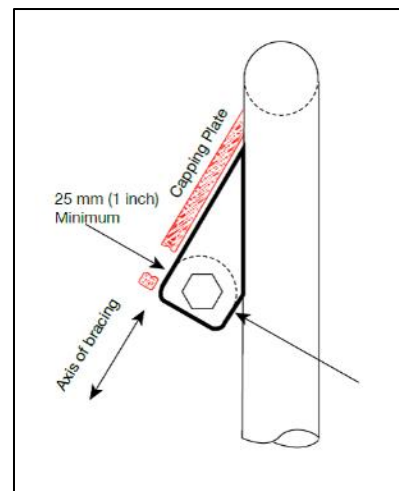


Figure 9 – Double Lug Joint

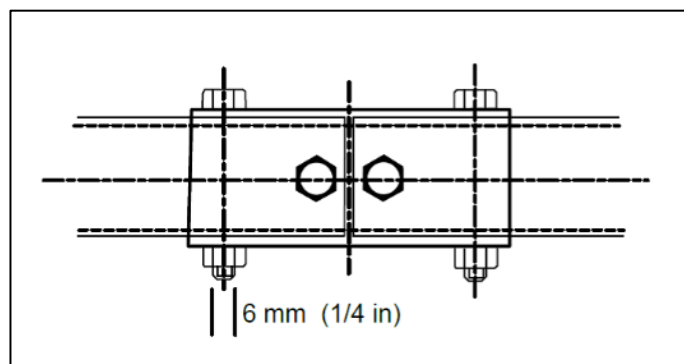


Figure 10 – Sleeved Butt Joint

- T3.16.3 The threaded fasteners used to secure non-permanent joints are considered critical fasteners and must comply with **ARTICLE T11**.
- T3.16.4 No spherical rod ends are allowed.
- T3.16.5 For double-lug joints, each lug must be at least 4.5 mm thick steel, measure 25 mm minimum perpendicular to the axis of the bracing and be as short as practical along the axis of the bracing.
- T3.16.6 All double-lug joints, whether fitted at the top or bottom of the tube, must include a capping arrangement. (See **Figure 8** and **Figure 9**)
- T3.16.7 In a double-lug joint the pin or bolt must be 10 mm Grade 9.8 or 3/8 inch SAE Grade 8 minimum. The attachment holes in the lugs and in the attached bracing must be a close fit with the pin or bolt.
- T3.16.8 For sleeved butt joints (**Figure 10**), the sleeve must have a minimum length of 76 mm (38 mm on either side of the joint) and be a close-fit around the base tubes. The wall thickness of the sleeve must be at least that of the base tubes. The bolts must be 6 mm Grade 9.8 or 1/4 inch SAE Grade 8 minimum. The holes in the sleeves and tubes must be a close-fit with the bolts.

T3.17 Frontal Impact Structure

- T3.17.1 The driver's feet *and* legs must be completely contained within the Major Structure of the Frame. While the driver's feet are touching the pedals, in side and front views no part of the driver's feet *or* legs can extend above or outside of the Major Structure of the Frame.
- T3.17.2 Forward of the Front Bulkhead must be an energy-absorbing Impact Attenuator.

T3.18 Bulkhead

- T3.18.1 The Front Bulkhead must be constructed of closed section tubing per Rule **T3.3.1**.
- T3.18.2 Except as allowed by **T3.22.2**, The Front Bulkhead must be located forward of all non-crushable objects, e.g. batteries, master cylinders, hydraulic reservoirs.
- T3.18.3 The Front Bulkhead must be located such that the soles of the driver's feet, when touching but not applying the pedals, are rearward of the bulkhead plane. (This plane is defined by the forward-most surface of the tubing.) Adjustable pedals must be in the forward most position.

T3.19 Front Bulkhead Support

- T3.19.1 The Front Bulkhead must be securely integrated into the Frame.
- T3.19.2 The Front Bulkhead must be supported back to the Front Roll Hoop by a minimum of three (3) Frame Members on each side of the vehicle with one at the top (within 50.8 mm of its top-most surface), one (1) at the bottom, and one (1) as a diagonal brace to provide triangulation.
- T3.19.3 The triangulation must be node-to-node, with triangles being formed by the Front Bulkhead, the diagonal and one of the other two required Front Bulkhead Support Frame Members.
- T3.19.4 All the Frame Members of the Front Bulkhead Support system listed above must be constructed of closed section tubing per Section **T3.3.1**.

T3.20 Impact Attenuator (IA)

- T3.20.1 On all cars there must be an Impact Attenuator and an Anti-Intrusion Plate forward of the Front Bulkhead, with the Anti-Intrusion Plate between the Impact Attenuator and the Front Bulkhead.

All methods of attachment of the IA to the Anti-Intrusion Plate and of the Anti-Intrusion Plate to the Front Bulkhead must provide adequate load paths for transverse and vertical loads in the event of off-axis impacts.

T3.20.2 The Anti-Intrusion Plate must:

- (a) Be a 1.5 mm (0.060 in) thick solid steel or 4.0 mm (0.157 in) thick solid aluminum plate. Monocoques may use an approved alternative as per **T3.38**.
- (b) Be attached securely and directly to the Front Bulkhead.
- (c) Have an outer profile that meets the requirements of **T3.20.3**.

T3.20.3 Alternative designs of the Anti-Intrusion Plate required by **T3.20.2** that do not comply with the minimum specifications given above require an approved “Structural Equivalency Spreadsheet”, per **T3.8**. Equivalency must also be proven for perimeter shear strength of the proposed design.

T3.20.4 The requirements for the outside profile of the Anti-Intrusion Plate are dependent on the method of attachment to the Front Bulkhead:

- For welded joints the profile must extend at least to the centerline of the Front Bulkhead tubes on all sides.
- For bolted joints the profile must match the outside dimensions of the Front Bulkhead around the entire periphery.

T3.20.5 For tube frame cars, the accepted methods of attaching the Anti-Intrusion Plate to the Front Bulkhead are:

- (a) Welding, where the welds are either continuous or interrupted. If interrupted, the weld/space ratio must be at least 1:1. All weld lengths must be greater than 25 mm (1”).
- (b) Bolted joints, using a minimum of eight (8) 8 mm Metric Grade 8.8 (5/16” SAE Grade 5) bolts with positive locking. The distance between any two bolt centers must be at least 50 mm (2”).

NOTE: Holes in mandated tubes will require appropriate measures to ensure compliance with **T3.3.1** Note 3, and **T3.3.2**

T3.20.6 For monocoque cars, the attachment of the Anti-Intrusion Plate to the monocoque structure must be documented in the team’s SES submission. This must prove the attachment method is equivalent to the bolted joints described in **T3.20.5** and that the attachment method utilized will fail before any other part of the monocoque.

T3.20.7 The Impact Attenuator must be:

- (a) At least 200 mm (7.8 in) long, with its length oriented along the fore/aft axis of the Frame.
- (b) At least 100 mm (3.9 in) high and 200 mm (7.8 in) wide for a minimum distance of 200 mm (7.8 in) forward of the Front Bulkhead.
- (c) Attached securely to the Anti-Intrusion Plate.

Segmented foam attenuators must have all segments bonded together to prevent sliding or parallelogramming.

T3.20.8 The accepted methods of attaching the Impact Attenuator to the Anti-Intrusion Plate are:

- (a) Bolted joints, using a minimum of four (4) 8 mm Metric Grade 8.8 (5/16" SAE Grade 5) bolts with positive locking. The distance between any two bolt centers must be at least 50 mm (2").
- (b) By the use of a structural adhesive. The adhesive must be appropriate for use with both substrate types. Appropriate adhesive choice, substrate preparation, and equivalency of this bonded joint to the bolted joint in **T3.20.8(a)** must be documented in the team's IAD Report.

Note: Foam IA's cannot be attached solely by the bolted method.

- T3.20.9 If a team uses the "standard" FSAE Impact Attenuator³, and the outside profile of the Anti-Intrusion Plate extends beyond the "standard" Impact Attenuator by more than 25 mm (1") on any side, a diagonal or X-brace made from 1.00" x 0.049" steel tube, or an approved equivalent per **T3.5**, must be included in the Front Bulkhead.

T3.21 Impact Attenuator Test Data Report Requirement

T3.21.1 Impact Attenuator Test Data Report Requirement

All teams, whether they are using their own design of Impact Attenuator (IA) or the "standard" FSAE Impact Attenuator, must submit an Impact Attenuator Data Report using the Impact Attenuator Data (IAD) Template found on the Formula Hybrid + Electric Document page at: <https://www.formula-hybrid.org/documents>

- T3.21.2 All teams must submit calculations and/or test data to show that their Impact Attenuator, when mounted on the front of their vehicle and run into a solid, non-yielding impact barrier with a velocity of impact of 7.0 meters/second, would give an average deceleration of the vehicle not to exceed 20 g, with a peak deceleration less than or equal to 40 g's.

NOTE 1: Quasi-static testing is allowed.

NOTE 2: The calculations of how the reported absorbed energy, average deceleration and peak deceleration figures have been derived from the test data **MUST** be included in the report and appended to the report template.

- T3.21.3 Calculations must be based on the actual vehicle mass with a 175 lb. driver, full fluids, and rounded up to the nearest 100 lb.

Note: Teams may only use the "Standard" FSAE impact attenuator design and data submission process if their vehicle mass with driver is 300 kgs (661 lbs) or less.

To be eligible to utilize the "Standard" FSAE impact attenuator, teams must either have previously brought a Formula Hybrid + Electric vehicle that weighs under 300 kgs (661 lbs) with driver to a Formula Hybrid + Electric competition, or submit a detailed mass measurement spreadsheet covering all the vehicle's components as part of the Impact Attenuator Data Report, showing that the new vehicle meets the above requirements.

- T3.21.4 Teams using a front wing must prove that the combination of the Impact Attenuator and front wing, when used together, do not exceed the peak deceleration of rule **T3.21.2**. Teams can use the following methods to show the design does not exceed the limits given in **T3.21.2**.

³ The officially approved "standard" Formula SAE Impact Attenuator may be [found here on the Formula SAE website](#)

- (a) Physical testing of the Impact Attenuator with wing mounts, links, vertical plates, and a structural representation of the airfoil section to determine the peak force. See <http://formula-hybrid.org/students/tech-support/> FAQs for an example of the structure to be included in the test. Or
 - (b) Combine the peak force from physical testing of the Impact Attenuator Assembly with the wing mount failure load calculated from fastener shear and/or link buckling. Or
 - (c) If they are using the Standard FSAE Impact Attenuator (see **T3.21.3** above), combine the peak load of 95 kN exerted by the Standard FSAE Impact Attenuator with the wing mount failure load calculated from fastener shear and/or buckling.
- T3.21.5 When using acceleration data, the average deceleration must be calculated based on the raw data. The peak deceleration can be assessed based on the raw data, and if peaks above the 40g limit are apparent in the data, it can then be filtered with a Channel Filter Class (CFC) 60 (100 Hz) filter per SAE Recommended Practice J211 “Instrumentation for Impact Tests”, or a 100 Hz, 3rd order, lowpass Butterworth (-3dB at 100 Hz) filter.
- T3.21.6 A schematic of the test method must be supplied along with photos of the attenuator before and after testing.
- T3.21.7 The test piece must be presented at technical inspection for comparison to the photographs and the attenuator fitted to the vehicle.
- T3.21.8 The report must be submitted electronically in Adobe Acrobat ® format (*.pdf file) to the address and by the date provided in the Action Deadlines provided on the Formula Hybrid + Electric website. This material must be a single file (text, drawings, data or whatever you are including).
- T3.21.9 The Impact Attenuator Data Report must be named as follows:
 carnumber_schoolname_competitioncode_IAD.pdf using the assigned car number, the complete school name and competition code e.g.
087_University of SAE_FHE_IAD.pdf
- T3.21.10 Teams that submit their Impact Attenuator Data Report after the due date will be penalized as listed in section **A9.2**.
- T3.21.11 Impact Attenuator Reports will be evaluated by the organizers and the evaluations will be passed to the Design Event Captain for consideration in that event.
- T3.21.12 During the test, the attenuator must be attached to the Anti-Intrusion Plate using the intended vehicle attachment method. The Anti-Intrusion Plate must be spaced at least 50 mm from any rigid surface. No part of the Anti-Intrusion Plate may permanently deflect more than 25.4 mm beyond the position of the Anti-Intrusion Plate before the test.
- Note:** The 25.4 mm spacing represents the front bulkhead support and insures that the plate does not intrude excessively into the cockpit.
- T3.21.13 Dynamic testing (sled, pendulum, drop tower, etc.) of the impact attenuator may only be done at a dedicated test facility. The test facility may be part of the University but must be supervised by professional staff or University faculty. Teams are not allowed to construct their own dynamic test apparatus. Quasi-static testing may be performed by teams using their universities facilities/equipment, but teams are advised to exercise due care when performing all tests.

T3.22 Non-Crushable Objects

- T3.22.1 Except as allowed by [T3.22.2](#), all non-crushable objects (e.g. batteries, master cylinders, hydraulic reservoirs) inside the primary structure must have 25 mm (1”) clearance to the rear face of the Impact Attenuator Anti-Intrusion Plate.
- T3.22.2 The front wing and wing supports may be forward of the Front Bulkhead, but may NOT be located in or pass through the Impact Attenuator. If the wing supports are in front of the Front Bulkhead, the supports must be included in the test of the Impact Attenuator. See [T3.21](#).

T3.23 Front Bodywork

- T3.23.1 Sharp edges on the forward facing bodywork or other protruding components are prohibited.
- T3.23.2 All forward facing edges on the bodywork that could impact people, e.g. the nose, must have forward facing radii of at least 38 mm. This minimum radius must extend to at least forty-five degrees (45°) relative to the forward direction, along the top, sides and bottom of all affected edges.

T3.24 Side Impact Structure for Tube Frame Cars

The Side Impact Structure must meet the requirements listed below.

- T3.24.1 The Side Impact Structure for tube frame cars must be comprised of at least three (3) tubular members located on each side of the driver while seated in the normal driving position, as shown in [Figure 11](#)

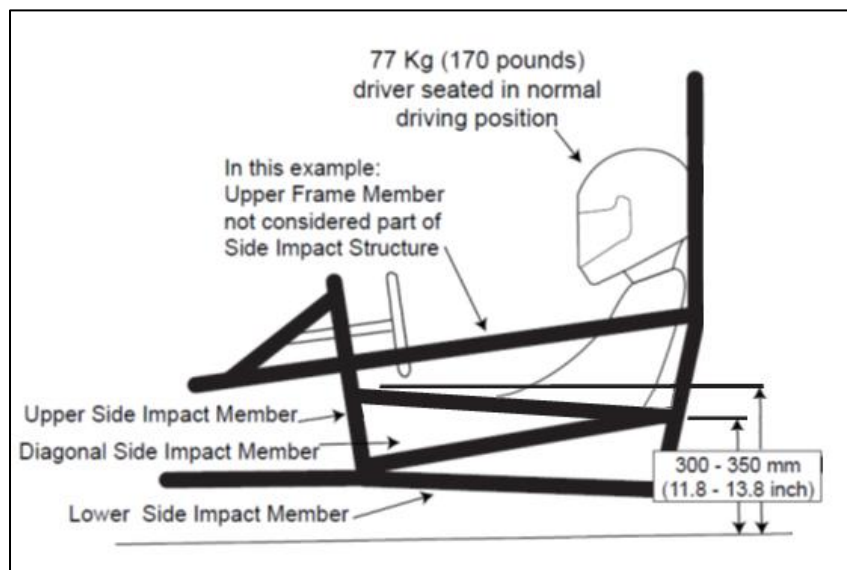


Figure 11 – Side Impact Structure

- T3.24.2 The three (3) required tubular members must be constructed of material per Section [T3.3](#).
- T3.24.3 The locations for the three (3) required tubular members are as follows:
- The upper Side Impact Structural member must connect the Main Hoop and the Front Hoop. With a 77 kg driver seated in the normal driving position all of the member must be

at a height between 300 mm and 350 mm above the ground. The upper frame rail may be used as this member if it meets the height, diameter and thickness requirements.

- (b) The lower Side Impact Structural member must connect the bottom of the Main Hoop and the bottom of the Front Hoop. The lower frame rail/frame member may be this member if it meets the diameter and wall thickness requirements.
- (c) The diagonal Side Impact Structural member must connect the upper and lower Side Impact Structural members forward of the Main Hoop and rearward of the Front Hoop.

T3.24.4 With proper gusseting and/or triangulation, it is permissible to fabricate the Side Impact Structural members from more than one piece of tubing.

T3.24.5 Alternative geometry that does not comply with the minimum requirements given above requires an approved “Structural Equivalency Spreadsheet” per Rule [T3.8](#).

T3.25 Inspection Holes

T3.25.1 To allow the verification of tubing wall thicknesses, 4.5 mm inspection holes must be drilled in a non-critical location of both the Main Hoop and the Front Hoop.

T3.25.2 In addition, the Technical Inspectors may check the compliance of other tubes that have minimum dimensions specified in [T3.3.1](#). This may be done by the use of ultra-sonic testing or by the drilling of additional inspection holes at the inspector’s request.

T3.25.3 Inspection holes must be located so that the outside diameter can be measured across the inspection hole with a caliper, i.e. there must be access for the caliper to the inspection hole and to the outside of the tube one hundred eighty degrees (180°) from the inspection hole.

T3.26 Composite Tubular Space Frames

T3.26.1 Composite tubular space frames are not permitted in the Primary Structure of the vehicle (See [T3.2\(g\)](#))

T3.26.2 Composite tubular structures may be used for other tubes regulated under [T3.3](#) provided the team receives prior approval from the Formula Hybrid + Electric chief technical examiner. This will require submission of the following data:

- (a) Test data on the joints used in the structure.
- (b) Static strength testing on all proposed configurations within the frame.
- (c) An assessment of the ability of all joints to handle cyclic loading.
- (d) The equivalency of the composite tubes to withstand maximum forces and moments (when compared to baseline materials).

This information must also be included in the structural equivalency submission.

T3.27 Monocoque General Requirements

T3.27.1 All equivalency calculations must prove equivalency relative to steel grade SAE/AISI 1010.

T3.27.2 All sections of the rules apply to monocoque structures except for the following sections which supplement or supersede other rule sections.

T3.27.3 Monocoque construction requires an approved Structural Equivalency Spreadsheet, per Section [T3.8](#). The form must demonstrate that the design is equivalent to a welded frame in terms of energy dissipation, yield and ultimate strengths in bending, buckling and tension. Information must include: material type(s), cloth weights, resin type, fiber orientation, number of layers,

core material, and lay-up technique. The 3 point bend test and shear test data and pictures must also be included as per **T3.30** Monocoque Laminate Testing. The Structural Equivalency must address each of the items below. Data from the laminate testing results must be used as the basis for any strength or stiffness calculations.

T3.27.4 Composite and metallic monocoques have the same requirements.

T3.27.5 Composite monocoques must meet the materials requirements in Rule **T3.7** Composite Materials.

T3.28 Monocoque Inspections

T3.28.1 Due to the monocoque rules and methods of manufacture it is not always possible to inspect all aspects of a monocoque during technical inspection. For items which cannot be verified by an inspector it is the responsibility of the team to provide documentation, both visual and/or written, that the requirements have been met. Generally the following items should be possible to be confirmed by the technical inspector:

- (a) Verification of the Main Hoop outer diameter and wall thickness where it protrudes above the monocoque
- (b) Visual verification that the Main Hoop goes to the lowest part of the tub, locally
- (c) Verify mechanical attachment of Main Hoop to tub exists and matches the SES, at all points shown on the SES.
- (d) Verify the outside diameter and wall thickness of the Front Hoop by providing access as required by Rule **T3.25.3**.
- (e) Verify visually or by feel that the Front Hoop is installed.
- (f) Verify that the Front Hoop goes to the lowest part of the tub, locally.
- (g) Verify mechanical attachment of the Front Hoop (if included) against the SES.

T3.29 Monocoque Buckling Modulus – Equivalent Flat Panel Calculation

When specified in the rules, the *EI* of the monocoque must be calculated as the *EI* of a flat panel with the same composition as the monocoque about the neutral axis of the laminate. The curvature of the panel and geometric cross section of the monocoque must be ignored for these calculations.

Note: Calculations of *EI* that do not reference **T3.29** may take into account the actual geometry of the monocoque.

T3.30 Monocoque Laminate Testing

T3.30.1 Teams must build a representative flat panel section of the monocoque side impact zone (“zone” is defined in **T3.32.2** and **T3.2(k)**) and perform a 3 point bending test on this panel. They must prove by physical test that a section 200 mm x 500 mm has at least the same properties as a baseline steel side impact tube (See **T3.3.1** “Baseline Steel Materials”) for bending stiffness and two side impact tubes for yield and ultimate strength.

The data from these tests and pictures of the test samples must be included in the SES, the test results will be used to derive strength and stiffness properties used in the SES formulae for all laminate panels. The test specimen must be presented at technical inspection. If the test specimen does not meet these requirements, then the monocoque side impact zone must be strengthened appropriately.

Note: Teams are advised to make an equivalent test with the base line steel tubes such that any compliance in the test rig can be accounted for.

- T3.30.2 If laminates with a lay-up different to that of the side-impact structure are used then additional physical tests must be completed for any part of the monocoque that forms part of the primary structure. The material properties derived from these tests must then be used in the SES for the appropriate equivalency calculations.

Note: A laminate with more or less plies, of the same lay-up as the side-impact structure, does not constitute a “different lay-up” and the material properties may be scaled accordingly.

- T3.30.3 Perimeter shear tests must be completed by measuring the force required to push or pull a 25 mm diameter flat punch through a flat laminate sample.

The sample, measuring at least 100 mm x 100 mm must have core and skin thicknesses identical to those used in the actual monocoque and be manufactured using the same materials and processes.

The fixture must support the entire sample, except for a 32 mm hole aligned co-axially with the punch. The sample must not be clamped to the fixture.

The force-displacement data and photos of the test setup must be included in the SES.

The first peak in the load-deflection curve must be used to determine the skin shear strength. This may be less than the minimum force required by [T3.32.3/T3.33.3](#).

The maximum force recorded must meet the requirements of [T3.32.3/T3.33.3](#).

Note: The edge of the punch and hole in the fixture may include an optional fillet up-to a maximum radius of 1 mm.

T3.31 Monocoque Front Bulkhead

See Rule [T3.27](#) for general requirements that apply to all aspects of the monocoque. In addition, when modeled as an “L” shaped section the EI of the front bulkhead about both vertical and lateral axis must be equivalent to that of the tubes specified for the front bulkhead under [T3.18](#). The length of the section perpendicular to the bulkhead may be a maximum of 25.4 mm measured from the rearmost face of the bulkhead.

Furthermore, any front bulkhead which supports the IA plate must have a perimeter shear strength equivalent to a 1.5 mm thick steel plate.

T3.32 Monocoque Front Bulkhead Support

- T3.32.1 In addition to proving that the strength of the monocoque is adequate, the monocoque must have equivalent EI to the sum of the EI of the six (6) baseline steel tubes that it replaces.

- T3.32.2 The EI of the vertical side of the front bulkhead support structure must be equivalent to at least the EI of one baseline steel tube that it replaces when calculated as per rule [T3.29](#) Monocoque Buckling Modulus.

- T3.32.3 The perimeter shear strength of the monocoque laminate in the front bulkhead support structure should be at least 4 kN for a section with a diameter of 25 mm. This must be proven by a physical test completed as per [T3.30.3](#) and the results include in the SES

T3.33 Monocoque Side Impact

- T3.33.1 In addition to proving that the strength of the monocoque is adequate, the side of the monocoque must have equivalent EI to the sum of the EI of the three (3) baseline steel tubes that it replaces.

- T3.33.2 The side of the monocoque between the upper surface of the floor and 350 mm above the ground (Side Impact Zone) must have an EI of at least 50% of the sum of the EI of the three (3) baseline steel tubes that it replaces when calculated as per Rule [T3.29](#) Monocoque Buckling Modulus.
- T3.33.3 The perimeter shear strength of the monocoque laminate should be at least 7.5 kN for a section with a diameter of 25 mm. This must be proven by physical test completed as per [T3.30.3](#) and the results included in the SES.

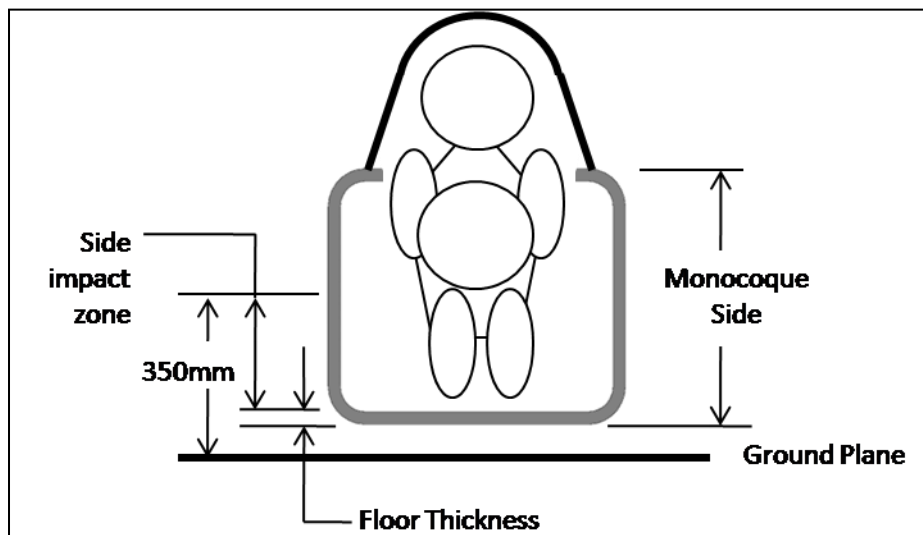


Figure 12 – Side Impact Zone Definition for a Monocoque

T3.34 Monocoque Main Hoop

- T3.34.1 The Main Hoop must be constructed of a single piece of uncut, continuous, closed section steel tubing per [T3.3.1](#) and extend down to the bottom of the monocoque.
- T3.34.2 The Main Hoop must be mechanically attached at the top and bottom of the monocoque and at intermediate locations as needed to show equivalency.
- T3.34.3 Mounting plates welded to the Roll Hoop must be at least 2.0 mm thick steel.
- T3.34.4 Attachment of the Main Hoop to the monocoque must comply with [T3.39](#).

T3.35 Monocoque Front Hoop

- T3.35.1 Composite materials are not allowed for the front hoop. See Rule [T3.27](#) for general requirements that apply to all aspects of the monocoque.
- T3.35.2 Attachment of the Front Hoop to the monocoque must comply with Rule [T3.39](#).

T3.36 Monocoque Front and Main Hoop Bracing

- T3.36.1 See Rule [T3.27](#) for general requirements that apply to all aspects of the monocoque.
- T3.36.2 Attachment of tubular Front or Main Hoop Bracing to the monocoque must comply with Rule [T3.39](#).

T3.37 Monocoque Impact Attenuator and Anti-Intrusion Plate Attachment

The attachment of the Impact Attenuator and Anti-Intrusion Plate to a monocoque structure requires an approved “Structural Equivalency Spreadsheet” per Rule **T3.8** that shows the equivalency to a minimum of eight (8) 8 mm Metric Grade 8.8 (5/16 inch SAE Grade 5) bolts for the Anti-Intrusion Plate attachment and a minimum of four (4) bolts to the same minimum specification for the Impact Attenuator attachment.

T3.38 Monocoque Impact Attenuator Anti-Intrusion Plate

T3.38.1 See Rule **T3.27** for general requirements that apply to all aspects of the monocoque and Rule **T3.20.3** for alternate Anti-Intrusion Plate designs.

T3.39 Monocoque Attachments

T3.39.1 In any direction, each attachment point between the monocoque and the other primary structure must be able to carry a load of 30 kN.

T3.39.2 The laminate, mounting plates, backing plates and inserts must have sufficient shear area, weld area and strength to carry the specified 30 kN load in any direction. Data obtained from the laminate perimeter shear strength test (**T3.33.3**) should be used to prove adequate shear area is provided

T3.39.3 Each attachment point requires a minimum of two (2) 8 mm Metric Grade 8.8 or 5/16 inch SAE Grade 5 bolts

T3.39.4 Each attachment point requires steel backing plates with a minimum thickness of 2.0 mm. Alternate materials may be used for backing plates if equivalency is approved.

T3.39.5 The Front Hoop Bracing, Main Hoop Bracing and Main Hoop Bracing Supports only may use one (1) 10 mm Metric Grade 8.8 or 3/8 inch SAE Grade 5 bolt as an alternative to **T3.39.3** if the bolt is on the centerline of tube similar to the figure below.

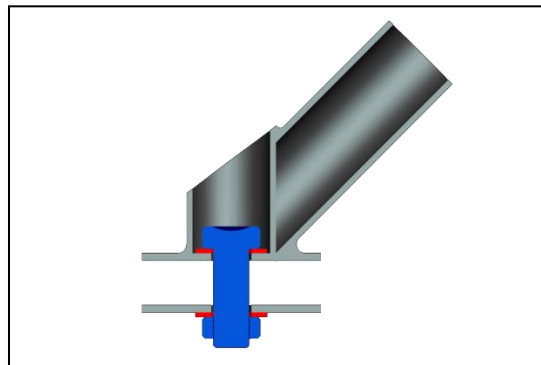


Figure 13 – Alternate Single Bolt Attachment

T3.39.6 No crushing of the core is permitted

T3.39.7 Main Hoop bracing attached to a monocoque (i.e. not welded to a rear space frame) is always considered “mechanically attached” and must comply with Rule **T3.16**.

T3.40 Monocoque Driver's Harness Attachment Points

- T3.40.1 The monocoque attachment points for the shoulder and lap belts must support a load of 13 kN before failure.
- T3.40.2 The monocoque attachment points for the anti-submarine belts must support a load of 6.5 kN before failure.
- T3.40.3 If the lap belts and anti-submarine belts are attached to the same attachment point, then this point must support a load of 19.5 kN before failure.
- T3.40.4 The strength of lap belt attachment and shoulder belt attachment must be proven by physical test where the required load is applied to a representative attachment point where the proposed layup and attachment bracket is used.

ARTICLE T4 COCKPIT

T4.1 Cockpit Opening

Important Note: Teams are advised that cockpit template and Percy (**Figure 5**) compliance will be strictly enforced during mechanical technical inspection. Check the Formula Hybrid + Electric website for an instructional video on template and Percy inspection procedures.



- T4.1.1 In order to ensure that the opening giving access to the cockpit is of adequate size, a template shown in **Figure 14** will be inserted downwards into the cockpit opening. It will be held horizontally and inserted vertically from a height above any Primary Structure or bodywork that is between the Front Hoop and the Main Hoop until it has passed below the top bar of the Side Impact Structure (or until it is 350 mm (13.8 inches) above the ground for monocoque cars). Fore and aft translation of the template (while maintaining its position parallel to the ground) will be permitted during insertion.

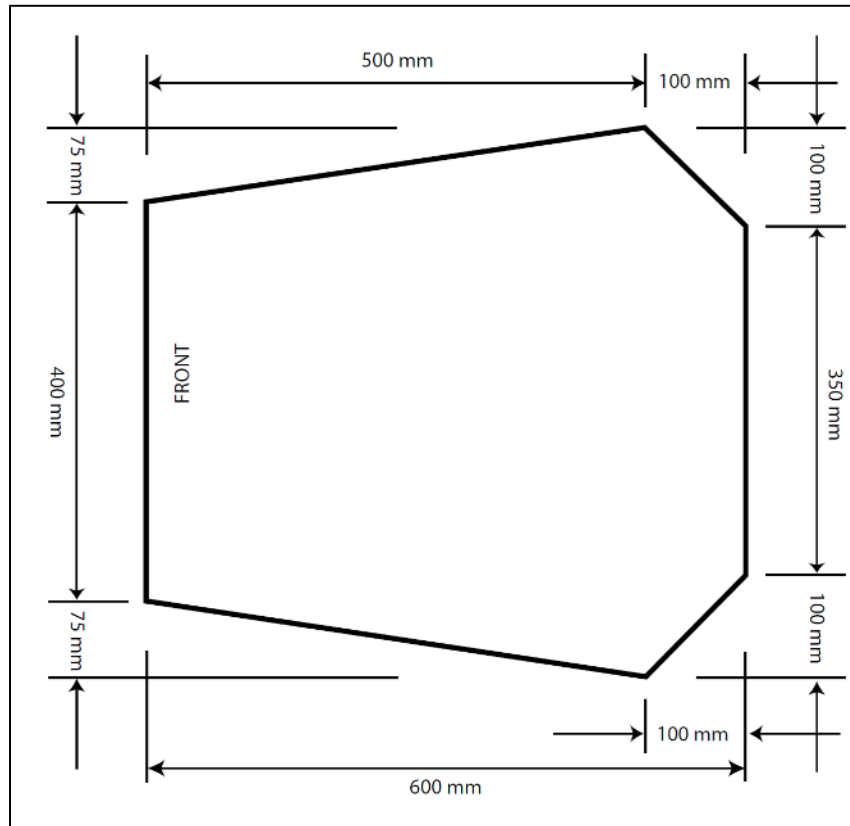


Figure 14 – Cockpit Opening Template

- T4.1.2 During this test, the steering wheel, steering column, seat and all padding may be removed. The shifter or shift mechanism may not be removed unless it is integral with the steering wheel and is removed with the steering wheel. The firewall may not be moved or removed.

Note: As a practical matter, for the checks, the steering column will not be removed. The technical inspectors will maneuver the template around the steering column shaft, but not the steering column supports.

T4.2 Cockpit Internal Cross Section:

- T4.2.1 A free vertical cross section, which allows the template shown in [Figure 15](#) to be passed horizontally through the cockpit to a point 100 mm rearwards of the face of the rearmost pedal when in the inoperative position, must be maintained over its entire length. If the pedals are adjustable, they will be put in their most forward position.

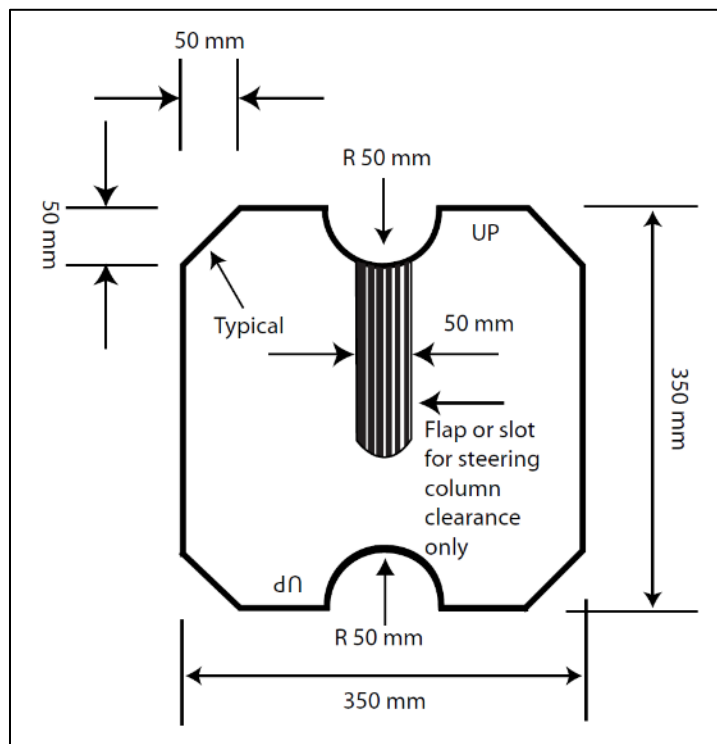


Figure 15 – Cockpit Internal Cross Section Template

- T4.2.2 The template, with maximum thickness of 7 mm, will be held vertically and inserted into the cockpit opening rearward of the rear-most portion of the steering column.

Note: At the discretion of the technical inspectors, the internal cross-section template may be moved vertically by small increments during fore and aft travel to clear height deviations in the floor of the vehicle (e.g. those caused by the steering rack, etc.). The template must still fit through the cross-section at the location of vertical deviation.

A video demonstrating the template procedure can be found on YouTube:

<https://youtu.be/azz5kbmiQbw>

- T4.2.3 The only items that may be removed for this test are the steering wheel, and any padding required by Rule **T5.8** “Driver’s Leg Protection” that can be easily removed without the use of tools with the driver in the seat. The seat and any seat insert that may be used by any team member must remain in the cockpit.
- T4.2.4 Teams whose cars do not comply with **T4.1** or **T4.2** will not be given a Technical Inspection Sticker and will not be allowed to compete in the dynamic events.



Note: Cables, wires, hoses, tubes, etc. must not impede the passage of the templates required by [T4.1](#) and [T4.2](#).

T4.3 Driver's Seat

- T4.3.1 In side view the lowest point of the driver's seat must be no lower than the top surface of the lower frame rails or by having a longitudinal tube (or tubes) that meets the requirements for Side Impact tubing, passing underneath the lowest point of the seat.
- T4.3.2 When seated in the normal driving position, adequate heat insulation must be provided to ensure that the driver will not contact any metal or other materials which may become heated to a surface temperature above sixty degrees C (60°C). The insulation may be external to the cockpit or incorporated with the driver's seat or firewall. The design must show evidence of addressing all three (3) types of heat transfer, namely conduction, convection and radiation, with the following between the heat source, e.g. an exhaust pipe or coolant hose/tube and the panel that the driver could contact, e.g. the seat or floor:
- (a) Conduction Isolation by:
 - (i) No direct contact between the heat source and the panel, or
 - (ii) a heat resistant, conduction isolation material with a minimum thickness of 8 mm between the heat source and the panel.
 - (b) Convection Isolation by a minimum air gap of 25 mm between the heat source and the panel.
 - (c) Radiation Isolation by:
 - (i) A solid metal heat shield with a minimum thickness of 0.4 mm or
 - (ii) reflective foil or tape when combined with [T4.3.2\(a\)\(ii\)](#) above.

T4.4 Floor Close-out

All vehicles must have a floor closeout made of one or more panels, which separate the driver from the pavement. If multiple panels are used, gaps between panels are not to exceed 3 mm. The closeout must extend from the foot area to the firewall and prevent track debris from entering the car. The panels must be made of a solid, non-brittle material.

T4.5 Firewall

- T4.5.1 Firewall(s) must separate the driver compartment from the following components:
- (a) Fuel Tanks.
 - (b) Accumulators.
 - (c) All components of the fuel supply.
 - (d) External engine oil systems including hoses, oil coolers, tanks, etc.
 - (e) Liquid cooling systems including those for I.C. engine and electrical components.
 - (f) Lithium-based GLV batteries.
 - (g) All tractive systems (TS) components
 - (h) All conductors carrying tractive system voltages (TSV) (Whether contained within conduit or not.)
- T4.5.2 The firewall(s) must be a rigid, non-permeable surface made from 1.5 mm or thicker aluminum or proven equivalent. See [Appendix H](#) – Firewall Equivalency Test.

- T4.5.3 The firewall(s) must seal completely against the passage of fluids and hot gasses, including at the sides and the floor of the cockpit, e.g. there can be no holes in a firewall through which seat belts pass.
- T4.5.4 Pass-throughs for GLV wiring, cables, etc. are allowable if grommets are used to seal the pass-throughs.
- Multiple panels may be used to form the firewall but must be mechanically fastened in place and sealed at the joints.
- T4.5.5 For those components listed in [T4.5.1](#) that are mounted behind the driver, the firewall(s) must extend sufficiently far upwards and/or rearwards such that a straight line from any part of any listed component to any part of the tallest driver that is more than 150 mm below the top of his/her helmet, must pass through the firewall.
- T4.5.6 For those components listed in section [T4.5.1](#) positioned under the driver, the firewall must extend:
- (a) Continuously rearwards the full width of the cockpit from the Front Bulkhead, under and up behind the driver to a point where the helmet of the 95th percentile male template ([T3.9.4](#)) touches the head restraint, and
 - (b) Alongside the driver, from the top of the Side Impact Structure down to the lower portion of the firewall required by [T4.5.6\(a\)](#) and from the rearmost front suspension mounting point to connect (without holes or gaps) behind the driver with the firewall required by [T4.5.6\(a\)](#).
- See [Figure 16\(a\)](#).
- T4.5.7 For those components listed in section [T4.5.1](#) that are mounted outboard of the Side Impact System (e.g. in side pods), the firewall(s) must extend from 100 mm forward to 100 mm rearward of the of the listed components and
- (a) alongside the driver at the full height of the listed component, and
 - (b) cover the top of the listed components and
 - (c) run either
 - (i) under the cockpit between the firewall(s) required by [T4.5.7\(a\)](#), or
 - (ii) extend 100 mm out under the listed components from the firewall(s) that are required by [T4.5.8](#)
- See [Figure 16\(b&c\)](#).
- T4.5.8 For the components listed in section [T4.5.1](#) that are mounted in ways that do not fall clearly under any of sections [T4.5.5](#), [T4.5.6](#) or [T4.5.7](#), the firewall must be configured to provide equivalent protection to the driver, and the firewall configuration must be approved by the Rules Committee.
- T4.5.9 Containers that meet the firewall requirements of [T4.5.2](#), including accumulator containers, may be accepted as part of the firewall system; redundant barriers are not required. Tractive System wiring and conduit must not be exposed to the driver while seated in or during egress from the vehicle.

Note: To ensure adequate time for consideration and possible re-designs, applications should be submitted at least 1 month in advance of the event.

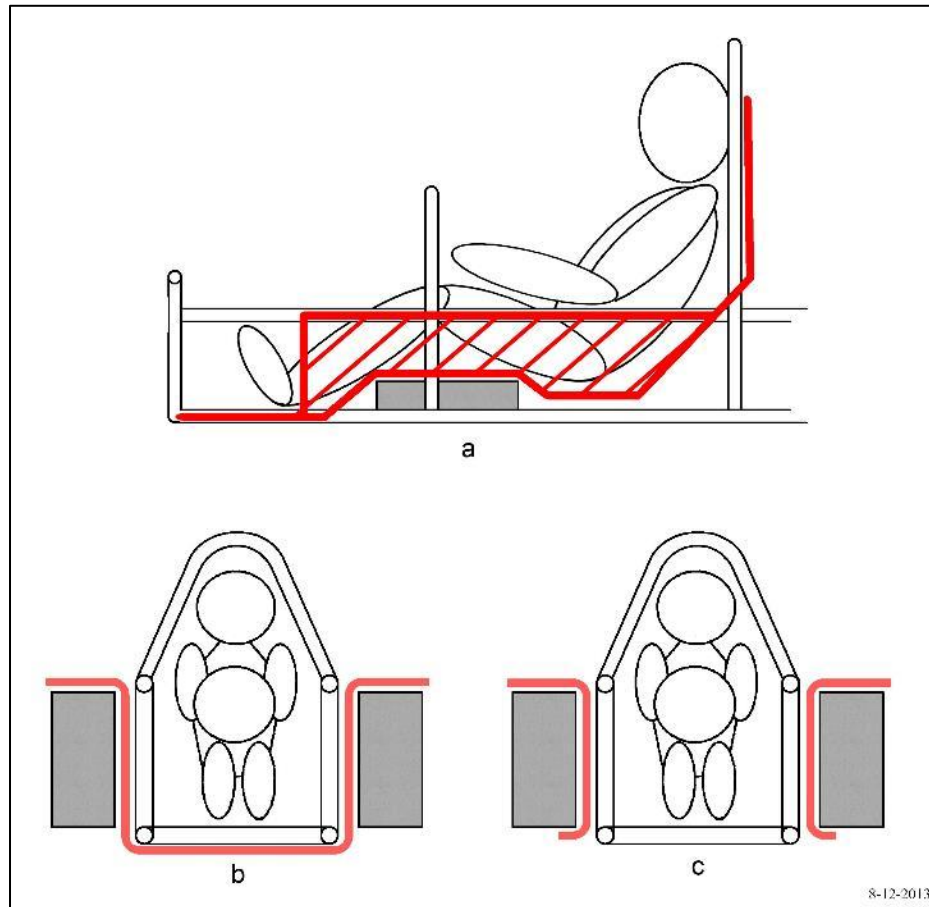


Figure 16 – Examples⁴ of firewall configurations

T4.6 Accessibility of Controls

All vehicle controls, including the shifter, must be operated from inside the cockpit without any part of the driver, e.g. hands, arms or elbows, being outside the planes of the Side Impact Structure defined in Rule [T3.24](#) and [T3.33](#).

T4.7 Driver Visibility

T4.7.1 The driver must have adequate visibility to the front and sides of the car. With the driver seated in a normal driving position he/she must have a minimum field of vision of two hundred degrees (200°) (a minimum one hundred degrees (100°) to either side of the driver). The required visibility may be obtained by the driver turning his/her head and/or the use of mirrors.

T4.7.2 If mirrors are required to meet Rule [T4.7.1](#), they must remain in place and adjusted to enable the required visibility throughout all dynamic events.

⁴ The firewalls shown in red in [Figure 16](#) are examples only and are not meant to imply that a firewall must lie outside the frame rails.

T4.8 Driver Egress

All drivers must be able to exit to the side of the vehicle in no more than 5 seconds. Egress time begins with the driver in the fully seated position, hands in driving position on the connected steering wheel and wearing the required driver equipment. Egress time will stop when the driver has both feet on the pavement.

T4.9 Emergency Shut Down Test

T4.9.1 With their vision obscured, all drivers must be able to operate the cockpit Big Red Button (BRB) in less than one second.

Time begins with the driver in the fully seated position, hands in driving position on the connected steering wheel, and wearing the required driver equipment.

ARTICLE T5 DRIVER'S EQUIPMENT (BELTS AND COCKPIT PADDING)

T5.1 Belts - General

T5.1.1 Definitions (**Note:** Belt dimensions listed are nominal widths.)

- (a) **5-point system** – consists of a lap belt, two (2) shoulder straps and a single anti-submarine strap.
- (b) **6-point system** – consists of a lap belt, two (2) shoulder straps and two (2) leg or anti-submarine straps.
- (c) **7-point system** – system is the same as the 6-point except it has three (3) anti-submarine straps, two (2) from the 6-point system and one (1) from the 5-point system.
- (d) **Upright driving position** - is defined as one with a seat back angled at thirty degrees (30°) or less from the vertical as measured along the line joining the two 200 mm circles of the template of the 95th percentile male as defined in [Table 6](#) and positioned per [T3.9.4](#).
- (e) **Reclined driving position** - is defined as one with a seat back angled at more than thirty degrees (30°) from the vertical as measured along the line joining the two 200 mm circles of the template of the 95th percentile male as defined in [Table 6](#) and positioned per [T3.9.4](#).
- (f) **Chest-groin line** - is the straight line that in side view follows the line of the shoulder belts from the chest to the release buckle.

T5.1.2 Harness Requirements

All drivers must use a 5, 6 or 7 point restraint harness meeting the following specifications:

- (a) SFI Specification 16.1, SFI Specification 16.5, SFI Specification 16.6, or FIA specification 8853/98 or FIA specification 8853-2016.

Note: FIA harnesses with 44 mm (“2 inch”) shoulder straps meeting T5.1.2 and T5.1.3 are approved for use at FH+E

- (b) To accommodate drivers of differing builds, all lap belts must incorporate a tilt lock adjuster (“quick adjuster”).

Note: Lap belts with “pull-up” adjusters are recommended over “pull-down” adjusters and a tilt lock adjuster in each portion of the lap belt is highly recommended.

- (c) Cars with a “reclined driving position” (see [T5.1.1\(e\)](#) above) must have either a 6 point or 7-point harness, AND have either anti-submarine belts with tilt lock adjusters (“quick adjusters”) or have two (2) sets of anti-submarine belts installed.
 - (d) The shoulder harness must be the over-the-shoulder type. Only separate shoulder straps are permitted (i.e. “Y”-type shoulder straps are not allowed). The “H”-type configuration is allowed.
 - (e) The belts must bear the appropriate dated labels.
 - (f) The material of all straps must be in perfect condition.
- T5.1.3 Harness Replacement - Harnesses must be replaced following December 31st of the year of the expiration date shown on the label. (**Note:** SFI belts are normally certified for two (2) years from the date of manufacture while FIA belts are normally certified for five (5) years from the date of manufacture.)
- T5.1.4 The restraint system must be worn tightly at all times.
- T5.2 Belt, Strap and Harness Installation - General**
- T5.2.1 The lap belt, shoulder harness and anti-submarine strap(s) must be securely mounted to the Primary Structure. Such structure and any guide or support for the belts must meet the minimum requirements of [T3.3.1](#).
- Note:** Rule [T3.4.5](#) applies to these tubes as well so a non-straight shoulder harness bar would require support per [T3.4.5](#)
- T5.2.2 The tab or bracket to which any harness is attached must fulfill the following requirements:
- (a) Have a minimum cross sectional area of 60 sq. mm (0.093 sq. in) of steel to be sheared or failed in tension at any point of the tab, and
 - (b) Have a minimum thickness of 1.6 mm (0.063 inch).
 - (c) Where lap belts and anti-submarine belts use the same attachment point, there must be a minimum cross sectional area of 90 sq. mm (0.140 sq. in) of steel to be sheared or failed in tension at any point of the tab.
 - (d) Where brackets are fastened to the chassis, two 6mm Metric Grade 8.8 (1/4 inch SAE Grade 5) fasteners or stronger must be used to attach the bracket to the chassis.
 - (e) Where a single shear tab is welded to the chassis, the tab to tube welding must be on both sides of the base of the tab.
 - (f) The bracket or tab should be aligned such that it is not put in bending when that portion of the harness is put under load.
- NOTE:** Double shear attachments are preferred. Where possible, the tabs and brackets for double shear mounts should also be welded on both sides.
- T5.2.3 Harnesses, belts and straps must not pass through a firewall, i.e. all harness attachment points must be on the driver’s side of any firewall.
- T5.2.4 The attachment of the Driver’s Restraint System to a monocoque structure requires an approved Structural Equivalency Spreadsheet per Rule [T3.8](#).
- T5.2.5 All adjusters must be threaded in accordance with manufacturer’s instructions. Examples are given in [Figure 17](#).

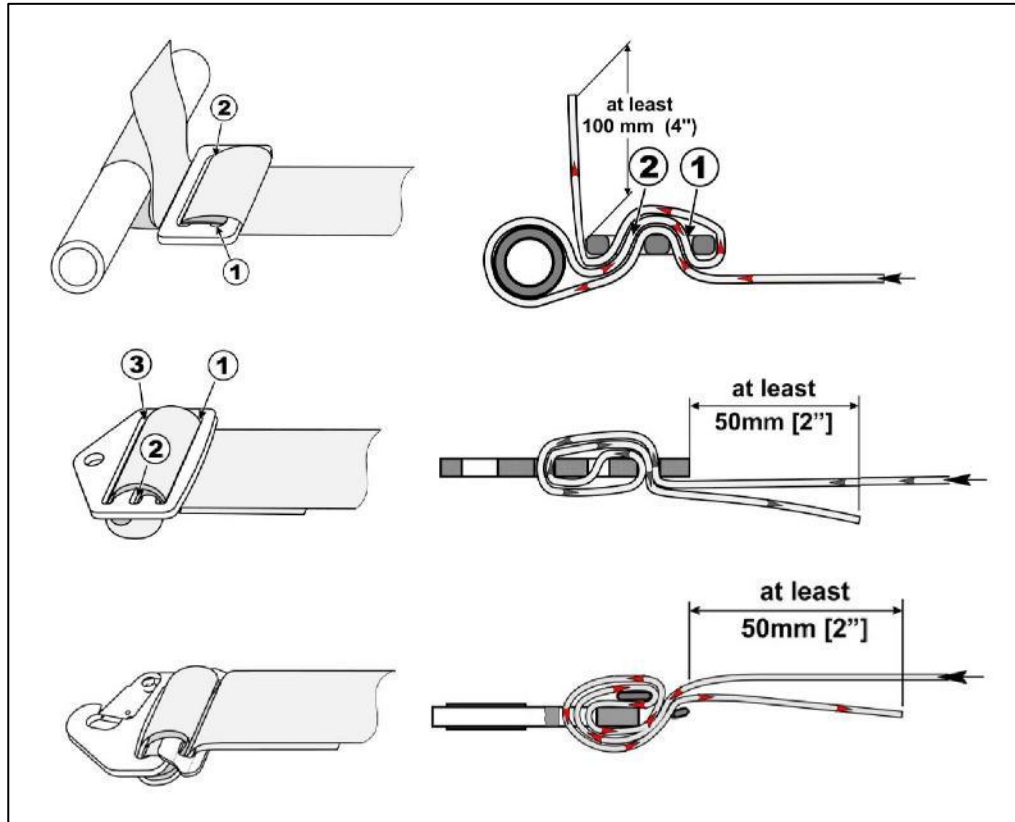


Figure 17 - Seat Belt Threading Examples

T5.2.6 The restraint system installation is subject to approval of the Chief Technical Inspector.

T5.3 Lap Belt Mounting

T5.3.1 The lap belt must pass around the pelvic area below the Anterior Superior Iliac Spines (the hip bones).

T5.3.2 The lap belts must not be routed over the sides of the seat. The lap belts must come through the seat at the bottom of the sides of the seat to maximize the wrap of the pelvic surface and continue in a straight line to the anchorage point.

T5.3.3 Where the belts or harness pass through a hole in the seat, the seat must be rolled or grommeted to prevent chafing of the belts.

T5.3.4 To fit drivers of differing statures correctly, in side view, the lap belt must be capable of pivoting freely by using either a shouldered bolt or an eye bolt attachment, i.e. mounting lap belts by wrapping them around frame tubes is no longer acceptable.

T5.3.5 With an “upright driving position”, in side view the lap belt must be at an angle of between forty-five degrees (45°) and sixty-five degrees (65°) to the horizontal. This means that the centerline of the lap belt at the seat bottom should be between 0 – 76 mm forward of the seat back to seat bottom junction. (See [Figure 18](#))

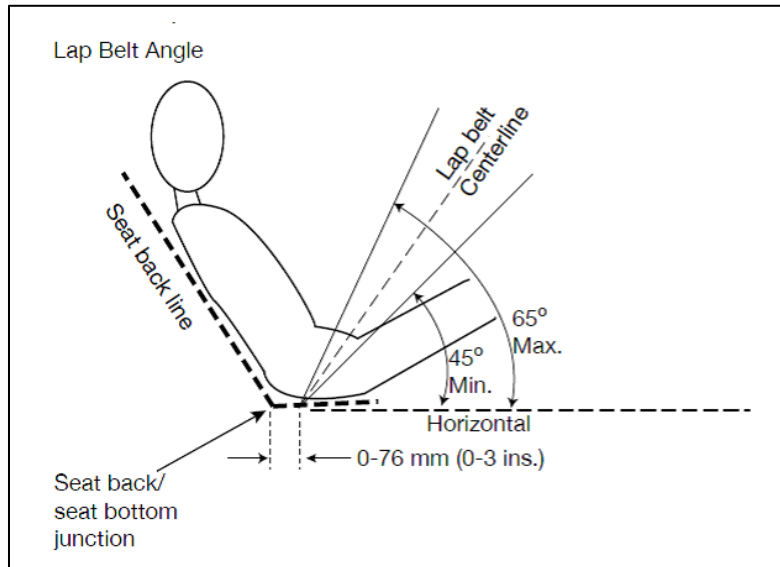


Figure 18 – Lap Belt Angles with Upright Driver

T5.3.6 With a “reclined driving position”, in side view the lap belt must be between an angle of sixty degrees (60°) and eighty degrees (80°) to the horizontal.

T5.3.7 Any bolt used to attach a lap belt, either directly to the chassis or to an intermediate bracket, must be a minimum of either:

- (a) 10mm Metric Grade 8.8 (3/8 inch SAE Grade 5) **OR**
- (b) The bolt diameter specified by the harness manufacturer.

T5.4 Shoulder Harness

T5.4.1 The shoulder harness must be mounted behind the driver to structure that meets the requirements of [T3.3.1](#). However, it cannot be mounted to the Main Roll Hoop Bracing or attendant structure without additional bracing to prevent loads being transferred into the Main Hoop Bracing.

T5.4.2 If the harness is mounted to a tube that is not straight, the joints between this tube and the structure to which it is mounted must be reinforced in side view by gussets or triangulation tubes to prevent torsional rotation of the harness mounting tube.

T5.4.3 The shoulder harness mounting points must be between 178 mm and 229 mm apart. (See [Figure 19](#))

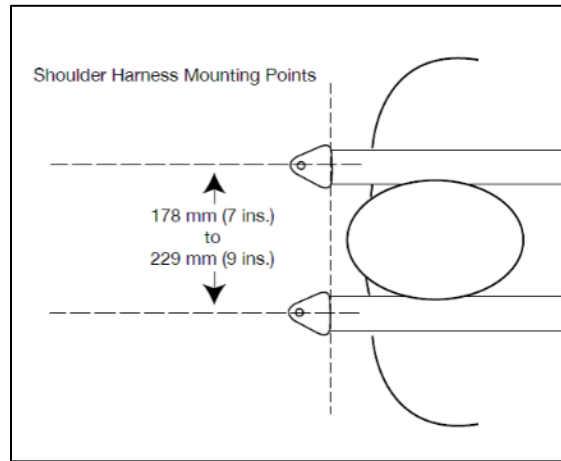


Figure 19 – Shoulder Harness Mounting – Top View

- T5.4.4 From the driver's shoulders rearwards to the mounting point or structural guide, the shoulder harness must be between ten degrees (10°) above the horizontal and twenty degrees (20°) below the horizontal. (See [Figure 20](#)).

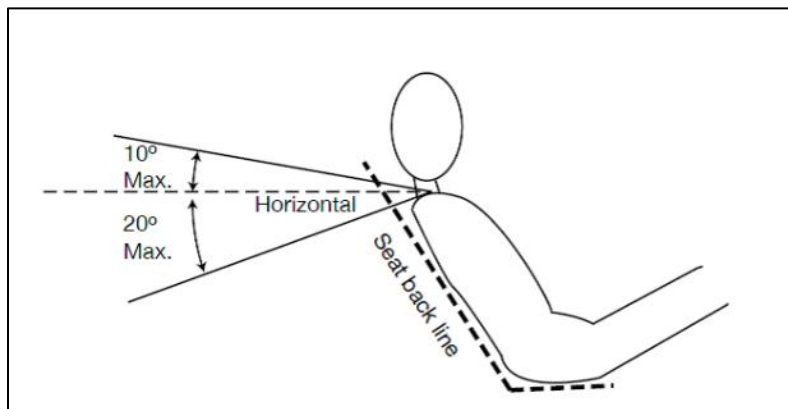


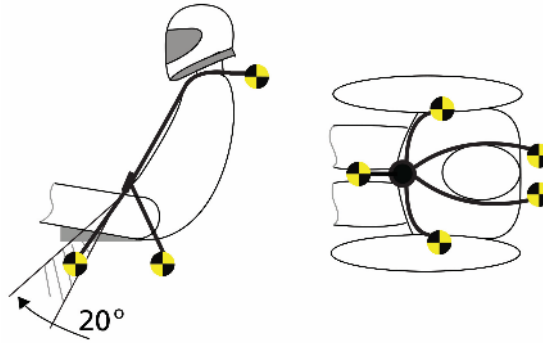
Figure 20 - Shoulder Harness Mounting – Side View

- T5.4.5 Any bolt used to attach a shoulder harness belt, either directly to the chassis or to an intermediate bracket, must be a minimum of either:

- (a) 10mm Metric Grade 8.8 (3/8 inch SAE Grade 5) **OR**
- (b) The bolt diameter specified by the harness manufacturer.

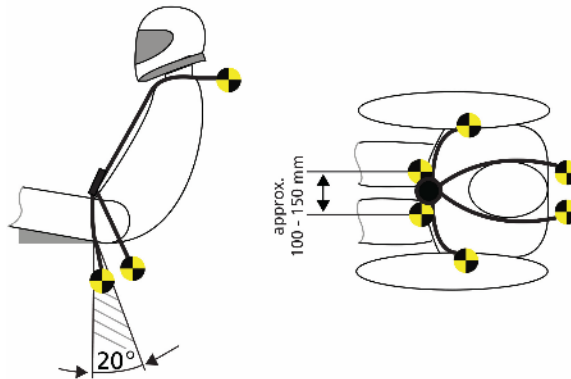
T5.5 Anti-Submarine Belt Mounting

- T5.5.1 The anti-submarine belt of a 5-point harness must be mounted so that the mounting point is in line with, or angled slightly forward (up to twenty degrees (20°)) of, the driver's chest-groin line.

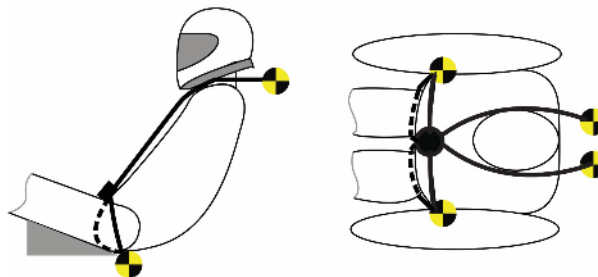


T5.5.2 The anti-submarine belts of a 6 point harness must be mounted either:

- (a) With the belts going vertically down from the groin, or angled up to twenty degrees (20°) rearwards. The anchorage points should be approximately 100 mm apart. Or



- (b) With the anchorage points on the Primary Structure at or near the lap belt anchorages, the driver sitting on the anti-submarine belts, and the belts coming up around the groin to the release buckle.



T5.5.3 All anti-submarine belts must be installed so that they go in a straight line from the anchorage point(s) to:

- Either the harness release buckle for the 5-point mounting per [T5.5.1](#),

- Or the first point where the belts touch the driver's body for the 6-point mounting per **T5.5.2(a)** or **T5.5.2(b)**.

without touching any hole in the seat or any other intermediate structure.

T5.5.4 Any bolt used to attach an anti-submarine belt, either directly to the chassis or to an intermediate bracket, must be a minimum of either:

- (a) 8mm Metric Grade 8.8 (5/16 inch SAE Grade 5) **OR**
- (b) The bolt diameter specified by the belt manufacturer.

T5.6 Head Restraint

T5.6.1 A head restraint must be provided on the car to limit the rearward motion of the driver's head.

T5.6.2 The restraint must:

- (a) Be vertical or near vertical in side view.
- (b) Be padded with a minimum thickness of 38 mm (1.5 inches) of an energy absorbing material that meets either **SFI Standard 45.2**, or is listed on the **FIA Technical List No. 17** as a "Type A or a Type B Material for single seater cars", i.e. CONFOR™ foam CF-42AC (pink) or CF-42M (pink) or CF45 (blue) or CF45M (blue) .
- (c) Have a minimum width of 15 cm (6 inches).
- (d) Have a minimum area of 235 sq. cms (36 sq. inches) **AND** have a minimum height adjustment of 17.5 cm (7 inches), **OR** have a minimum height of 28 cm (11 inches).
- (e) Be covered in a thin, flexible cover with a hole with a minimum diameter of 20 mm in a surface other than the front surface, through which the energy absorbing material can be seen.
- (f) Be located so that for each driver:
 - (i) The restraint is no more than 25 mm (1 inch) away from the back of the driver's helmet, with the driver in their normal driving position.
 - (ii) The contact point of the back of the driver's helmet on the head restraint is no less than 50 mm (2 inches) from any edge of the head restraint.

Note 1: Head restraints may be changed to accommodate different drivers (See **T1.2.2(d)**).

Note 2: The above requirements must be met for all drivers.

Note 3: Approximately 100 mm (4 inches) longitudinal adjustment is required to accommodate 5th to 95th Percentile drivers. This is not a specific rules requirement, but teams must have sufficient longitudinal adjustment and/or alternative thickness head restraints available, such that the above requirements are met by all their drivers.

T5.6.3 The restraint, its attachment and mounting must be strong enough to withstand a force of 890 Newtons applied in a rearward direction.

T5.7 Roll Bar Padding

Any portion of the roll bar, roll bar bracing or frame which might be contacted by the driver's helmet must be covered with a minimum thickness of 12 mm (0.5 inches) of padding which meets SFI spec 45.1 or FIA 8857-2001.

T5.8 Driver's Leg Protection

T5.8.1 To keep the driver's legs away from moving or sharp components, all moving suspension and steering components, and other sharp edges inside the cockpit between the front roll hoop and a vertical plane 100 mm rearward of the pedals, must be shielded with a shield made of a solid material. Moving components include, but are not limited to springs, shock absorbers, rocker arms, anti-roll/sway bars, steering racks and steering column CV joints.

T5.8.2 Covers over suspension and steering components must be removable to allow inspection of the mounting points.

ARTICLE T6 GENERAL CHASSIS RULES

T6.1 Suspension

T6.1.1 The car must be equipped with a fully operational suspension system with shock absorbers, front and rear, with usable wheel travel of at least 50.8 mm, 25.4 mm jounce and 25.4 mm rebound, with driver seated. The judges reserve the right to disqualify cars which do not represent a serious attempt at an operational suspension system or which demonstrate handling inappropriate for an autocross circuit.

T6.1.2 All suspension mounting points must be visible at Technical Inspection, either by direct view or by removing any covers.

T6.2 Ground Clearance

The ground clearance must be sufficient to prevent any portion of the car (other than tires) from touching the ground during track events, and with the driver aboard there must be a minimum of 25.4 mm of static ground clearance under the complete car at all times.

T6.3 Wheels

T6.3.1 The wheels of the car must be 8 inches (203.2 mm) or more in diameter.

T6.3.2 Any wheel mounting system that uses a single retaining nut must incorporate a device to retain the nut and the wheel in the event that the nut loosens. A second nut ("jam nut") does not meet these requirements.

T6.3.3 Standard wheel lug bolts are considered engineering fasteners and any modification will be subject to extra scrutiny during technical inspection. Teams using modified lug bolts or custom designs will be required to provide proof that good engineering practices have been followed in their design.

T6.3.4 Aluminum wheel nuts may be used, but they must be hard anodized and in pristine condition.

T6.4 Tires

T6.4.1 Vehicles may have two types of tires as follows:

- (a) Dry Tires – The tires on the vehicle when it is presented for technical inspection are defined as its "Dry Tires". The dry tires may be any size or type. They may be slicks or treaded.

- (b) Rain Tires – Rain tires may be any size or type of treaded or grooved tire provided:
 - (i) The tread pattern or grooves were molded in by the tire manufacturer, or were cut by the tire manufacturer or his appointed agent. Any grooves that have been cut must have documentary proof that it was done in accordance with these rules.
 - (ii) There is a minimum tread depth of 2.4 mm.

Note: Hand cutting, grooving or modification of the tires by the teams is specifically prohibited.

- T6.4.2 Within each tire set, the tire compound or size, or wheel type or size may not be changed after static judging has begun. Tire warmers are not allowed. No traction enhancers may be applied to the tires after the static judging has begun.

T6.5 Steering

- T6.5.1 The steering wheel must be mechanically connected to the wheels, i.e. “steer-by-wire” or electrically actuated steering is prohibited.
- T6.5.2 The steering system must have positive steering stops that prevent the steering linkages from locking up (the inversion of a four-bar linkage at one of the pivots). The stops may be placed on the uprights or on the rack and must prevent the tires from contacting suspension, body, or frame members during the track events.
- T6.5.3 Allowable steering system free play is limited to seven degrees (7°) total measured at the steering wheel.
- T6.5.4 The steering wheel must be attached to the column with a quick disconnect. The driver must be able to operate the quick disconnect while in the normal driving position with gloves on.
- T6.5.5 Rear wheel steering, which can be electrically actuated, is permitted but only if mechanical stops limit the range of angular movement of the rear wheels to a maximum of six degrees (6°). This must be demonstrated with a driver in the car and the team must provide the facility for the steering angle range to be verified at Technical Inspection.
- T6.5.6 The steering wheel must have a continuous perimeter that is near circular or near oval, i.e. the outer perimeter profile can have some straight sections, but no concave sections. “H”, “Figure 8”, or cutout wheels are not allowed.
- T6.5.7 In any angular position, the top of the steering wheel must be no higher than the top-most surface of the Front Hoop. See [Figure 7](#).
- T6.5.8 Steering systems using cables for actuation are not prohibited by [T6.5.1](#) but additional documentation must be submitted. The team must submit a failure modes and effects analysis report with design details of the proposed system as part of the structural equivalency spreadsheet (SES).

The report must outline the analysis that was done to show the steering system will function properly, potential failure modes and the effects of each failure mode and finally failure mitigation strategies used by the team. The organizing committee will review the submission and advise the team if the design is approved. If not approved, a non-cable based steering system must be used instead.
- T6.5.9 The steering rack must be mechanically attached to the frame. If fasteners are used they must be compliant with [ARTICLE T11](#).

- T6.5.10 Joints between all components attaching the steering wheel to the steering rack must be mechanical and be visible at Technical Inspection. Bonded joints without a mechanical backup are not permitted.

T6.6 Jacking Point

- T6.6.1 A jacking point, which is capable of supporting the car's weight and of engaging the organizers' "quick jacks", must be provided at the rear of the car.
- T6.6.2 The jacking point is required to be:
- (a) Visible to a person standing 1 meter behind the car.
 - (b) Painted orange.
 - (c) Oriented horizontally and perpendicular to the centerline of the car
 - (d) Made from round, 25–29 mm O.D. aluminum or steel tube
 - (e) A minimum of 300 mm long
 - (f) Exposed around the lower 180 degrees (180°) of its circumference over a minimum length of 280 mm
 - (g) The height of the tube is required to be such that:
 - (i) There is a minimum of 75 mm clearance from the bottom of the tube to the ground measured at tech inspection.
 - (ii) With the bottom of the tube 200 mm above ground, the wheels do not touch the ground when they are in full rebound.

Comment on Disabled Cars – The organizers and the Rules Committee remind teams that cars disabled on course must be removed as quickly as possible. A variety of tools may be used to move disabled cars including quick jacks, dollies of different types, tow ropes and occasionally even boards. We expect cars to be strong enough to be easily moved without damage. Speed is important in clearing the course and although the course crew exercises due care, parts of a vehicle can be damaged during removal. The organizers are not responsible for damage that occurs when moving disabled vehicles. Removal/recovery workers will jack, lift, carry or tow the car at whatever points they find easiest to access. Accordingly, we advise teams to consider the strength and location of all obvious jacking, lifting and towing points during the design process.

T6.7 Rollover Stability

- T6.7.1 The track and center of gravity of the car must combine to provide adequate rollover stability.
- T6.7.2 Rollover stability will be evaluated on a tilt table using a pass/fail test. The vehicle must not roll when tilted at an angle of sixty degrees (60°) to the horizontal in either direction, corresponding to 1.7 G's. The tilt test will be conducted with the tallest driver in the normal driving position.

ARTICLE T7 BRAKE SYSTEM

T7.1 Brake System - General

- T7.1.1 The car must be equipped with a braking system that acts on all four wheels and is operated by a single control.
- T7.1.2 It must have two (2) independent hydraulic circuits such that in the case of a leak or failure at any point in the system, effective braking power is maintained on at least two (2) wheels. Each

hydraulic circuit must have its own fluid reserve, either by the use of separate reservoirs or by the use of a dammed, OEM-style reservoir.

- T7.1.3 A single brake acting on a limited-slip differential is acceptable.
- T7.1.4 The brake system must be capable of locking all four (4) wheels during the test specified in section [T7.2](#).
- T7.1.5 “Brake-by-wire” systems are prohibited.
- T7.1.6 Unarmored plastic brake lines are prohibited.
- T7.1.7 The braking systems must be protected with scatter shields from failure of the drive train (see [T8.4](#)) or from minor collisions.
- T7.1.8 In side view no portion of the brake system that is mounted on the sprung part of the car can project below the lower surface of the frame or the monocoque, whichever is applicable.
- T7.1.9 The brake pedal must be designed to withstand a force of 2000 N without any failure of the brake system or pedal box. This may be tested by pressing the pedal with the maximum force that can be exerted by any official when seated normally.
- T7.1.10 The brake pedal must be fabricated from steel or aluminum or machined from steel, aluminum or titanium.
- T7.1.11 The first 50% of the brake pedal travel may be used to control regeneration without necessarily actuating the hydraulic brake system.

The remaining brake pedal travel must directly actuate the hydraulic brake system, but brake energy regeneration may remain active.

Note: Any strategy to regenerate energy while coasting or braking must be covered by the ESF.

T7.2 Brake Test

- T7.2.1 The brake system will be dynamically tested and must demonstrate the capability of locking all four (4) wheels and stopping the vehicle in a straight line at the end of an acceleration run specified by the brake inspectors⁵.
- T7.2.2 After accelerating, the tractive system must be switched off by the driver and the driver has to lock all four wheels of the vehicle by braking. The brake test is passed if all four wheels simultaneously lock while the tractive system is shut down.

Note: It is acceptable if the Tractive System Active Light switches off shortly after the vehicle has come to a complete stop as the reduction of the system voltage may take up to 5 seconds.

T7.3 Brake Over-Travel Switch

- T7.3.1 A brake pedal over-travel switch must be installed on the car as part of the shutdown system and wired in series with the shutdown buttons ([EV7.1](#)). This switch must be installed so that in the event of brake system failure such that the brake pedal over travels it will result in the shutdown system being activated.



⁵ It is a persistent source of mystery to the organizers, how a small percentage of teams, after passing all the required inspections, appear to have never checked to see if they can lock all four wheels. [Check this early!](#)

- T7.3.2 Repeated actuation of the switch must not restore power to these components, and it must be designed so that the driver cannot reset it.
- T7.3.3 The brake over-travel switch must not be used as a mechanical stop for the brake pedal and must be installed in such a way that it and its mounting will remain intact and operational when actuated.
- T7.3.4 The switch must be implemented directly. i.e. It may not operate through programmable logic controllers, engine control units, or digital controllers
- T7.3.5 The Brake Over-Travel switch must be a mechanical single pole, single throw (commonly known as a two-position) switch (push-pull or flip type) as shown below.

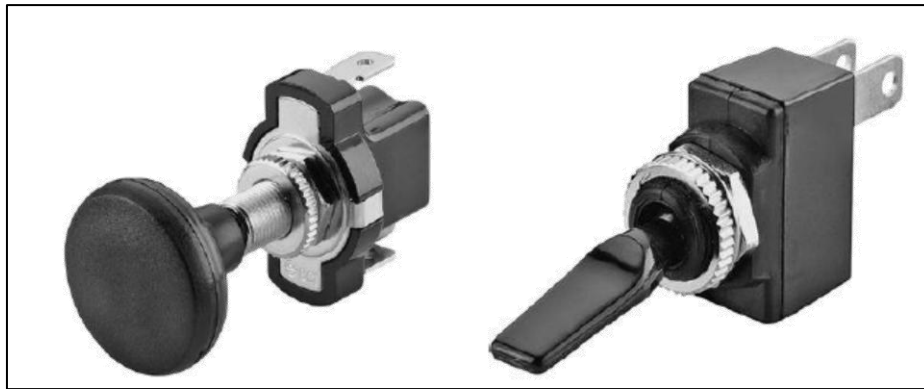


Figure 21 – Over-travel Switches

T7.4 Brake Light

- T7.4.1 The car must be equipped with a red brake light.
- T7.4.2 The brake light itself must have a black background and a rectangular, triangular or near round shape with a minimum shining surface of at least 15 cm².
- T7.4.3 The brake light must be clearly visible from the rear in bright sunlight.
- T7.4.4 When LED lights are used without an optical diffuser, they may not be more than 20 mm apart. If a single line of LEDs is used, the minimum length is 150 mm.
- T7.4.5 The light must be mounted between the wheel centerline and driver's shoulder level vertically and approximately on vehicle centerline laterally.

ARTICLE T8 POWERTRAIN

T8.1 Coolant Fluid Limitations

- T8.1.1 Water-cooled engines must use only plain water. Glycol-based antifreeze, “water wetter”, water pump lubricants of any kind, or any other additives are strictly prohibited.

T8.1.2 Electric motors, accumulators or electronics must use plain water or approved⁶ fluids as the coolant.

T8.2 System Sealing

T8.2.1 Any cooling or lubrication system must be sealed to prevent leakage.

T8.2.2 Any vent on a cooling or lubrication system must employ a catch-can to retain any fluid that is expelled. A separate catch-can is required for each vent.

T8.2.3 Each catch-can on a cooling or lubrication system must have a minimum volume of ten (10) percent of the fluid being contained, or 0.9 liter whichever is greater.

T8.2.4 Any vent on other systems containing liquid lubricant, e.g. a differential or gearbox, etc., must have a catch-can with a minimum volume of ten (10) percent of the fluid being contained or 0.5 liter, whichever is greater.

T8.2.5 Catch-cans must be capable of containing liquids at temperatures in excess of 100 deg. C without deformation, be shielded by a firewall, be below the driver's shoulder level, and be positively retained, i.e. no tie-wraps or tape as the primary method of retention.

T8.2.6 Any catch-can for a cooling system must vent through a hose with a minimum internal diameter of 3 mm down to the bottom levels of the Frame.

T8.3 Transmission and Drive

Any transmission may be used.

T8.4 Drive Train Shields and Guards

T8.4.1 Exposed high-speed final drivetrain equipment such as Continuously Variable Transmissions (CVTs), sprockets, gears, pulleys, torque converters, clutches, belt drives and clutch drives, must be fitted with scatter shields in case of failure. The final drivetrain shield must cover the chain or belt from the drive sprocket to the driven sprocket/chain wheel/belt or pulley. The final drivetrain shield must start *and end* parallel to the lowest point of the chain wheel/belt/pulley. (See **Figure 22**) Body panels or other existing covers are not acceptable unless constructed from approved materials per **T8.4.3** or **T8.4.4**.

Note: If equipped, the engine drive sprocket cover may be used as part of the scatter shield system.

⁶ "Opticool" (<https://soltexinc.com/product/products/thermal-fluids/opticool-fluids/opticool-a-fluid/>) is permitted.

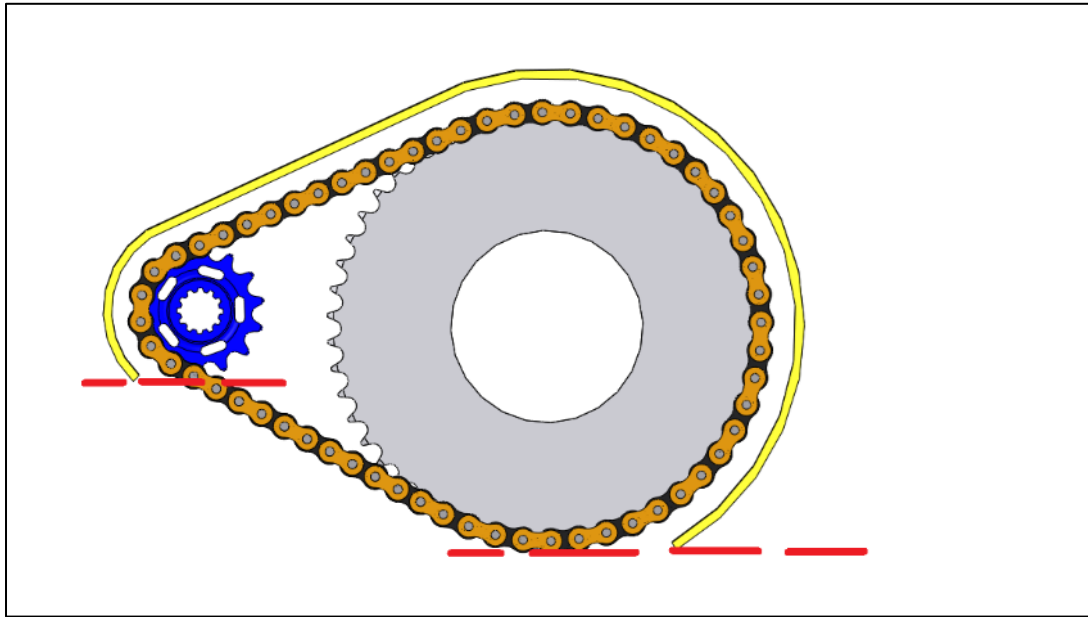


Figure 22 - Final Drive Scatter Shield Example

Comment: Scatter shields are intended to contain drivetrain parts which might separate from the car.

- T8.4.2 Perforated material may not be used for the construction of scatter shields.
- T8.4.3 **Chain Drive** - Scatter shields for chains must be made of at least 2.6 mm steel or stainless steel (no alternatives are allowed), and have a minimum width equal to three (3) times the width of the chain. The guard must be centered on the center line of the chain and remain aligned with the chain under all conditions.
- T8.4.4 **Non-metallic Belt Drive** - Scatter shields for belts must be made from at least 3.0 mm Aluminum Alloy 6061-T6, and have a minimum width that is equal to 1.7 times the width of the belt.
- T8.4.5 The guard must be centered on the center line of the belt and remain aligned with the belt under all conditions.
- T8.4.6 Attachment Fasteners - All fasteners attaching scatter shields and guards must be a minimum 6mm Metric Grade 8.8 or 1/4 inch SAE Grade 5 or stronger.
- T8.4.7 **Finger Guards** – Finger guards are required to cover any drivetrain parts that spin while the car is stationary with the engine running. Finger guards may be made of lighter material, sufficient to resist finger forces. Mesh or perforated material may be used but must prevent the passage of a 12 mm diameter object through the guard.

Comment: Finger guards are intended to prevent finger intrusion into rotating equipment while the vehicle is at rest.

T8.5 Integrity of systems carrying fluids – Tilt Test

- T8.5.1 During technical inspection, the car must be capable of being tilted to a forty-five degree (45°) angle without leaking fluid of any type.

- T8.5.2 The tilt test will be conducted on hybrid cars with the fuel tank filled with either 3.7 litres of fuel or to 50 mm below the top of the filler neck (whichever is less), and on all cars with all other vehicle systems that contain fluids filled to their normal maximum capacity.

ARTICLE T9 AERODYNAMIC DEVICES

T9.1 Aero Dynamics and Ground Effects - General

All aerodynamic devices must satisfy the following requirements:

T9.2 Location

In plain view, no part of any aerodynamic device, wing, under tray or splitter can be:

- (a) Further forward than 460 mm forward of the fronts of the front tires
- (b) No further rearward than the rear of the rear tires.
- (c) No wider than the outside of the front tires or rear tires measured at the height of the hubs, whichever is wider.

T9.3 Wing Edges - Minimum Radii

All wing leading edges must have a minimum radius 12.7 mm. Wing leading edges must be as blunt or blunter than the required radii for an arc of plus or minus 45 degrees ($\pm 45^\circ$) centered on a plane parallel to the ground or similar reference plane for all incidence angles which lie within the range of adjustment of the wing or wing element. If leading edge slats or slots are used, both the fronts of the slats or slots and of the main body of the wings must meet the minimum radius rules.

T9.4 Other Edge Radii Limitations

All wing edges, end plates, Gurney flaps, wicker bills, splitters undertrays and any other wing accessories must have minimum edge radii of at least 3 mm i.e., this means at least a 6 mm thick edge.

T9.5 Ground Effect Devices

No power device may be used to move or remove air from under the vehicle except fans designed exclusively for cooling. Power ground effects are prohibited.

T9.6 Driver Egress Requirements

- T9.6.1 Egress from the vehicle within the time set in Rule [T4.8](#) “Driver Egress,” must not require any movement of the wing or wings or their mountings.
- T9.6.2 The wing or wings must be mounted in such positions, and sturdily enough, that any accident is unlikely to deform the wings or their mountings in such a way to block the driver’s egress.

ARTICLE T10 COMPRESSED GAS SYSTEMS AND HIGH PRESSURE HYDRAULICS

T10.1 Compressed Gas Cylinders and Lines

Any system on the vehicle that uses a compressed gas as an actuating medium must comply with the following requirements:

- (a) Working Gas -The working gas must be nonflammable, e.g. air, nitrogen, carbon dioxide.

- (b) Cylinder Certification - The gas cylinder/tank must be of proprietary manufacture, designed and built for the pressure being used, certified by an accredited testing laboratory in the country of its origin, and labeled or stamped appropriately.
- (c) Pressure Regulation -The pressure regulator must be mounted directly onto the gas cylinder/tank.
- (d) Protection – The gas cylinder/tank and lines must be protected from rollover, collision from any direction, or damage resulting from the failure of rotating equipment.
- (e) Cylinder Location - The gas cylinder/tank and the pressure regulator must be located either rearward of the Main Roll Hoop and within the envelope defined by the Main Roll Hoop and the Frame (See **T3.2**), or in a structural side-pod. In either case it must be protected by structure that meets the requirements of **T3.24** or **T3.33**. It must not be located in the cockpit.
- (f) Cylinder Mounting - The gas cylinder/tank must be securely mounted to the Frame, engine or transmission.
- (g) Cylinder Axis - The axis of the gas cylinder/tank must not point at the driver.
- (h) Insulation - The gas cylinder/tank must be insulated from any heat sources, e.g. the exhaust system.
- (i) Lines and Fittings - The gas lines and fittings must be appropriate for the maximum possible operating pressure of the system.

T10.2 High Pressure Hydraulic Pumps and Lines

The driver and anyone standing outside the car must be shielded from any hydraulic pumps and lines with line pressures of 300 psi (2100 kPa) or higher. The shields must be steel or aluminum with a minimum thickness of 1 mm.

Note: Brake and hydraulic clutch lines are not classified as “hydraulic pump lines” and as such, are excluded from **T10.2**.

ARTICLE T11 FASTENERS

T11.1 Fastener Grade Requirements

T11.1.1 All threaded fasteners utilized in the driver’s cell structure, plus the steering, braking, driver’s harness and suspension systems must meet or exceed, SAE Grade 5, Metric Grade 8.8 and/or AN/MS specifications.

T11.1.2 The use of button head cap, pan head, flat head, round head or countersunk screws or bolts in ANY location in the following systems is prohibited:



- (a) Driver’s cell structure,
- (b) Impact attenuator attachment
- (c) Driver’s harness attachment
- (d) Steering system
- (e) Brake system
- (f) Suspension system.

Note: Hexagonal recessed drive screws or bolts (sometimes called Socket head cap screws or Allen screws/bolts) are permitted.

T11.2 Securing Fasteners



T11.2.1 All critical bolt, nuts, and other fasteners on the steering, braking, driver's harness, and suspension must be secured from unintentional loosening by the use of positive locking mechanisms.

Positive locking mechanisms are defined as those that:

- (a) The Technical Inspectors (and the team members) are able to see that the device/system is in place, i.e., it is visible, **AND**
- (b) The “positive locking mechanism” does not rely on the clamping force to apply the “locking” or anti-vibration feature. In other words, if it loosens a bit, it still prevents the nut or bolt from coming completely loose.

See [Figure 23](#)

Positive locking mechanisms include:

- (a) Correctly installed safety wiring
- (b) Cotter pins
- (c) Nylon lock nuts (where the temperature does not exceed 80°C)
- (d) Prevailing torque lock nuts

Note: Lock washers, bolts with nylon patches, and thread locking compounds, e.g. Loctite®, DO NOT meet the positive locking requirement.



Figure 23 - Examples of positive locking nuts

T11.2.2 There must be a minimum of two (2) full threads projecting from any lock nut.

T11.2.3 All spherical rod ends and spherical bearings on the steering or suspension must be in double shear or captured by having a screw/bolt head or washer with an O.D. that is larger than spherical bearing housing I.D.

T11.2.4 Adjustable tie-rod ends must be constrained with a jam nut to prevent loosening.

ARTICLE T12 TRANSPONDERS

T12.1 Transponders

- T12.1.1 Transponders will be used as part of the timing system for the Formula Hybrid + Electric competition.
- T12.1.2 Each team is responsible for having a functional, properly mounted transponder of the specified type on their vehicle. Vehicles without a specified transponder will not be allowed to compete in any event for which a transponder is used for timing and scoring.
- T12.1.3 All vehicles must be equipped with at least one MYLAPS Car/Bike Rechargeable Power Transponder or MYLAPS Car/Bike Direct Power Transponder⁷.

Note 1: Except for their name, AMB TranX260 transponders are identical to MYLAPS Car/Bike Transponders and comply with this rule. If you own a functional AMB TranX260 it does not need to be replaced.

Note 2: It is the responsibility of the team to ensure that electrical interference from their vehicle does not stop the transponder from functioning correctly



T12.2 Transponder Mounting – All Events

The transponder mounting requirements are:

- (a) **Orientation** – The transponder must be mounted vertically and orientated so the number can be read “right-side up”.
- (b) **Location** – The transponder must be mounted on the driver’s right side of the car forward of the front roll hoop. The transponder must be no more than 60 cm above the track.
- (c) **Obstructions** – There must be an open, unobstructed line between the antenna on the bottom of the transponder and the ground. Metal and carbon fiber may interrupt the transponder signal. The signal will normally transmit through fiberglass and plastic. If the

⁷ Transponders are usually available for loan at the competition. Please ask the organizers well in advance to confirm availability.

signal will be obstructed by metal or carbon fiber, a 10.2 cm diameter opening can be cut, the transponder mounted flush with the opening, and the opening covered with a material transparent to the signal.

- (d) **Protection** – Mount the transponder where it will be protected from obstacles.

ARTICLE T13 VEHICLE IDENTIFICATION

T13.1 Car Number

T13.1.1 Each car will be assigned a number at the time of its entry into a competition.

T13.1.2 Car numbers must appear on the vehicle as follows:

- (a) **Locations:** In three (3) locations: the front and both sides;
- (b) **Height:** At least 20.0 cm (8 inches) high;
- (c) **Font:** Block numbers (i.e. sans-serif characters). Italic, outline, serif, shadow, or cursive numbers are prohibited.
- (d) **Stroke Width and Spacing between Numbers:** At least 2.5 cm. (1.0 inches)
- (e) **Color:** Either white numbers on a black background or black numbers on a white background. No other color combinations will be approved.
- (f) **Background shape:** The number background must be one of the following: round, oval, square or rectangular. There must be at least 2.5 cm between the edge of the numbers and the edge of the background.
- (g) **Clear:** The numbers must not be obscured by parts of the car, e.g. wheels, side pods, exhaust system, etc.

T13.1.3 While car numbers for teams registered for Formula Hybrid + Electric can be found on the “Registered Teams” section of the SAE Collegiate Design Series website, please be sure to check with Formula Hybrid +Electric officials to ensure these numbers are correct.

Comment: Car numbers must be quickly read by course marshals when your car is moving at speed. Make your numbers easy to see and easy to read.



Figure 24 - Example Car Number

T13.2 School Name

T13.2.1 Each car must clearly display the school name (or initials – if unique and generally recognized) in roman characters at least 5 cm high on both sides of the vehicle. The characters must be placed on a high contrast background in an easily visible location.

T13.2.2 The school name may also appear in non-roman characters, but the roman character version must be uppermost on the sides.

T13.3 SAE & IEEE Logos

T13.3.1 SAE and IEEE logos must be prominently displayed on the front and/or both sides of the vehicle. Each logo must be at least 7 cm x 20 cm. The organizers will provide the following decals at the competition:

- (a) SAE, 7.6 cm x 20.3 cm in either White or Black.
- (b) IEEE, 11.4 cm x 30.5 cm (Blue only).

Actual-size JPEGs may be downloaded from the Formula Hybrid + Electric website.

T13.4 Technical Inspection Sticker Space

Technical inspection stickers will be placed on the upper nose of the vehicle. Cars must have a clear and unobstructed area at least 25.4 cm wide x 20.3cm high on the upper front surface of the nose along the vehicle centerline.

ARTICLE T14 EQUIPMENT REQUIREMENTS

T14.1 Driver's Equipment

The equipment specified below must be worn by the driver anytime he or she is in the cockpit with the engine running or with the tractive system energized.

T14.2 Helmet

T14.2.1 A well-fitting, closed face helmet that meets one of the following certifications and is labeled as such:

(a) Snell K2015, K2020, K2025, M2015, M2020D, M2020R, M2025D, M2025R, SA2015, SA2020, SA2025, EA2016

(b) SFI 31.1/2015 or newer, SFI 41.1/2015 or newer.

(c) FIA 8859-2015, FIA 8859-2024, FIA 8860-2010, FIA 8860-2018, FIA 8860-2018 ABP.

Open faced helmets and off-road/motocross helmets (helmets without integrated face shields) are not approved.

T14.2.2 All helmets to be used in the competition must be presented during Technical Inspection where approved helmets will be stickered. The organizer reserves the right to impound all non-approved helmets until the end of the competition.

T14.3 Balaclava

A balaclava which covers the driver's head, hair and neck, made from acceptable fire resistant material as defined in [T14.12](#), or a full helmet skirt of acceptable fire resistant material. The balaclava requirement applies to drivers of either gender, with any hair length.

T14.4 Eye Protection

An impact resistant face shield, made from approved impact resistant materials. The face shields supplied with approved helmets (See [T14.2](#) above) meet this requirement.

T14.5 Suit

A fire-resistant suit that covers the body from the neck down to the ankles and the wrists. One (1) piece suits are required. The suit must be in good condition, i.e. it must have no tears or open seams, or oil stains that could compromise its fire-resistant capability.

The suit must be certified to one of the following standards and be labeled as such:

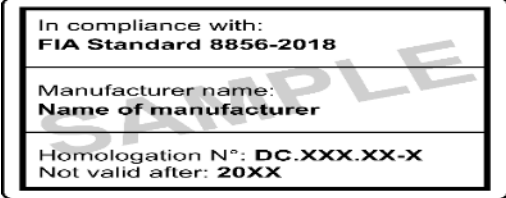
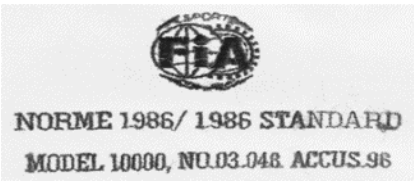
-FIA 8856-2018	 In compliance with: FIA Standard 8856-2018 Manufacturer name: Name of manufacturer Homologation N°: DC.XXX.XX-X Not valid after: 20XX
-SFI 3.4/5	 sfi MANUFACTURER CERTIFIED 3.4/5
-FIA Standard 8856-1986	  NORME 1986/ 1986 STANDARD MODEL 10000, NO.03.048. ACCUS.96
-SFI 3-2A/1 but only when used with fire resistant, e.g. Nomex, underwear that covers the body from wrist to ankles. -SFI 3-2A/5 (or higher)	 sfi MANUFACTURER CERTIFIED 3-2A/1
- FIA Standard 8856-2000	  Standard 8856-2000 RS.000.00 Manufacturer name Year of manufacture: 0000 Actual size 100x40mm

Table 7 – SFI / FIA Standards Logos

Note: An SFI 3-2A/1 or 3.4/1 (single layer) suit is ONLY allowed WITH fire resistant underwear.

T14.6 Underclothing

All drivers should wear fire resistant underwear (long pants and long sleeve top) under their approved driving suit. This fire-resistant underwear must be made from acceptable fire-resistant material and cover the driver's body completely from the neck down to the ankles and wrists.

Note: If drivers do not wear fire resistant long underwear, they should wear cotton underwear under the approved driving suit. Tee-shirts, or other undergarments made from Nylon or any other synthetic materials may melt when exposed to high heat.

T14.7 Socks

Socks made from an accepted fire-resistant material, e.g. Nomex, which cover the bare skin between the driver's suit and the boots or shoes. Socks made from wool or cotton are acceptable. Socks of nylon or polyester are not acceptable.

T14.8 Shoes

Shoes of durable fire-resistant material and which are in good condition (no holes worn in the soles or uppers).

T14.9 Gloves

Fire resistant gloves made from made from acceptable fire-resistant material as defined in [T14.12](#). Gloves of all leather construction or fire-resistant gloves constructed using leather palms with no insulating fire resisting material underneath are not acceptable.

T14.10 Arm Restraints

Arm restraints certified and labeled to SF1 standard 3.3, or a commercially manufactured equivalent, must be worn such that the driver can release them and exit the vehicle unassisted regardless of the vehicle's position.

T14.11 Driver's Equipment Condition

All driving apparel covered by [ARTICLE T14](#) must be in good condition. Specifically, driving apparel must not have any tears, rips, open seams, areas of significant wear or abrasion or stains which might compromise fire resistant performance.

T14.12 Fire Resistant Material

For the purpose of this section some, but not all, of the approved fire-resistant materials are: Carbon X, Indura, Nomex, Polybenzimidazole (commonly known as PBI) and Proban.

T14.13 Synthetic Material – Prohibited

T-shirts, socks or other undergarments (not to be confused with FR underwear) made from nylon or any other synthetic material which will melt when exposed to high heat are prohibited.

ARTICLE T15 OTHER REQUIRED EQUIPMENT

T15.1 Fire Extinguishers

T15.1.1 Each team must have at least two (2) 2.3 kg (5 lb.) dry chemical (Min. 3-A:40-B:C) Fire extinguishers.

T15.1.2 Extinguishers of larger capacity (higher numerical ratings) are acceptable.

T15.1.3 All extinguishers must be equipped with a manufacturer-installed pressure/charge gauge that shows “FULL”.

T15.2 Special Requirements

Teams must identify any fire hazards specific to their vehicle’s components and if fire extinguisher/fire extinguisher material other than those required in section **T15.1** are needed to suppress such fires, then at least two (2) additional extinguishers/material (at least 5 lb or equivalent) of the required type must be procured and accompany the car at all times.

As recommendations vary, teams are advised to consult the rules committee before purchasing expensive extinguishers that may not be necessary.

T15.3 Chemical Spill Absorbent

Teams must have chemical spill absorbent at hand, appropriate to their specific risks. This material must be presented at technical inspection.

T15.4 Insulated Gloves

Insulated gloves are required, rated for at least the voltage in the TSV system, with protective over-gloves.

T15.4.1 Electrical gloves require testing by a qualified company and must have a test date printed on them that is within 14 months of the competition.

T15.5 Safety Glasses

Safety glasses must be worn as specified in section **D10.7**

T15.6 MSDS Sheets

Materials Safety Data Sheets (MSDS) for the accumulator devices are required and must be included in the ESF.

T15.7 Additional

Any special safety equipment called for in the MSDS, for example correct gloves recommended for handling any electrolyte material in the accumulator.

ARTICLE T16 ON-BOARD CAMERAS

T16.1 Mounts

The mounts for video/photographic cameras must be of a safe and secure design.

T16.1.1 All camera installations must be approved at Technical Inspection.

T16.1.2 Helmet mounted cameras are prohibited.

T16.1.3 The body of a camera or recording unit that weighs more than 0.25 kg must be secured at a minimum of 2 points on different sides of the camera body. Plastic or elastic attachments are not permitted. If a tether is used to restrain the camera, the tether length must be limited so that the camera cannot contact the driver.

PART IC - INTERNAL COMBUSTION ENGINE

ARTICLE IC1 INTERNAL COMBUSTION ENGINE

IC1.1 Engine Limitation

Engines must be Internal Combustion, four-stroke piston engines, with a maximum displacement of 250cc for spark ignition engines and 310cc for diesel engines and be either:

- (a) Modified or custom fabricated. (See section [IC1.2](#))

OR

- (b) Stock – defined as:

- (i) Any single cylinder engine,

OR

- (ii) Any twin cylinder engine from a motorcycle approved for licensed use on public roads,

OR

- (iii) Any commercially available “industrial” IC engine meeting the above displacement limits.

Note: If you are not sure whether or not your engine qualifies as “stock”, contact the organizers.

IC1.2 Permitted modifications to a stock engine are:

- (a) Modification or removal of the clutch, primary drive and/or transmission.
- (b) Changes to fuel mixture, ignition or cam timings.
- (c) Replacement of camshaft. (Any lobe profile may be used.)
- (d) Replacement or modification of any exhaust system component.
- (e) Replacement or modification of any intake system component; i.e., components upstream of (but NOT including) the cylinder head. The addition of forced induction will move the engine into the modified category.
- (f) Modifications to the engine casings. (This does not include the cylinders or cylinder head).
- (g) Replacement or modification of crankshafts for the purpose of simplifying mechanical connections. (Stroke must remain stock.)

IC1.3 Engine Inspection

The organizers reserve the right to tear down any number of engines to confirm conformance to the rules. The initial measurement will be made externally with a measurement accuracy of one (1) percent. When installed to and coaxially with spark plug hole, the measurement tool has dimensions of 381 mm long and 30 mm diameter. Teams may choose to design in access space for this tool above each spark plug hole to reduce time should their vehicle be inspected.

IC1.4 Starter

Each car must be equipped with an on-board starter or equivalent, and must be able to move without any outside assistance at any time during the competition. Specifically, push starts are not permitted.

IC1.4.1 A hybrid may use the forward motion of the vehicle derived from the electric drive to start the I.C. engine, except that this starting technique may not be used until after the car receives the “green flag” in any event.

IC1.4.2 A manual starting system operable by the driver while belted in is permissible.

IC1.5 Air Intake System

IC1.5.1 Air Intake System Location

All parts of the engine air and fuel control systems (including the throttle or carburetor, and the complete air intake system, including the air cleaner and any air boxes) must lie within the surface defined by the top of the roll bar and the outside edge of the four tires. (See [Figure 25](#))

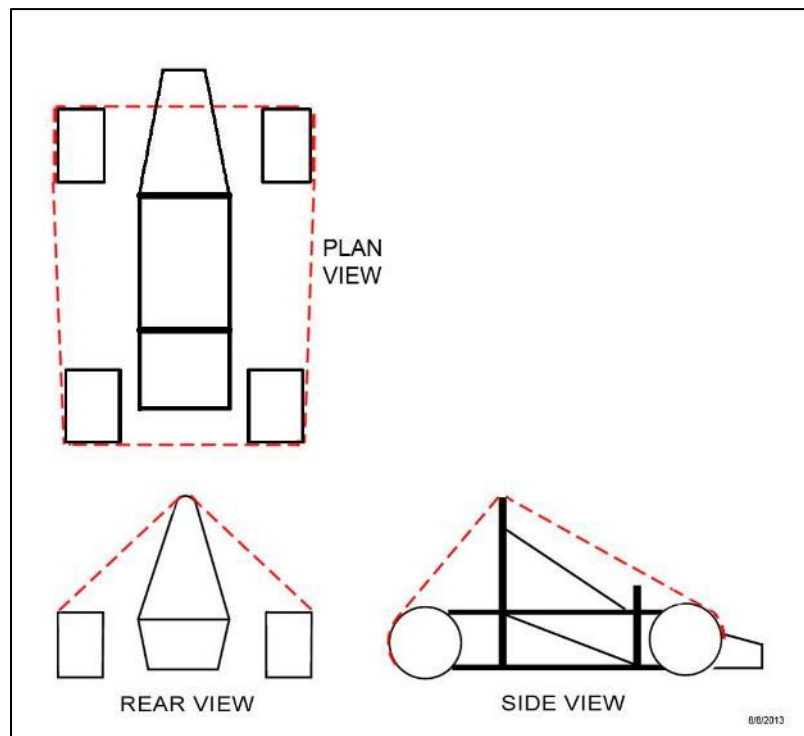


Figure 25- Surface Envelope

IC1.5.2 Any portion of the air intake system that is less than 350 mm above the ground must be shielded from side or rear impact collisions by structure built to Rule [T3.24](#) or [T3.33](#) as applicable.

IC1.5.3 Intake Manifold

If an intake manifold is used, it must be securely attached to the engine crankcase, cylinder, or cylinder head with brackets and mechanical fasteners. This precludes the use of hose clamps, plastic ties, or safety wires.

Original equipment rubber parts that bolt or clamp to the cylinder head and to the throttle body or carburetor are acceptable.

Note: These rubber parts are referred to by various names by the engine manufacturers; e.g., “insulators” by Honda, “joints” by Yamaha, and “holders” by Kawasaki.

Other than such original equipment parts the use of rubber hose is not considered a structural attachment. Intake systems with significant mass or cantilever from the cylinder head must be supported to prevent stress to the intake system.

Supports to the engine must be rigid.

Supports to the frame or chassis must incorporate some isolation to allow for engine movement and chassis flex.

IC1.5.4 Air boxes and filters

Large air boxes must be securely mounted to the frame or engine and connections between the air box and throttle must be flexible. Small air cleaners designed for mounting to the carburetor or throttle body may be cantilevered from the throttle body.

IC1.6 Accelerator and Accelerator Actuation

IC1.6.1 Carburetor/Throttle Body

All spark ignition engines must be equipped with a carburetor or throttle body. The carburetor or throttle body may be of any size or design.

IC1.6.2 Accelerator Actuation - General

All systems that transmit the driver’s control of the speed of the vehicle, commonly called “Accelerator systems”, must be designed and constructed as “fail safe” systems, so that the failure of any one component, be it mechanical, electrical or electronic, will not result in an uncontrolled acceleration of the vehicle. This applies to both IC engines and to electric motors that power the vehicle.

The Accelerator control may be actuated mechanically, electrically or electronically, i.e. electrical Accelerator control (ETC) or “drive-by-wire” is acceptable.

Drive-by-wire controls of the electric motor controller must comply with TS isolation requirements. See [EV5.1.1](#)

Any Accelerator pedal must have a positive pedal stop incorporated on the Accelerator pedal to prevent over stressing the Accelerator cable or any part of the actuation system.

IC1.6.3 Mechanical Accelerator Actuation

If mechanical Accelerator actuation is used, the Accelerator cable or rod must have smooth operation, and must not have the possibility of binding or sticking.

The Accelerator actuation system must use at least two (2) return springs located at the throttle body, so that the failure of any component of the Accelerator system will not prevent the Accelerator returning to the closed position.

Note: Springs in Throttle Position Sensors (TPS) are NOT acceptable as return springs.

Accelerator cables must be at least 50 mm from any exhaust system component and out of the exhaust stream.

Any Accelerator pedal cable must be protected from being bent or kinked by the driver’s foot when it is operated by the driver or when the driver enters or exits the vehicle.

If the Accelerator system contains any mechanism that could become jammed, for example a gear mechanism, then this must be covered to prevent ingress of any debris.

The use of a push-pull type Accelerator cable with an Accelerator pedal that is capable of forcing the Accelerator closed (e.g. toe strap) is recommended.

Electrical actuation of a mechanical throttle is permissible, provided releasing the Accelerator pedal will override the electrical system and cause the throttle to close.

IC1.6.4 Electrical Accelerator Actuation

When electrical or electronic throttle actuation is used, the throttle actuation system must be of a fail-safe design to assure that any single failure in the mechanical or electrical components of the Accelerator actuation system will result in the engine returning to idle (IC engine) or having zero torque output (electric motor).

Teams should use commercially available electrical Accelerator actuation systems.

The methodology used to ensure fail-safe operation must be included as a required appendix to the Design Report.

IC1.7 Intake System Restrictor

IC1.7.1 Non-stock engines (See [IC1.1](#)) must be fitted with an air inlet restrictor as listed below. All the air entering the engine must pass through the restrictor which must be located downstream of any engine throttling device.

IC1.7.2 The restrictor must be circular with a maximum diameter of:

(a) Gasoline fueled cars – 12.90 mm

IC1.7.3 The restrictor must be located to facilitate measurement during the inspection process.

IC1.7.4 The circular restricting cross section may NOT be movable or flexible in any way, e.g. the restrictor may not be part of the movable portion of a barrel throttle body.

IC1.7.5 Any device that has the ability to throttle the engine downstream of the restrictor is prohibited.

IC1.7.6 If more than one engine is used, the intake air for all engines must pass through the one restrictor.

Note: Section [IC1.7](#) applies only to those engines that are not on the approved stock engine list, or that have been modified beyond the limits specified in [IC1.2](#).

IC1.8 Turbochargers & Superchargers

Turbochargers or superchargers are permitted. The compressor must be located downstream of the inlet restrictor. The addition of a Turbo or Supercharger will move the engine into the Modified category.

IC1.9 Fuel Lines

IC1.9.1 Plastic fuel lines between the fuel tank and the engine (supply and return) are prohibited.

IC1.9.2 If rubber fuel line or hose is used, the components over which the hose is clamped must have annular bulb or barbed fittings to retain the hose. Also, clamps specifically designed for fuel lines must be used. These clamps have three (3) important features; (See [Figure 26](#))

(a) A full 360 degree (360°) wrap,

(b) a nut and bolt system for tightening, and

(c) rolled edges to prevent the clamp cutting into the hose.

Worm-gear type hose clamps are not approved for use on any fuel line.





Figure 26 – Hose Clamps

IC1.9.3 Fuel lines must be securely attached to the vehicle and/or engine.

IC1.9.4 All fuel lines must be shielded from possible rotating equipment failure or collision damage.

IC1.10 Fuel Injection System Requirements

IC1.10.1 Fuel Lines – Flexible fuel lines must be either

- (a) Metal braided hose with either crimped-on or reusable, threaded fittings, **OR**
- (b) Reinforced rubber hose with some form of abrasion resistant protection with fuel line clamps per [IC1.9.2](#).

Note: Hose clamps over metal braided hose will not be accepted.

IC1.10.2 Fuel Rail – If used, a fuel rail must be securely attached to the engine cylinder block, cylinder head, or intake manifold with brackets and mechanical fasteners. This precludes the use of hose clamps, plastic ties, or safety wire.

IC1.11 Crankcase / Engine lubrication venting

IC1.11.1 Any crankcase or engine lubrication vent lines routed to the intake system must be connected upstream of the intake system restrictor, if fitted.

IC1.11.2 Crankcase breathers that pass through the oil catch tank(s) to exhaust systems, or vacuum devices that connect directly to the exhaust system, are prohibited.

ARTICLE IC2 FUEL AND FUEL SYSTEM

IC2.1 Fuel

IC2.1.1 All fuel at the Formula Hybrid + Electric Competition will be provided by the organizer.

IC2.1.2 During all performance events the cars must be operated with the fuels provided by the organizer.

IC2.1.3 The fuels provided at the Formula Hybrid + Electric Competition are:

- (a) Gasoline (Sunoco Optima)

Note: More information including the fuel energy equivalencies, and a link for the Sunoco fuel specifications are given in [Appendix A](#)

IC2.2 Fuel Additives - Prohibited

- IC2.2.1 Nothing may be added to the provided fuels. This prohibition includes nitrous oxide or any other oxidizing agent.
- IC2.2.2 No agents other than fuel and air may be induced into the combustion chamber. Non-adherence to this rule will be reason for disqualification.
- IC2.2.3 Officials have the right to inspect the oil.

IC2.3 Fuel Temperature Changes - Prohibited

The temperature of fuel introduced into the fuel system may not be changed with the intent to improve calculated fuel efficiency.

IC2.4 Fuel Tanks

- IC2.4.1 The fuel tank is defined as that part of the fuel containment device that is in contact with the fuel. It may be made of a rigid material or a flexible material.
- IC2.4.2 Fuel tanks made of a rigid material cannot be used to carry structural loads, e.g. from roll hoops, suspension, engine or gearbox mounts, and must be securely attached to the vehicle structure with mountings that allow some flexibility such that chassis flex cannot unintentionally load the fuel tank.
- IC2.4.3 Any fuel tank that is made from a flexible material, for example a bladder fuel cell or a bag tank, must be enclosed within a rigid fuel tank container which is securely attached to the vehicle structure. Fuel tank containers (containing a bladder fuel cell or bag tank) may be load carrying.
- IC2.4.4 Any size fuel tank may be used.
- IC2.4.5 The fuel system must have a drain fitting for emptying the fuel tank. The drain must be at the lowest point of the tank and be easily accessible. It must not protrude below the lowest plane of the vehicle frame, and must have provision for safety wiring.

IC2.5 Fuel System Location Requirements

- IC2.5.1 All parts of the fuel storage and supply system must lie within the surface defined by the top of the roll bar and the outside edge of the four tires. (See [Figure 25](#)).
- IC2.5.2 All fuel tanks must be shielded from side or rear impact collisions. Any fuel tank which is located outside the Side Impact Structure required by [T3.24](#) or [T3.33](#) must be shielded by structure built to [T3.24](#) or [T3.33](#).
- IC2.5.3 A firewall must separate the fuel tank from the driver, per [T4.5](#).

IC2.6 Fuel Tank Filler Neck

- IC2.6.1 All fuel tanks must have a filler neck⁸:
 - (a) With a minimum inside diameter of 38 mm
 - (b) That is vertical (with a horizontal filler cap) or angled at no more than forty-five degrees (45°) from the vertical.

⁸ Some flush fillers may be approved by contacting the Formula Hybrid + Electric rules committee.

IC2.6.2 If a sight tube is fitted, it may not run below the top surface of the fuel tank.

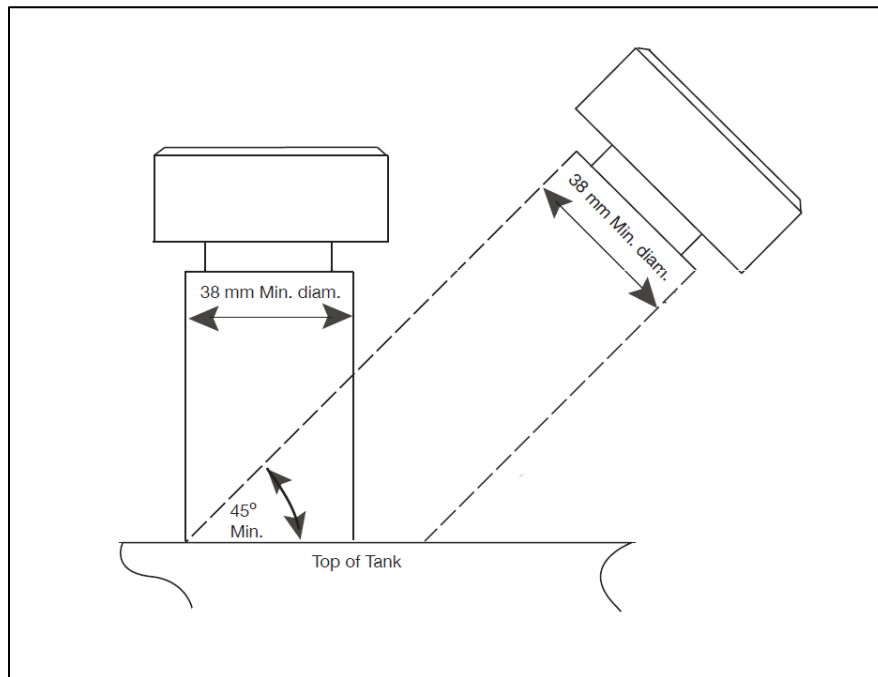


Figure 27 - Filler Neck

IC2.7 Tank Filling Requirement

- IC2.7.1 The tank must be capable of being filled to capacity without manipulating the tank or vehicle in any way (shaking vehicle, etc.).
- IC2.7.2 The fuel system must be designed such that the spillage during refueling cannot contact the driver position, exhaust system, hot engine parts, or the ignition system.
- IC2.7.3 Belly pans must be vented to prevent accumulation of fuel.

IC2.8 Venting Systems

- IC2.8.1 The fuel tank and carburetor venting systems must be designed such that fuel cannot spill during hard cornering or acceleration. This is a concern since motorcycle carburetors normally are not designed for lateral accelerations.
- IC2.8.2 All fuel vent lines must be equipped with a check valve to prevent fuel leakage when the tank is inverted. All fuel vent lines must exit outside the bodywork.

ARTICLE IC3 EXHAUST SYSTEM AND NOISE CONTROL

IC3.1 Exhaust System General

IC3.1.1 Exhaust Outlet

The exhaust must be routed so that the driver is not subjected to fumes at any speed considering the draft of the car.

- IC3.1.2 The exhaust outlet(s) must not extend more than 45 cm behind the centerline of the rear wheels, and must be no more than 60 cm above the ground.
- IC3.1.3 Any exhaust components (headers, mufflers, etc.) that protrude from the side of the body in front of the main roll hoop must be shielded to prevent contact by persons approaching the car or a driver exiting the car.
- IC3.1.4 The application of fibrous material, e.g. “header wrap”, to the outside of an exhaust manifold or exhaust system is prohibited.

IC3.2 Noise Measuring Procedure

- IC3.2.1 The sound level will be measured during a static test. Measurements will be made with a free-field microphone placed free from obstructions at the exhaust outlet level, 0.5 m from the end of the exhaust outlet, at an angle of forty-five degrees (45°) with the outlet in the horizontal plane. The test will be run with the gearbox in neutral at the engine speed defined below. Where more than one exhaust outlet is present, the test will be repeated for each exhaust and the highest reading will be used.

Vehicles that do not have manual throttle control must provide some means for running the engine at the test RPM.

- IC3.2.2 The car must be compliant at all engine speeds up to the test speed defined below.
- IC3.2.3 If the exhaust has any form of movable tuning or throttling device or system, it must be compliant with the device or system in all positions. The position of the device must be visible to the officials for the noise test and must be manually operable by the officials during the noise test.

IC3.2.4 Test Speeds

The test speed for a given engine will be the engine speed that corresponds to an average piston speed of 914.4 m/min for automotive or motorcycle engines, and 731.5 m/min for Diesels and “Industrial” engines. The calculated speed will be rounded to the nearest 500 rpm. The test speeds for typical engines will be published by the organizers.

- IC3.2.5 An “industrial engine” is defined as an engine which, according to the manufacturers’ specifications and without the required restrictor, is not capable of producing more than 5 hp per 100cc. To have an engine classified as “an industrial engine”, approval must be obtained from organizers prior to the Competition.

- IC3.2.6 Vehicles not equipped with engine tachometers must provide some external means for measuring RPM, such as a hand-held meter or lap top computer.



Note: Teams that do not provide the means to measure engine speed will not pass the noise test, will not receive the sticker and hence will not be eligible to compete in any dynamic event.

- IC3.2.7 Engines with mechanical, closed loop speed control will be tested at their maximum (governed) speed.

IC3.3 Maximum Sound Level

The maximum permitted sound level is 110 dB A, fast weighting.

IC3.4 Noise Level Re-testing

At the option of the officials, noise can be measured at any time during the competition. If a car fails the noise test, it will be withheld from the competition until it has been modified and re-passes the noise test.

PART EV - ELECTRICAL POWERTRAINS AND SYSTEMS

ARTICLE EV1 ELECTRICAL SYSTEMS OVERVIEW

EV1.1 Definitions

- **Accumulator** - Batteries and/or capacitors that store the electrical energy to be used by the tractive system. This term includes both electrochemical batteries and ultracapacitor devices.
- **Accumulator Container** - A housing that encloses the accumulator devices, isolating them both physically and electrically from the rest of the vehicle.
- **Accumulator Segment** - A subgroup of accumulator devices that must adhere to the voltage and energy limits listed in [Table 8](#).
- **AMS – Accumulator Monitoring System.** [EV9.6](#)
- **Barrier** – A material, usually rigid, that resists a flow of charge. Most often used to provide a structural barrier and/or increase the creepage distance between two conductors. See [Figure 36](#).
- **BRB – Big Red Button.** [EV7.5](#) and [EV7.6](#)
- **Creepage Distance** - The shortest distance measured along the surface of the insulating material between two conductors. See [Figure 36](#).
- **Enclosure** – An insulated housing containing electrical circuitry.
- **GLV Grounded Low Voltage system** - Every conductive part that is not part of the tractive system. (This includes the GLV electrical system, frame, conductive housings, carbon-fiber components etc.)
- **GLVMS – Grounded Low Voltage Master Switch.** [EV7.3](#)
- **IMD – Insulation Monitoring Device.** [EV9.4](#)
- **Insulation** – A material that physically resists a flow of charge. May be rigid or flexible.
- **Isolation** - Electrical, or “Galvanic” isolation between two or more electrical conductors such that if a voltage potential exists between them, no current will flow.
- **Region** – A design/construction methodology wherein enclosures are divided by insulating barriers and/or spacing into TS and GLV sections, simplifying the electrical isolation of the two systems.
- **Separation** – A physical distance (“spacing”) maintained between conductors.
- **SMD – Segment Maintenance Disconnect.** [EV2.7](#)
- **ESOK – Electrical Systems OK.** [EV9.3](#)
- **Tractive System (TS)** - The drive motors, the accumulators and every part that is electrically connected to either of those components.

- **Tractive System Enclosure** - A housing that contains TS components other than accumulator devices.
- **TSSI – Tractive System Status Indicator.** [EV2.9.1](#)
- **TS/GLV** – The relationship between two electrical conductors; one being part of the TS system and the other GLV.
- **TS/TS** – The relationship between two TS conductors. This designation implies that they are at different potentials and/or polarities.
- **TSMP – Tractive System Measuring Points.** [EV10.3](#)
- **TSMS – Tractive System Master Switch.** [EV7.4](#)

EV1.2 Maximum System Voltages

EV1.2.1 The maximum permitted operating voltage and energy limits are listed in [Table 8](#) below.

Formula Hybrid + Electric voltage and energy limits	
Maximum operating voltage for Hybrid ⁹ (TSV)	300 V
Maximum operating voltage for EV	600V
Maximum GLV	60 VDC or 50 VAC
Maximum accumulator segment voltage	120 V
Maximum accumulator segment energy ¹⁰	6 MJ

Table 8 - Voltage and Energy Limits

EV1.2.2 Maximum operating voltage for hybrid vehicles above those shown in [Table 8](#) may be permitted via a request for a special allowance to the FH+E rules committee. Teams requesting such an allowance must demonstrate an appropriate level of electrical engineering knowledge and experience.

EV1.2.3 Vehicles with a maximum operating voltage over 300V will be required to complete a form demonstrating electrical safety competency including:

- Written vehicle electrical safety procedures
- A list of electrical safety team members
- Review of team experience and competency in high-voltage electrical systems

⁹ The maximum operating voltage is defined as the maximum measured accumulator voltage during normal charging conditions. (A maximum capacity accumulator will require at least three segments.)

¹⁰ Segment energy is calculated as Energy (MJ) = $(V \cdot Ah \cdot 3600) / 1e^6$. (Note that this is different from the fuel allocation energy in [Appendix A](#)).

- Acceptance of this document is a requirement for participation in the event.
- (a) This form will be an addendum in ESF-1

ARTICLE EV2 ENERGY STORAGE (ACCUMULATORS)

EV2.1 Permitted Devices

EV2.1.1 The following accumulator devices are acceptable: batteries (e.g. lithium-ion, NiMH, lead acid and similar battery chemistries) and capacitors, such as super caps or ultracaps.

The following accumulator devices are not permitted: molten salt batteries, thermal batteries, fuel cells, mechanical storage such as flywheels or hydraulic accumulators.

EV2.1.2 Accumulator systems using pouch-type lithium-ion cells must be commercially constructed and specifically approved by the Formula Hybrid + Electric rules committee or must comply with the requirements detailed in **ARTICLE EV11**.

EV2.1.3 Manufacturer's data sheets showing the rated specification of the accumulator cell(s) which are used must be provided in the ESF along with their number and configurations.



EV2.2 Accumulator Segments

EV2.2.1 The accumulator segments must be separated so that the segment limits in **Table 8** are met by an electrically insulating barrier meeting the TS/GLV requirements in **EV5.4**. For all lithium based cell chemistries, these barriers must also be fire resistant (according to UL94-V0, FAR25 or equivalent).

EV2.2.2 The barrier must be non-conducting, fire retardant (UL94V0) and provide a complete barrier to the spread of arc or fire. It must have a fire resistance equal to 1/8" FR4 fiberglass, such as Garolite G-9¹¹.

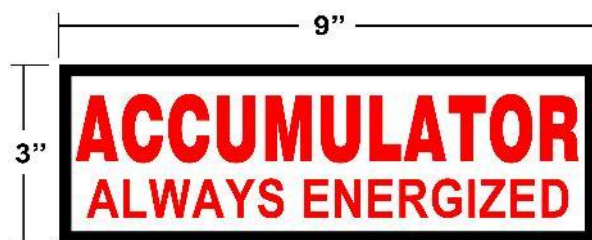
EV2.2.3 UL-94-HB sealants may be used for small gaps.

EV2.3 Accumulator Containers

EV2.3.1 All devices which store the tractive system energy must be enclosed in (an) accumulator container(s).

EV2.3.2 Each accumulator container must be labeled with the words "**ACCUMULATOR – ALWAYS ENERGIZED**". (This is in addition to the TSV label requirements in **EV3.1.5**).

Labels must be 3 inches by 9 inches with bold red lettering on a white background and be clearly visible on all sides of the accumulator container that could be exposed during operation



and/or maintenance of the car and at least one such label must be visible with the bodywork in place¹².

¹¹ <https://www.mcmaster.com/grade-g-9-garolite/>

¹² Self-adhesive stickers are available from the organizers on request.

Figure 28 - Accumulator Sticker

EV2.3.3 If the accumulator container(s) is not easily accessible during Electrical Tech Inspection, detailed pictures of the internals taken during assembly must be provided. If the pictures do not adequately depict the accumulator, it may be necessary to disassemble the accumulator to pass Electrical Tech Inspection.

EV2.3.4 The accumulator container may not contain circuitry or components other than the accumulator itself and necessary supporting circuitry such as the AIRs, AMS, IMD and pre-charge circuitry. For example, the accumulator container may not contain the motor controller, TSV or any GLV circuits other than those required for necessary accumulator functions.

Note 1: The purpose of this requirement is to allow work on other parts of the tractive system without opening the accumulator container and exposing (always-live) high voltage.

Note 2: It is possible to meet this requirement by dividing a large box into an accumulator section and a non-accumulator section, with an insulating barrier between them. In this case, it must be possible to open the non-accumulator section while keeping the accumulator section closed, meeting the requirements of the “finger probe” test. See: [EV3.1.3](#).

EV2.3.5 If spare accumulators, segments, modules, or cells are brought to the competition, then they all must be of the same size, weight and type as those that they are intended to replace. Spare components must be presented at Electrical Tech Inspection.

EV2.4 Accumulator Container –Construction

EV2.4.1 The accumulator container(s) must be built of mechanically robust material.

EV2.4.2 The container material must be fire resistant according to UL94-V0, FAR25 or equivalent.

EV2.4.3 If any part of the accumulator container is made of electrically conductive material, an insulating barrier meeting the TS/GLV requirements in [EV5.4](#) must be affixed to the inside wall of the accumulator container such that the barrier provides 100% coverage of all electrically conductive container components.

EV2.4.4 All conductive penetrations (mounting hardware, hinges, latches etc.) must be covered on the inside of the accumulator container by an insulating material meeting [EV5.4](#).

EV2.4.5 All conductive surfaces on the outside of the container must have a low-resistance connection to the GLV system ground.

EV2.4.6 The cells and/or segments must be appropriately secured against loosening inside the container. Internal structural elements must be either non-thermoplastic or have a melting temperature of greater than 150 degrees C.

EV2.4.7 All accumulator devices must be attached to the accumulator container(s) with mechanical fasteners.

EV2.4.8 Holes in the container are only allowed for the wiring-harness, ventilation, cooling or fasteners. All wiring harness holes must be sealed according to [EV3.3.1](#). Openings for ventilation must be designed to prevent water entry from rain or road spray, and to minimize spread of fire. Completely open side pods are not allowed.

EV2.4.9 Any accumulator that may vent an explosive gas must have a ventilation system or pressure relief valve to prevent the vented gas from reaching an explosive concentration.

EV2.4.10 Every accumulator container which is completely sealed must have a pressure relief valve to prevent high-pressure in the container.

EV2.4.11 An Accumulator Container that is built to an approved 2023 or 2024 FSAE Structural Equivalency EV-HV Enclosure Spreadsheet will be considered to be in compliance with Formula Hybrid + Electric Rules [EV2.4.1-EV2.4.7](#).

EV2.5 Accumulator Container – Mounting

EV2.5.1 All accumulator containers must lie within the surface envelope as defined by [IC1.5.1](#)

EV2.5.2 All accumulator containers must be rigidly mounted to the chassis to prevent the containers from loosening during the dynamic events or possible accidents.

EV2.5.3 The mounting system for the accumulator container must be designed to withstand forces from a 40g deceleration in the horizontal plane and 20 g deceleration in the vertical plane. The calculations/tests proving this must be part of the SES.

EV2.5.4 For tube frame cars, each accumulator container must be attached to the Frame by a minimum of four (4) 8 mm Metric Grade 8.8 or 5/16 inch Grade 5 bolts.

EV2.5.5 For monocoques:

1. Each accumulator container must be attached to the Frame at a minimum of four (4) points, each capable of carrying a load in any direction of 400 Newtons x the mass of the accumulator in kgs, i.e. if the accumulator has a mass of 50 kgs, each attachment point must be able to carry a load of 20kN in any direction.
2. The laminate, mounting plates, backing plates and inserts must have sufficient shear area, weld area and strength to carry the specified load in any direction. Data obtained from the laminate perimeter shear strength test ([T3.30](#)) should be used to prove adequate shear area is provided***
3. Each attachment point requires a minimum of one (1) 8 mm Metric Grade 8.8 or 5/16 inch SAE Grade 5 bolt.
4. Each attachment point requires steel backing plates with a minimum thickness of 2 mm. Alternate materials may be used for backing plates if equivalency is approved.
5. The calculations/tests must be included in the SES.

EV2.5.6 An Accumulator Mounting system that is built to an approved 2023 or 2024 FSAE Structural Equivalency EV-HV Enclosure Spreadsheet will be considered to be in compliance with Formula Hybrid + Electric Rules EV2.5, and the FSAE SES may be submitted in place of the Formula Hybrid + Electric specific SES required by [EV2.5.3](#).

EV2.6 Accumulator Fusing

EV2.6.1 Every accumulator container must contain at least one fuse in the high-current TS path, located on the same side of the AIRs as the battery or capacitor. These fuses must comply with [EV5.5.4](#).

EV2.6.2 All details and documentation for fuse, fusible link and/or internal over current protection must be included in the ESF.

EV2.6.3 Parallel then Series (nPmS) connections.

If more than one battery cell or capacitor is used to form a set of cells in parallel and those parallel groups are then combined in series (**Figure 29**) then either:

- a) Each cell must be protected with a fuse or fusible link with a current rating less than or equal to 50% of the calculated short-circuit current (I_{sc}) of the cell or capacitor, where $I_{sc} = V_{nom} / IR(\Omega)$. (The nominal cell voltage divided by its dc internal resistance).. The fuse or fusible link must be rated for the full tractive system voltage, unless the special conditions in **EV2.6.5** (Fuse Voltage Ratings) are met.

OR

- b) Manufacturer's documentation must be provided that certifies that it is acceptable to use this number of single cells in parallel without fusing. This certification must be included in the ESF. (Commercially assembled packs or modules installed per manufacturer's instructions may be exempt from this requirement upon application to the rules committee.)

Note: if option (a) is used, fuse j in **Figure 27** may be omitted if all conductors carrying the entire pack current are adequately sized for the sum of the parallel fuse current ratings (i.e. for n fuses in parallel, each with current rating i , the conductors must be sized for a total current $i_{total} = n \cdot i$)

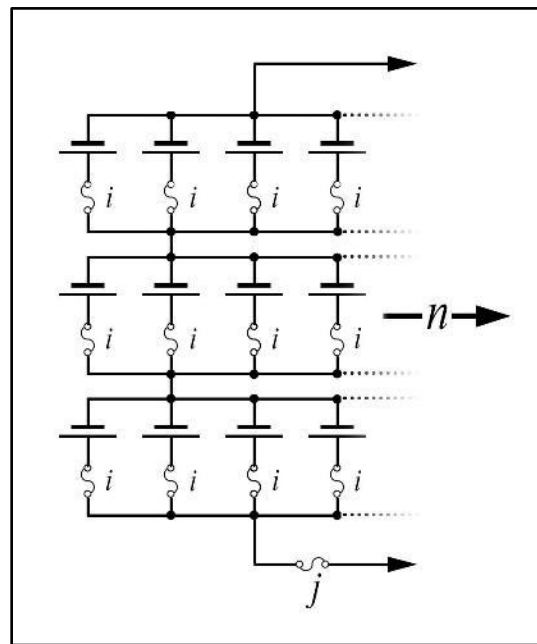


Figure 29 - Example nP3S Configuration

EV2.6.4 Series then Parallel (nSmP) connections.

If strings of batteries or capacitors in series are then combined in parallel (**Figure 30**) then each string must be individually fused per **EV5.5.4**.

Fuse j in **Figure 30** may be omitted if all conductors carrying the entire pack current are adequately sized for the sum of the parallel fuse current ratings (i.e. for n fuses in parallel, each with current rating i , the conductors must be sized for a total current $i_{total} = n \cdot i$ without fuse j , or may be sized for a lower current j if fuse j is included.)

All fuses used in **nSmP** configurations must be rated for the full tractive system voltage.

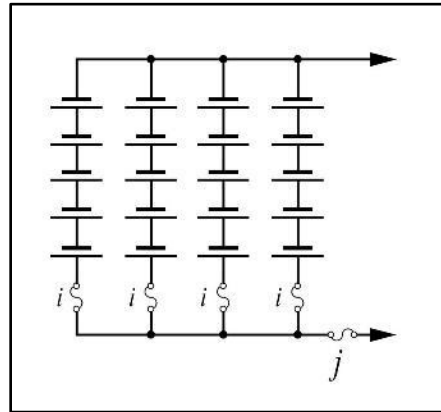


Figure 30 – Example 5S4P Configuration

EV2.6.5 Fuse Voltage Ratings.

Although fuses in the tractive system must normally be rated for the full tractive system voltage, under certain conditions an exception can be made for fuses or fusible links for individual cells or capacitors. These conditions, **defined in EV2.6.1**, apply to fuses or fusible links used to meet **EV2.6.3** and to **any** fusible links that are included in the accumulator construction. **These requirements are intended to ensure coordination between cell fuses and pack master fuses.**

The following conditions must be met to allow reduced voltage ratings:

The series fuse used to satisfy **EV2.6.1** is rated at a current less than or equal to **one half** of the sum of the parallel low-voltage fuses or fusible links, **AND**

1. fuse or fusible link current rating is specified in manufacturer's data

OR

2. suitable team generated test data is provided, demonstrating the ability to:
 - (i) Carry full rated current for at least 10 minutes with less than 50° C temperature rise.
 - (ii) Trip at 300% of rated current.
 - (iii) Interrupt current equal to the maximum short circuit current expected without producing heat, sparks, or flames that might damage nearby cells.

For example, in **Figure 29**, this requirement is met if $j \leq n \cdot \frac{i}{3}$, where i is the cell fuse or link rating, n is the number of cells in parallel and j is the master series fuse rating.

EV2.7 Accumulator - Segment Maintenance Disconnect

EV2.7.1 A Segment Maintenance Disconnect (SMD) must be installed between each accumulator segment See [EV2.2](#)). The SMD must be used whenever the accumulator containers are opened for maintenance and whenever accumulator segments are removed from the container.

Note: If the high-voltage disconnect (HVD, section [EV2.9](#)) is located between segments, it satisfies the requirement for an SMD between those segments.

EV2.7.2 The SMD may be implemented with a switch or a removable maintenance plug. Devices capable of remote operation such as relays or contactors may not be used. There must be a positive means of securing the SMD in the disconnected state; for example, a lockable switch can be secured with a zip-tie or a clip.

EV2.7.3 SMD methods requiring tools to isolate the segments are not permitted.

EV2.7.4 If the SMD is operated with the accumulator container open, any removable part used with the SMD (e.g., a removable plug or clip) must be non-conductive on its external surfaces. i.e. it will not cause a short if dropped into the accumulator. **This requirement may be waived on application to the rules committee if connection to battery terminals is prevented by barriers or location.**

EV2.7.5 Devices used for SMDs must be rated for the expected battery current and voltages

EV2.8 Accumulator - Isolation Relays

EV2.8.1 At least two isolation relays (AIRs) must be installed in every accumulator container, or in the accumulator section of a segmented container (See [EV2.3.4](#) Note 2) such that no TS voltage will be present outside the accumulator or accumulator section when the TS is shut down.

Note: AIRs must be before TSMPs, such that TSMPs de-energize when AIRs are open.

EV2.8.2 The accumulator isolation relays must be of a normally open (N.O.) type which are held in the closed position by the current flowing through the shutdown loop ([EV7.1](#)). When this flow of current is interrupted, the AIRs must disconnect both poles of the accumulator such that no TS voltage is present outside of the accumulator container(s).



EV2.8.3 When the AIRs are opened, the DC voltage in the tractive system must drop to under 60 VDC in less than five seconds.

EV2.8.4 The AIR contacts must be protected by Pre-Charge and Discharge circuitry, See [EV2.10](#).

EV2.8.5 If the AIR coils are not equipped with transient suppression by the manufacturer, then Transient Voltage suppressors¹³ must be added in parallel with the AIR coils.

EV2.8.6 AIRs containing mercury are not permitted.

¹³ Transient suppressors protect the circuitry in the shutdown loop from di/dt voltage spikes. The TVS voltage rating should be chosen appropriately.

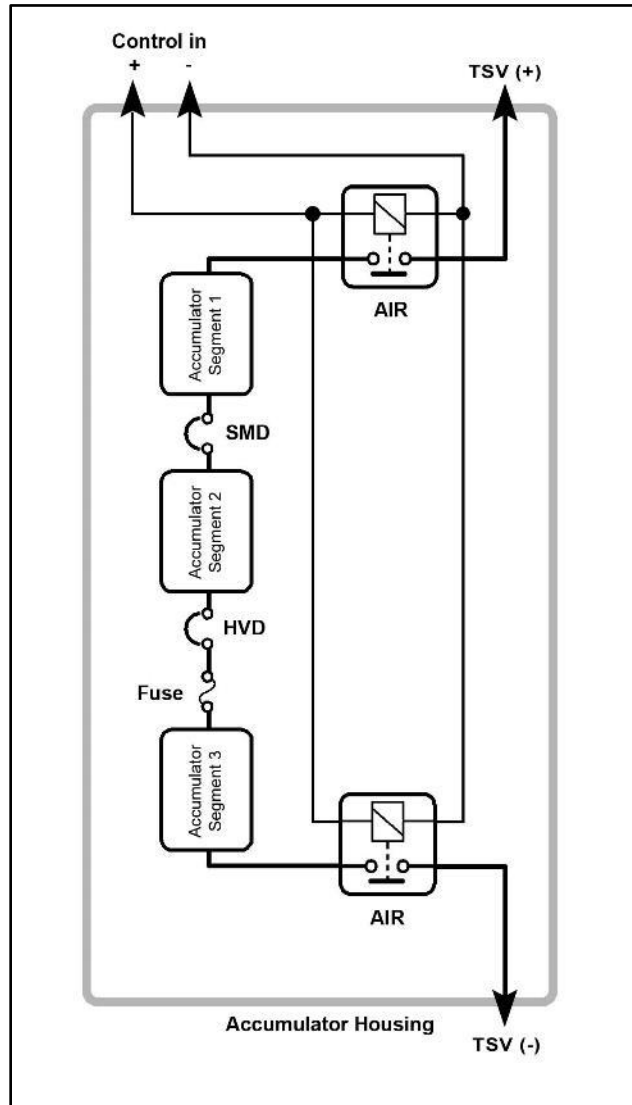


Figure 31 - Example Accumulator Segmenting

EV2.9 Accumulator - High Voltage Disconnect

EV2.9.1 Each vehicle must be fitted with a High Voltage Disconnect (HVD) making it possible to quickly and positively break the current path of the tractive system accumulator. This can be accomplished by turning off a disconnect switch, disconnecting the main connector or removing an accessible element such as a fuse. **A contactor or AIR device may not be used as an HVD.**

EV2.9.2 The HVD must be operable without the use of tools.

EV2.9.3 It must be possible to disconnect the HVD within 10 seconds in ready-to-race condition.

Note: Ready-to-race means that the car is fully assembled, including having all body panels in position, with a driver seated in the vehicle and without the car jacked up.

EV2.9.4 The team must demonstrate this during Electrical Tech Inspection.

- EV2.9.5 The disconnect must be clearly marked with "HVD". Multiple disconnects must be marked "HVD *n* of *m*", such as "HVD 1 of 3", HVD 2 of 3", etc.
- EV2.9.6 There must be a positive means of securing the HVD in the disconnected state; for example, a lockable switch can be secured with a zip-tie or a clip.
- Note:** A removable plug will meet this requirement if the plug is secured or fully removed such that it cannot accidentally reconnect.
- EV2.9.7 The HVD must be opened as part of the Lockout/Tagout procedure. See [EV12.1.1](#).
- EV2.9.8 The recommended electrical location for the HVD is near the middle of the accumulator string. In this case, it can serve as one of the SMDs. (See [Figure 31](#)).
- Note:** The HVD must be prior to TSMPs such that TSMPs are de-energized when the HVD is open.

EV2.10 Accumulator - Pre-Charge and Discharge Circuits

One particularly dangerous failure mode in an electric vehicle is AIR contacts that have welded shut. When the current through the AIR coils is interrupted, the AIRs should open, isolating the potentially lethal voltages within the accumulator container. If the AIRs are welded shut, these voltages will be present outside the accumulator even though the system is presumably shut down.

Welding is caused by high instantaneous currents across the contacts. This can be avoided by a correctly functioning pre-charge circuit.

EV2.10.1 Pre-charge

The AIR contacts must be protected by a circuit that will pre-charge the intermediate circuit to at least 90% of the rated accumulator voltage before completing the intermediate circuit by closing the second AIR.

The pre-charge circuit must be disabled if the shutdown circuit is deactivated; see [EV7.1](#). i.e. the pre-charge circuit must not be able to pre-charge the system if the shutdown circuit is open.

Pre-charge timing must be based on a voltage measurement of the intermediate circuit voltage compared to the voltage of the accumulator. The intermediate circuit voltage is typically the motor controller input voltage

- EV2.10.2 The pre-charge circuit must operate regardless of the sequence of operations used to energize the vehicle, including, for example, restarting after being automatically shut down by a safety circuit or the TSMS being toggled on or off repeatedly.

Note: Teams should consider failure models when there is a low/medium-impedance short downstream of the pre-charge circuitry, including a welded discharge relay

EV2.10.3 Discharge

If a discharge circuit is needed to meet the requirements of [EV2.8.3](#), it must be designed to handle the maximum expected discharge current for at least 15 seconds. The calculations determining the component values must be part of the ESF.

- EV2.10.4 The discharge circuit must be fail-safe. i.e. wired in a way that it is always active whenever the shutdown circuit is open or de-energized.

EV2.10.5 For always-on discharge circuits and other circuits that dissipate significant power for extended time periods, calculations of the maximum operating temperature of the power dissipating components (e.g., resistors) must be included in the ESF.

Note: Resistors operating at their rated power level often operate at 200° C or more. Insulating materials in proximity to resistors require care to ensure that they are not overheated. Resistors that dissipate more than 50% of their power rating and resistors that dissipate significant power without much open space around them will require additional care to ensure they are safe.

It is recommended that teams use resistors rated for at least double the maximum predicted power dissipation, document these design details in the ESF, and mount them such that heat dissipation is not impeded.

Note also that some resistors require external heat sinks in order to dissipate their rated power. Using these resistors for significant power dissipation will require additional documentation of the heat sink design or testing.

EV2.10.6 Fuses in the accumulator discharge current paths are not permitted.

EV2.10.7 Fuses in the accumulator pre-charge current path are permitted

EV2.11 Accumulator – Accumulator Management System (AMS)

EV2.11.1 Each accumulator must be monitored by an accumulator management system (AMS) whenever the tractive system is active or the accumulator is connected to a charger.

Note: Some parts of [EV2.11](#) may be waived for commercially manufactured accumulator assemblies. Requests must be submitted to the Formula Hybrid + Electric rules committee at least one month before the competition.

EV2.11.2 The AMS must monitor all critical voltages and temperatures in the accumulator as well the integrity of all its voltage and temperature inputs. If an out-of-range or a malfunction is detected, it must shut down the electrical systems, open the AIRs and shut down the I.C. drive system.¹⁴ (Some GLV systems may remain energized – See [Figure 37](#))

EV2.11.3 The tractive system must remain disabled until manually reset by a person other than the driver. It must not be possible for the driver to re-activate the tractive system from within the car in case of an AMS fault. AMS faults must be latched using a mechanical relay circuit (see [Appendix G](#)). Digital logic or microcontrollers may not be used for this function.

EV2.11.4 The AMS must continuously measure cell voltages in order to keep those voltages inside the allowed minimum and maximums stated in the cell data-sheet. (See [Table 9](#))

Notes:

1. If individual cells are directly connected in parallel, only one voltage measurement is required for that group. (Measured at the parallel connections, outside of the cell fuses. (See [Figure 29](#))

¹⁴ Teams may wish to also use the AMS to detect a low voltage or high temperature before these cross the critical threshold, and to alert the driver and/or decrease the power drawn from the accumulator, so as to mitigate the problem before the vehicle must be shut down.

2. Exemptions may be granted for commercial accumulator assemblies that do not meet these requirements.

Chemistry	Maximum number of cells per voltage measurement
PbAcid	6
NiMh	6
Lithium based	1

Table 9 - AMS Voltage Monitoring

EV2.11.5 The AMS must monitor the temperature of the minimum number of cells in the accumulator as specified in **Table 10** below. The monitored cells must be equally distributed over the accumulator container(s).

Chemistry	Cells monitored
PbAcid	5%
UltraCap	10%
NiMh	10%
Li-Ion	30%

Table 10 – AMS Temperature Monitoring

NOTE: It is acceptable to monitor multiple cells with one sensor if this sensor has direct contact to all monitored cells. It is also acceptable to use maximum-temperature detection schemes such as diode-paralleled thermistors or zener diodes to monitor multiple cells on a single circuit.

NOTE: Teams should monitor the temperature of all cells.

NOTE: AMS temperature monitoring at levels less than the requirements in Table 10 may be permitted, on application to the tech committee, for commercial accumulator modules contain integrated temperature sensors.

EV2.11.6 All voltage sense wires to the AMS must be protected by fuses or resistors (located as close as possible to the energy source) so that they cannot exceed their current carrying capacity in the event of a short circuit.

Note: If the AMS monitoring board is directly connected to the cell, it is acceptable to have a fuse or resistor integrated into the monitoring board.

EV2.11.7 Input channels of the AMS used for different segments of the accumulator must be isolated from one another with isolation rated for at least the maximum tractive system voltage. This isolation is also required between channels or sections of the AMS that are connected to different sides of a SMD, HVD, fuse, or AIR.

EV2.11.8 Any GLV connection to the AMS must be galvanically isolated from the TSV. This isolation must be documented in the ESF.

Note: Per [EV2.8.2](#), AMS connections that are not isolated, such as cell sense wires, cannot exit the accumulator container, unless they are isolated by additional relays when the AIRs are off. This requirement should be considered in the selection of an AMS system for a vehicle that uses more than one accumulator container. The need for additional isolation relays may also be avoided by utilizing a virtual accumulator. See [EV2.12](#)

EV2.11.9 Team-Designed Accumulator Management Systems.

Teams may design and build their own Accumulator Management Systems. However, microprocessor-based accumulator management systems are subject to the following restrictions:

1. The processor must be dedicated to the AMS function only. However, it may communicate with other systems through shared peripherals or other physical links.
2. The AMS circuit board must include a watchdog timer. It is strongly recommended that teams include the ability to test the watchdog function in their designs.

EV2.11.10 The status of the AMS must be shown to the driver by a red indicator light in the cockpit that is easily visible even in bright sunlight. This indicator must show the latched fault rather than instantaneous status of the AMS. Other indication methods may be used upon application for a variance with the rules committee

EV2.12 Virtual Accumulators.

In many cases, difficulties can be encountered when multiple accumulator containers are monitored by a single AMS. Therefore, a vehicle may use a single AMS with more than one accumulator container by defining a ‘Virtual Accumulator’ as provided below. See: [Figure 32](#).

EV2.12.1 Each housing of the virtual accumulator container must be permanently installed in the vehicle. i.e. not designed to be removed for charging or exchange.

EV2.12.2 The conduit(s) connecting accumulator housings must be flexible metallic liquid-tight steel electrical conduit (NEC type LFMC) securely fastened at each end with fittings rated for metallic LFMC.

EV2.12.3 Connectors between the interconnecting conduit and the housings containing the accumulators, and along the length of the interconnecting conduit must be rated for LFMC conduit.

EV2.12.4 The conduit must be red, or painted red¹⁵.

EV2.12.5 Any unsupported length of the interconnect conduit may be no greater than 150 mm. i.e. it must be physically supported at least every 150 mm to ensure that it cannot droop or be snagged by something on the track. The interconnect conduit must be contained entirely within the chassis structure.

EV2.12.6 Separate conduits must be provided between housings for:

¹⁵ LFMC conduit is available with a red jacket - see for example: <http://www.afcweb.com/liquid-tuff-conduit/ul-liquidtight-flexible-steel-conduit-type-lfmc/>

1. Individual tractive System conductors. (Only one high-current TS conductor may pass through any one conduit.)
2. AMS wiring such as cell voltage sense wires that are at TS potential.

Note: GLV wiring may be run in its own conduit or outside of conduit.

- EV2.12.7 All rules relating to accumulator housings (including, but not limited to firewalls, location etc.) also apply to the interconnect conduit.
- EV2.12.8 If the interconnect conduit is the lowest point in the virtual housing it must have a 3-5 mm drain hole in its lowest point to allow accumulated fluids to drain.
- EV2.12.9 Segmentation requirements must be met considering the housings individually, and as an interconnected system. For example, AMS wires must be grouped according to segment and must maintain that grouping through to the AMS.
- EV2.12.10 Each individual housing must comply with TS fusing requirements. (See [EV2.6](#)) i.e., there must be at least one TS fuse in each container (See Figure 32).
- EV2.12.11 A High Voltage Disconnect (HVD) can be installed in the conductors connecting accumulator housings if mounted in a suitable enclosure. In this case, the accumulator boundary extends to include conduit and the HVD enclosure.

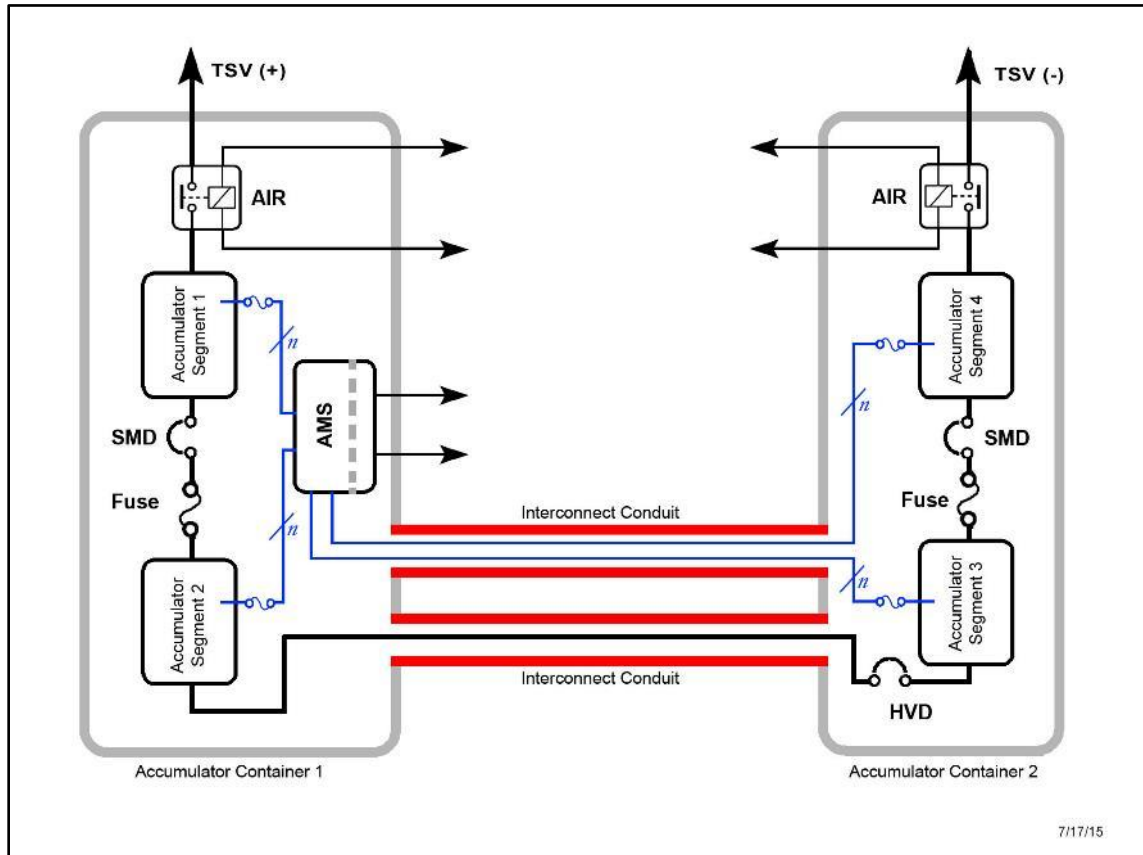


Figure 32 - Virtual Accumulator example

ARTICLE EV3 TRACTIVE SYSTEM WIRING AND CONSTRUCTION

EV3.1 Positioning of tractive system parts

EV3.1.1 Housings and/or covers must prevent inadvertent human contact with any part of the tractive system circuitry. This includes people working on or inside the vehicle. Covers must be secure and adequately rigid.

Body panels that must be removed to access other components, etc. are not a substitute for enclosing tractive system conductors.

EV3.1.2 Tractive system components and wiring must be physically protected from damage by rotating and/or moving parts and must be isolated from fuel system components.

EV3.1.3 **Finger Probe Test:** Inspectors must not be able to touch any tractive system electrical connection using a 10 cm long, 0.6 cm diameter non-conductive test probe.

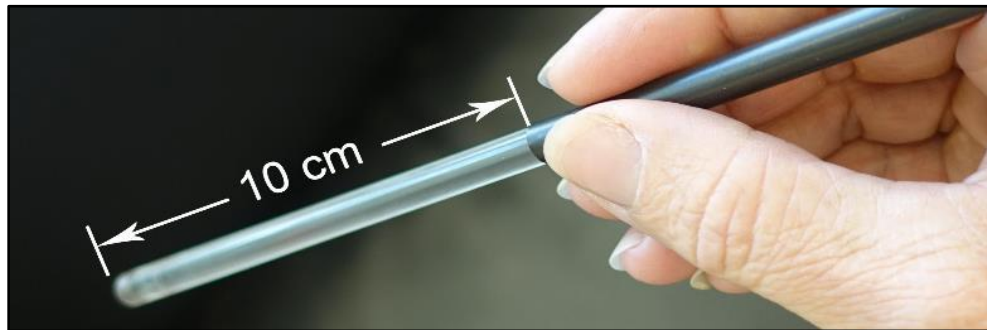


Figure 33 - Finger Probe

EV3.1.4 Housings constructed of electrically conductive material must have a minimum-resistance connection to GLV system ground. (See: [ARTICLE EV8](#)).

EV3.1.5 Every housing or enclosure containing parts of the tractive system must be labeled with the words “Danger”, “High Voltage” and a black lightning bolt on a yellow background. The label must be at least 4 x 6 cm. (See [Figure 34](#) below.)



Figure 34 - High Voltage Label

- EV3.1.6 All parts belonging to the tractive system including conduit, cables and wiring must be contained within the Surface Envelope of the vehicle (See [Figure 25](#)) such that they are protected against being damaged in case of a crash or roll-over situation or being caught (snagged) by road hazards.
- EV3.1.7 If tractive system parts are mounted in a position where damage could occur from a rear or side impact (below 350 mm from the ground), such as side mounted accumulators or rear mounted motors, they must be protected by a fully triangulated structure meeting the requirements of [T3.3](#), or approved equivalent per [T3.3.2](#) or [T3.7](#).
- EV3.1.8 Drive motors that are not located fully within the frame, must be protected by an interlock. This interlock must be configured such that the Shutdown Circuit, [EV7.1](#), will be opened if any part of the driven wheel assembly is dislocated from the frame. Drive motors located outside the frame must still comply with [EV7.1.3](#)
- EV3.1.9 No part of the tractive-system may project below the lower surface of the frame or the monocoque in either side or front view.
- EV3.1.10 Tractive systems and containers must be protected from moisture in the form of rain or puddles.
- EV3.1.11 Electric motors, accumulators or electronics that use fluid coolants must comply with [T8.1](#) and [T8.2](#).

EV3.2 TS wiring and conduit

- EV3.2.1 All tractive system wiring must be done to professional standards with adequate strain relief and protection from loosening due to vibration etc.
- EV3.2.2 Soldering in the high current path is prohibited. Exception: surface-mount fuses and similar components that are designed for soldering and the rated current.
1. Fuses must be mechanically supported by a PCB or equivalent, following manufacturer's instructions (e.g. recommended footprint). Free-hanging fuses connected by wires are not allowed.
 2. Team must submit a design document showing that PCB traces on these boards are properly designed for the current carried on all circuits. The battery design must still comply with [EV2.6](#), Accumulator Fusing.
- EV3.2.3 All wires, terminals and other conductors used in the tractive system must be sized appropriately for the continuous rating of the fuse which protects them. Wire size must comply with the table in Appendix E or alternately, with cable manufacturer's published current rating, if greater. Wires must be marked with wire gauge, temperature rating and insulation voltage rating¹⁶.
- EV3.2.4 The minimum acceptable temperature rating for TS wiring is 90°C. This requirement applies to wire and cable; it does not apply to conduit.
- EV3.2.5 All tractive system wiring that runs outside of electrical enclosures must be either:

¹⁶ Alternatively a manufacturer's part number printed on the wire will be sufficient if this number can be referenced to a manufacturer's data sheet.

1. Orange shielded, dual-insulated cable rated for automotive application, at least 5 mm overall cable diameter

OR

2. Enclosed in ORANGE non-conductive conduit (except for Virtual Accumulator systems which must use RED conduit. (See [EV2.12](#))

Note: UL Listed Conduit of other colors may be painted or wrapped with colored tape.

EV3.2.6 Conduit must be non-metallic and UL Listed¹⁷ as:

1. Conduit under UL1660, UL651 or UL651A

OR

2. Non-Metallic Protective Tubing (NMPT) under UL1696.

EV3.2.7 Conduit runs must be one piece. Conduit splices and/or transitions between conduit and shielded, dual-insulated cable may only occur within an enclosure and must comply with section [EV3.3](#).

EV3.2.8 Wiring to outboard wheel motors may be in conduit or may use shielded dual insulated cables. All other tractive system wiring that runs outside the vehicle frame (but within the roll envelope) must be within conduit.

EV3.2.9 When shielded dual-insulated cable is used, the shield must be grounded at both ends of the cable.

EV3.2.10 Wiring from motor controller to motor does not require fusing but must be sized for the controller maximum continuous output current, or per manufacturers instructions.

EV3.3 TS Cable Strain Relief

EV3.3.1 Cable or conduit exiting or entering a tractive system enclosure or the drive motor must use a liquid-tight fitting proving strain relief to the cable or conduit such that it will withstand a force of 200N without straining the cable¹⁸.

The fitting must be one of the following:

1. For conduit: A conduit fitting rated for use with the conduit used.
2. For shielded, dual insulated cable:
 - (i) A cable gland rated for use with the cable used

OR

- (ii) A connector rated for use with the cable used. The connector must provide termination of the shield to ground and latch in place securely enough to meet the strain-relief requirements listed above. Both portions of the connector must meet or exceed IEC standards IP53 (mated) and IP20 (unmated).

¹⁷ "UL Recognized" is not the same as "UL Listed".

¹⁸ This will be tested during the electrical tech inspection by pulling on the conduit using a spring scale.

EV3.3.2 Connections to drive motors that do not have provision for conduit connection are allowed to separate the strain-relief and insulation requirements. Specifically:

1. Cable strain relief must be capable of withstanding a 200N force, and must be within 15 cm of the motor terminals.

Note: Conventional cable strain relief fittings, mechanical clamps or saddles may be used, however the integrity of the cable insulation must be protected.

AND

2. the TS wiring must pass [EV3.1.3](#) (finger probe test)

For example, a motor with stud terminals for power input could use a 3D printed plastic cover on the terminals and suitable cable clamps within 15 cm to provide strain relief.

EV3.4 TS Electrical Connections

EV3.4.1 Tractive system connections must be designed so that they use intentional current paths through conductors such as copper or aluminum¹⁹. Steel is not permitted.

EV3.4.2 Tractive system connections must not include compressible material such as plastic or phenolic in the stack-up. (i.e. components identified as #1 in [Figure 35](#))

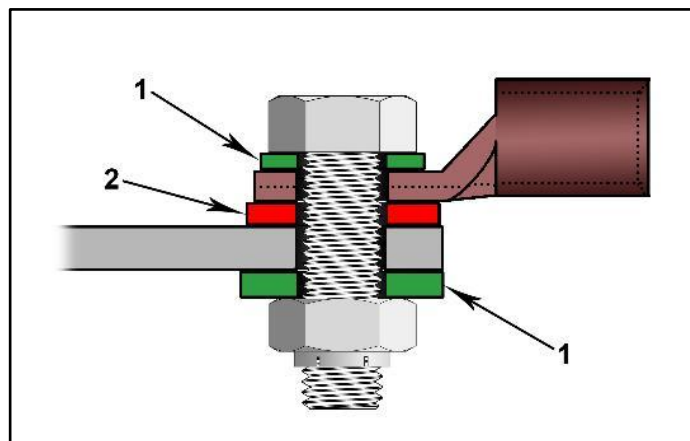


Figure 35 - Connection Stack-up

Note: The bolt shown in [Figure 35](#) can be steel since it is not the primary conductor. However the washer (#2), if used, must not be steel, since it is in the primary conduction path. If possible, no washer should be used in this location, so that the two primary conductors are directly clamped together.

EV3.4.3 Conductors and terminals must not be modified from their original size/shape and must be appropriate for the connection being made.

¹⁹ Aluminum conductors may be used, but require specific approval from the Formula Hybrid + Electric rules committee.

- EV3.4.4 Bolts with nylon inserts (Nylocks, etc.) and thread-locking compounds (Loctite, etc.) are not permitted.
- EV3.4.5 All bolted or threaded connections must be torqued properly. The use of a contrasting indicator paste²⁰ is strongly recommended to indicate completion of proper torqueing procedures.

Recommended TS Connection Fasteners
Fasteners specified and/or supplied by the component manufacturer.
Bolts with Belleville (conical) metal locking washers.
All-metal positive locking nuts. (See Figure 23)

Table 11 - Recommended TS Connection Fasteners

EV3.5 Motor Controllers

Note: Commercially available motor controllers containing boost converters that have internal voltages greater than 300 VDC may be used provided the unit is approved in writing by the rules committee.

- EV3.5.1 The tractive system motor(s) must be connected to the accumulator through a motor controller. Bypassing the control system and connecting the tractive system accumulator directly to the motor(s) is prohibited.
- EV3.5.2 The accelerator control must be a right-foot-operated foot pedal. Refer to [IC1.6](#) for specific requirements of the accelerator control
- EV3.5.3 The foot pedal must return to its original, rearward position when released. The foot pedal must have positive stops at both ends of its travel, preventing its sensors from being damaged or overstressed.
- EV3.5.4 All acceleration control signals (between the accelerator pedal and the motor controller) must have error checking.
- For analog acceleration control signals, this error checking must detect open circuit, short to ground and short to sensor power
- For digital acceleration control signals, this error checking must detect a loss of communication.
- EV3.5.5 TS circuitry, even at low voltage or current levels, is not allowed in the cockpit. All control signals that are referenced to TS and not GLV, such as non-isolated motor controller inputs, must be galvanically isolated and referenced to GLV ground.
- Note:** If these capabilities are built into the motor controller, then no additional error-checking circuitry is required.

²⁰ Such as DYKEM® Cross-Check Torque Seal®. See: <https://www.youtube.com/watch?v=W80C4XfyCQE>

- EV3.5.6 The accelerator signal limit shutoff may be tested during electrical tech inspection by replicating any of the fault conditions listed in [EV3.5.4](#)
- EV3.5.7 TS circuitry, even at low voltage or current levels, is not allowed in the cockpit. All control signals that are referenced to TS and not GLV, such as non-isolated motor controller inputs, must be galvanically isolated and referenced to GLV ground..²¹.
- EV3.5.8 Motor controller inputs that are galvanically isolated from the TSV may be run throughout the vehicle, but must be positively bonded to GLV ground.
- EV3.5.9 TS drive motors must spin freely when the TS system is in deactivated state, and when transitioned to a deactivated state.

EV3.6 Energy Meter

- EV3.6.1 Vehicles with accumulator energy capacities in excess of the limits in [Table 1](#) must be fitted with an Energy Meter provided by the organizer to compete in the endurance event.
- EV3.6.2 All power flowing between the accumulator and the Tractive System must pass through the Energy Meter.
- EV3.6.3 FH+E does not impose POWER limits and so the only function of the energy meter is to determine valid laps for the endurance event.

²¹ For commercial controllers that do not provide isolated throttle inputs, a remote throttle actuator can be fabricated with a (grounded) Bowden cable and non-conductive linkages.

ARTICLE EV4 GROUNDED LOW VOLTAGE SYSTEM

EV4.1 Grounded Low Voltage System (GLV)

EV4.1.1 The GLV system may not have a voltage greater than that listed in [Table 8](#).

EV4.1.2 All GLV batteries must be attached securely to the frame.

EV4.1.3 The hot (ungrounded) terminal of the battery must be insulated.

EV4.1.4 One terminal of the GLV battery or other GLV power source must be connected to the chassis by a stranded ground wire or flexible strap, with a minimum size of 12 AWG or equivalent cross-section. For vehicles without metal chassis members, the GLV system must be connected to conductive parts to meet grounding requirements.

EV4.1.5 The ground wire must run directly from the battery to the nearest frame ground and must be properly secured at both ends.

Note: Through-bolting a ring terminal to a gusset plate or dedicated tab welded to the frame is highly recommended.

EV4.1.6 Any wet-cell battery located in the driver compartment must be enclosed in a nonconductive marine-type container or equivalent and include a layer of 1.5 mm aluminum or equivalent between the container and driver.

EV4.1.7 GLV Battery Pack

1. Team constructed GLV batteries are allowed
2. The GLV battery must have appropriate over-current circuit protection
3. Cells wired in parallel must have branch-level overcurrent protection
4. GLV batteries that use lithium chemistries must have protection circuitry that prevents battery cells from over-discharging
 - (i) Team-implemented protective circuitry must monitor the voltages and temperatures of every cell group and disconnect the GLV battery if any voltage or temperature reading falls outside manufacturer's specifications.
 - (ii) Commercial OTS battery assemblies are acceptable if they have integrated protective circuitry for over-discharge, and cell voltage and temperature limits.
5. Note: GLV/TSV isolation rules still apply

EV4.1.8 Orange wire and/or conduit may not be used in GLV wiring. Exception: multi-conductor telecom-type cables containing orange color-coded wires may be used.

ARTICLE EV5 TRACTIVE SYSTEM VOLTAGE ISOLATION

Most Formula Hybrid + Electric vehicles contain voltages that could cause injury or death if they came in contact with a human body. In addition, all Formula Hybrid + Electric accumulator systems are capable of storing enough energy to cause injury, blindness or death if that energy is released unintentionally.

To minimize these risks, all tractive system components and wiring must at a minimum comply with the following rules.

EV5.1 Isolation Requirements

- EV5.1.1 All TS wiring and components must be galvanically (electrically) isolated from GLV by separation and/or insulation.
- EV5.1.2 All interaction between TS and GLV must be by means of galvanically isolated devices such as opto-couplers, transformers, digital isolators or isolated dc-dc converters.
- EV5.1.3 All connections from external devices such as laptops to a tractive system component must be galvanically isolated, and include a connection between the external device ground and the vehicle frame ground.
- EV5.1.4 All isolation devices must be rated for an isolation voltage of at least twice the maximum TS voltage.

EV5.2 General

- EV5.2.1 Tractive system and GLV conductors must not run through the same conduit.
- EV5.2.2 Tractive system and GLV wiring must not both be present in one connector. Connectors with an integral interlock such as the Amphenol SurLok Plus are exempted from this requirement.
- EV5.2.3 TS wiring must be separated from the driver's compartment by a firewall.
- EV5.2.4 TS wiring must not be present behind the instrument panel. TS wiring must not be present in the cockpit unless enclosed in conduit and separated from the driver by a firewall. TS potential must not be present on accelerator pedal or other controls in the cockpit.

EV5.3 Insulation, Spacing and Segregation

- EV5.3.1 Tractive system main current path wiring and any TS fused over 600W must be constructed using spacing, insulation, or both, in order to prevent short circuits between TS conductors. Minimum spacings are listed in [Table 12](#). Insulation used to meet this requirement must adhere to [EV5.4](#).
- EV5.3.2 Where GLV and TS circuits are present in the same enclosure, they must be segregated (in addition to any insulating covering on the wire) by:
 - 1. at least the distance specified in [Table 12](#)

OR

 - 2. a barrier material meeting the TS/GLV requirements of [EV5.4](#)
- EV5.3.3 All required spacings must be clearly defined. Components and cables must be securely restrained to prevent movement and maintain spacing.

Note: Grouping TS and GLV wiring into separate regions of an enclosure makes it easier to implement spacing or barriers to meet [EV5.3.2](#)

Maximum Vehicle TS Voltage	TS/TS ²²		TS/GLV
	Over Surface (Creepage)	Through Air	
V < 100 VDC	5.3 mm	2.1 mm	10.0 mm
100 VDC < V < 200 VDC	7.5 mm	4.3 mm	20.0 mm
V > 200 VDC	9.6 mm	6.4 mm	30.0 mm

Table 12 – Minimum Spacings

EV5.4 Insulation.

EV5.4.1 All electrical insulating material must be appropriate and adequately robust for the application in which it is used.

EV5.4.2 Insulating materials used for TS insulation or TS/GLV segregation must:

1. Be UL recognized (i.e., have an Underwriters Laboratories²³ or equivalent rating and certification).
2. Must be rated for the maximum expected operating temperatures at the location of use, or the temperatures listed in **Table 13** (whichever is greater).
3. Must meet the minimum thickness requirements listed in **Table 13**
4. The TS/GLV requirement also applies to TS to chassis ground insulation.

²² Outside of the accumulator container TS/TS spacing should comply with standard industry practice.

²³ Circuits fused under 600W should comply with standard industry practice for example:

UL1741 -

Section 26 - Spacings

Section 27 - Alternate Spacings – Clearances and Creepage Distances (which refers to UL840)

UL840 - Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment

	Minimum Temperature	Minimum Thickness
TS / GLV (see Note below)	150° C	0.25 mm
TS / TS TS/Chassis Ground	90° C	As required to be rated for the full TS voltage

Table 13 - Insulating Material - Minimum Temperatures and Thicknesses

Note: For TS/GLV isolation, insulating material must be used in addition to any insulating covering provided by the wire manufacturer.

- EV5.4.3 Insulating materials must extend far enough at the edges to meet spacing and creepage requirements between conductors.
- EV5.4.4 Thermoplastic materials such as vinyl insulation tape may not be used. Thermoset materials such as heat-shrink and self-fusing tapes (typically silicone) are acceptable.

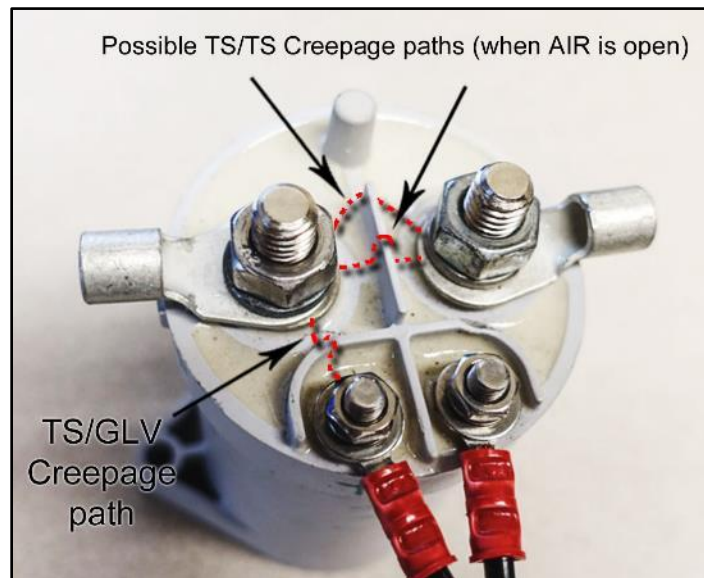


Figure 36 – Creepage Distance Example

EV5.5 Printed circuit board (PCB) isolation

Printed circuit boards designed and/or fabricated by teams must comply with the following:

- EV5.5.1 If tractive system circuits and GLV circuits are on the same circuit board they must be on separate, clearly defined areas of the board. Furthermore, the tractive system and **GLV** areas must be clearly marked on the PCB.
- EV5.5.2 Prototyping boards having plated holes and/or generic conductor patterns may not be used for applications where both GLV and TS circuits are present on the same board. Bare perforated board may be used if the spacing and marking requirements in [EV5.5.3](#) and [EV5.5.1](#) are met, and if the board is removable for inspection.

EV5.5.3 Required spacings between TS and GLV conductors are shown in **Table 14**. If a cut or hole in the PC board is used to allow the “through air” spacing, the cut must not be plated with metal, and the distance around the cut must satisfy the “over surface” spacing requirement. Spacings between TS and GLV conductors on an inner layer of PCBs may be reduced to the “through air” spacings. Prepreg layers used to reduce spacing must have manufacturer voltage ratings.²⁴



Note: Conformal coat must fully coat both TS and GLV conductors. Slots may be required to apply conformal coat to the undersides of pins on some components.

Maximum Vehicle TS Voltage	Spacing		Under Conformal Coating
	Over surface	Through air	
0-50 V	1.6 mm	1.6 mm	1 mm
50-150 V	6.4 mm	3.2 mm	2 mm
150-300 V	8.5 mm	6.4 mm	3mm
301-600V	12.7 mm	9.5 mm	4 mm

Table 14 – PCB TS/GLV Spacings

Note: Teams must be prepared to demonstrate spacings on team-built equipment. Information on this must be included in the ESF (**EV13.1**). If integrated circuits are used such as opto-couplers which are rated for the respective maximum tractive system voltage, but do not fulfill the required spacing, then they may still be used and the given spacing do not apply. This applies to the pin-to-pin through air and the pin-to-pin spacing over the package surface, but does not exempt the need to slot the board where pads would otherwise be too close to fulfill the creepage over PCB surface requirement.

1. Teams must supply high resolution (min. 300 dpi at 1:1) digital photographs of team-designed boards showing:
 - (i) All layers of unpopulated boards (inner layers or top/bottom layers that don't photograph well can be provided as copies of artwork files.)
 - (ii) Both top and bottom of fully populated and soldered boards.

EV5.5.4 If dimensional information is not obvious (i.e. 0.1 in x 0.1 in spacing) then a dimensional reference must be included in the photo. Spare boards should be made available for inspection. Teams should also be prepared to remove boards for direct inspection if asked to do so during the technical inspection.

²⁴ Example: <https://www.protoexpress.com/blog/high-voltage-circuit-boards/>

EV5.5.5 Printed circuit boards located inside the accumulator container and having tractive system connections on them must be fused to limited the power on the board to 600 watt or less, with the exception of pre-charge and discharge circuits.

ARTICLE EV6 FUSING - GENERAL

- EV6.1.1 All electrical systems (including tractive system, grounded low voltage system and charging system) must be appropriately fused.
- EV6.1.2 The continuous current rating of a fuse must not be greater than the continuous current rating of any electrical component that it protects. This includes wires, bus bars, battery cells or other conductors. See **Appendix E** for ampacity rating of copper wires. A manufacturer's current rating may be used for TS wiring if greater than the Appendix E value.

Note: Many fuses have a peak current capability significantly higher than their continuous current capability. Using such a fuse allows using a relatively small wire for a high peak current, because the rules only require the wire to be sized for the continuous current rating of the fuse.

- EV6.1.3 All fuses must be rated for the highest voltage in the systems they protect.
- EV6.1.4 Fuses used for DC must be rated for DC, and must carry a DC voltage rating equal to or greater than the maximum voltage of the system in which they are used²⁵. 
- EV6.1.5 All fuses must have an interrupt current rating which is higher than the theoretical short circuit current of the system that it protects.
- EV6.1.6 The fuse protecting a circuit **or** must be physically located at the end of the wiring closest to an uncontrolled energy source (e.g., a battery).

Note: For this rule, a battery is considered an energy source even for wiring intended for charging the battery, because current could flow in the opposite direction in a fault scenario.

- EV6.1.7 Circuits with branches using smaller wire than the main circuit require fuses or other current limiting means (such as circuit breakers or suitably-rated current-limiting resistors) located at the branching point, if the branch wire is too small to be protected by the main fuse for the circuit. 

Note: visit the <https://www.formula-hybrid.org/documents> and open the “Electrical Tech Documents” Folder.

²⁵ Note that where a voltage rating is not specified as AC or DC it is assumed to be an AC rating. The DC rating must be specifically called out in order for the fuse to be accepted as DC rated.

ARTICLE EV7 SHUTDOWN CIRCUIT AND SYSTEMS

EV7.1 Shutdown Circuit

The shutdown circuit is the primary safety system within a Formula Hybrid + Electric vehicle. It consists of a current loop that holds the Accumulator Isolation Relays (AIRs) closed. If the flow of current through this loop is interrupted, the AIRs will open, disconnecting the vehicle's high voltage systems from the source of that voltage within the accumulator container.

Shutdown may be initiated by several devices having different priorities as shown in the following table.

		Controlled Systems			
		Engine Starter (High Current)	GLV Supply to: Instrumentation Brake Light Data Acquisition Computers Telemetry Etc.	I.C. Engine Ignition Fuel pumps Starter solenoid Etc.	AIRs (TS Voltage)
Shutdown Sources	TSMS				OFF
	Cockpit BRB			OFF	
	Interlocks*			OFF	
	AMS			OFF	
	IMD			OFF	
	CAN Watchdog Failure			OFF	
	Brake Over-travel			OFF	
	Side-mounted BRBs Electric				OFF
	Side-mounted BRBs			OFF	
	GSVMS	OFF (Vehicle is COMPLETELY de-energized)			

*Optional interlocks as required. (such as wheel motors, etc.)

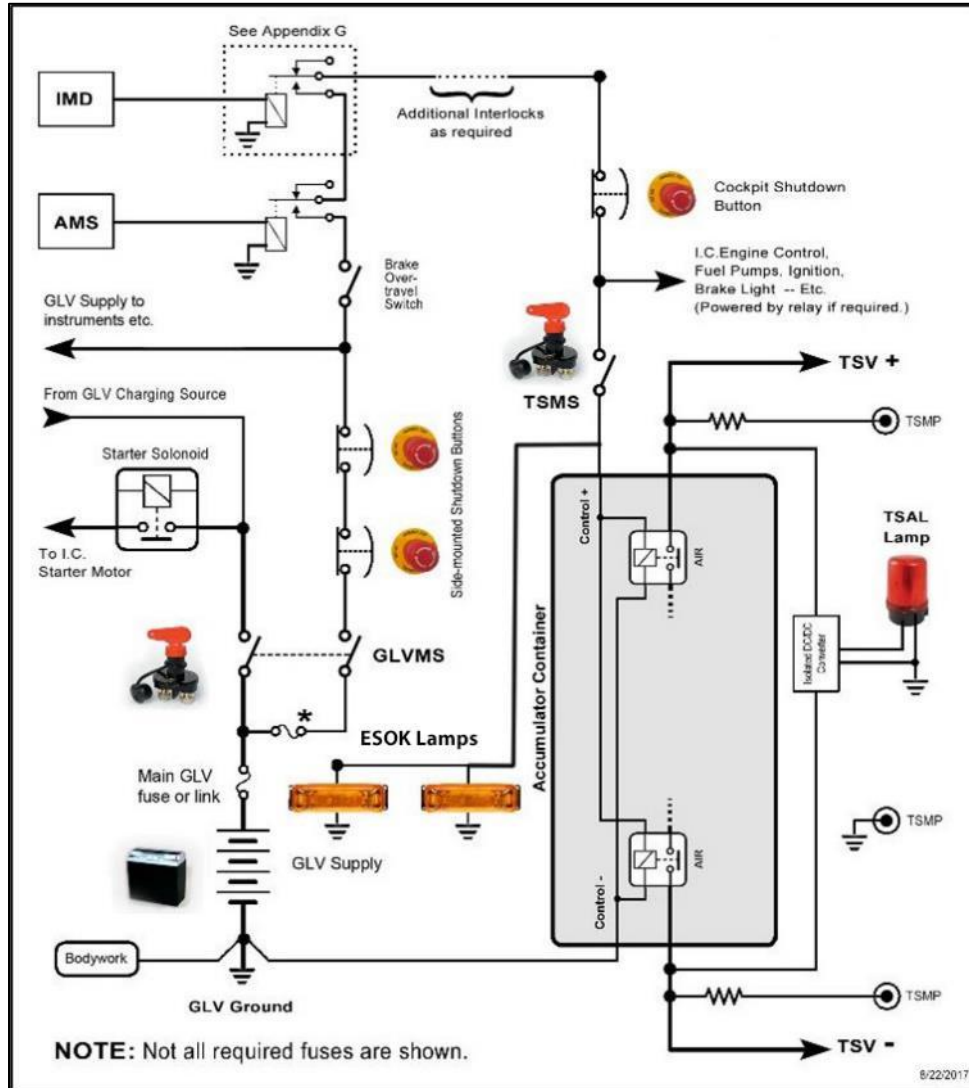
Figure 37 - Priority of shutdown sources

EV7.1.1 The shutdown circuit must consist of at least:

1. Grounded Low Voltage Master Switch (GLVMS) See: [EV7.3](#)
2. Tractive System Master Switch (TSMS) See: [EV7.4](#)
3. Two side mounted shutdown buttons (BRBs) See: [EV7.5](#)
4. Cockpit-mounted shutdown button. See: [EV7.6](#)
5. Brake over-travel switch. See: [T7.3](#)
6. A normally open (N.O.) relay controlled by the insulation monitoring device (IMD). See: [EV7.9](#) and [Figure 38](#).
7. A normally open (N.O.) relay controlled by the accumulator management system (AMS). See: [EV2.11](#) and [Figure 38](#).
8. All required interlocks.

9. Both IMD and AMS must use mechanical latching means as described in [Appendix G](#). Latch reset must be by manual push button. Logic-controlled reset is not allowed.
 10. If CAN communication are used for safety functions, a relay controlled by an independent CAN watchdog device that opens if relevant CAN messages are not received. This relay is not required to be latching.
- EV7.1.2 Any failure causing the GLV system to shut down must immediately deactivate the tractive system as well.
- EV7.1.3 The safety shutdown loop must be implemented as a series connection (logical AND) of all devices included in [EV7.1.1](#). Digital logic or microcontrollers may not be used for this function. Normally open auxiliary relays may be used to power high-current devices such as fans, pumps etc. The AIRs must be powered directly by the current flowing through the loop.
- EV7.1.4 All components in the shutdown circuit must be rated for the maximum continuous current in the circuit (i.e. AIR and relay current).
- Note:** A normally-open relay may be used to control AIR coils upon application to the rules committee.
- EV7.1.5 In the event of an AMS, IMD or Brake over-travel fault, it must not be possible for the driver to re-activate the tractive system from within the cockpit. This includes “cycling power” through the use of the cockpit shutdown button.
- Note:** Resetting or re-activating the tractive system by operating controls which cannot be reached by the driver²⁶ is considered to be working on the car.
- EV7.1.6 Electronic systems that contain internal energy storage must be prevented from feeding power back into the vehicle GLV circuits in the event of GLV shutdown.

²⁶ This would include the use of a wireless link remote from the vehicle.



*GLV fuse to comply with [EV6.1.7](#)

Figure 38 - Example Master Switch and Shutdown Circuit Configuration

EV7.2 Master Switches

EV7.2.1 Each vehicle must have two Master Switches:

1. Grounded Low Voltage Master Switch (GLVMS)
2. Traction System Master Switch (TSMS).

EV7.2.2 Both master switches must be located on the right side of the vehicle, in proximity to the Main Hoop, at the driver's shoulder height and be easily actuated from outside the car.

EV7.2.3 Both master switches must be of the rotary type, with a red, removable key, similar to the one shown in [Figure 39](#).

EV7.2.4 Both master switches must be direct acting. i.e. they may not operate through a relay.

EV7.2.5 Removable master switches are not allowed, e.g. mounted onto removable body work.

EV7.2.6 The function of each switch must be clearly marked with “GLV” and “TSV”.

EV7.2.7 The “ON” position of both switches must be parallel to the fore-aft axis of the vehicle

EV7.3 Grounded Low Voltage Master Switch (GLVMS)

EV7.3.1 The GLVMS is the highest priority shutdown and must disable power to all GLV electrical circuits. This includes the alternator, lights, fuel pump(s), I.C. engine ignition and electrical controls.

EV7.3.2 All GLV current must flow through the GLVMS.

EV7.3.3 Vehicles with GLV charging systems such as alternators or DC/DC converters must use a multi-pole switch to isolate the charging source from the GLV as illustrated in **Figure 38**²⁷.

EV7.3.4 The GLVMS must be identified with a label with a red lightning bolt in a blue triangle. (See **Figure 40**)

EV7.4 Tractive System Master Switch (TSMS)

EV7.4.1 The TSMS must open the Tractive System shutdown circuit.

EV7.4.2 The TSMS must be the last switch in the loop carrying the holding current to the AIRs. (See **Figure 38**)



Figure 39 - Typical Master Switch

²⁷ <https://www.pegasusautoracing.com/productdetails.asp?RecID=1464>



Figure 40 - International Kill Switch Symbol

EV7.5 Side-mounted Shutdown Buttons (BRB)

The side-mounted shutdown buttons (Big Red Buttons – or BRBs) are the first line of defense for a vehicle that is malfunctioning or in trouble. Corner and safety workers are instructed to push the BRB first when responding to an emergency.

- EV7.5.1 One button must be located on each side of the vehicle behind the driver's compartment at approximately the level of the driver's head. They must be installed facing outward and be easily visible from the sides of the car.
- EV7.5.2 The side-mounted BRBs must be red and a minimum of 38 mm. in diameter.
- EV7.5.3 The side-mounted shut-down buttons must be a push-pull or push-rotate emergency switch where pushing the button opens the shutdown circuit.
- EV7.5.4 The side-mounted shutdown buttons must shut down all electrical systems with the exception of the high-current connection to the engine starter motor.
- EV7.5.5 The shut-down buttons may not act through logic such as a micro-controller or relays.
- EV7.5.6 The shutdown buttons may not be easily removable, e.g. they may not be mounted onto removable body work.
- EV7.5.7 The Side-mounted BRBs must be identified with a label with a red lightning bolt in a blue triangle. (See [Figure 40](#))

EV7.6 Cockpit Shutdown Button (BRB)

- EV7.6.1 One shutdown button must be mounted in the cockpit and be easily accessible by the driver while fully belted in and with the steering wheel in any position.
- EV7.6.2 The cockpit shut-down button must be a push-pull or push-rotate emergency switch where pushing the button opens the shutdown circuit. The cockpit shutdown button must be red and at least 24 mm in diameter.
- EV7.6.3 Pushing the cockpit mounted button must open the AIRs and shut down the I.C. engine. (See: [Figure 37](#))

EV7.6.4 The cockpit shutdown button must be driver resettable. i.e. if the driver disables the system by pressing the cockpit-mounted shutdown button, the driver must then be able to restore system operation by pulling the button back out.

Note: There must still be one additional action by the driver after pulling the button back out to reactivate the motor controller per [EV7.7.2](#).

EV7.6.5 The cockpit shut-down buttons may not act through logic such as a micro-controller or relays.

EV7.7 Vehicle Start button

EV7.7.1 The GLV system must be powered up before it is possible to activate the tractive system shutdown loop.

EV7.7.2 After enabling the shutdown circuit, at least one action, such as pressing a “start” button must be performed by the driver before the vehicle is “ready to drive”. i.e. it will respond to any accelerator input.

EV7.7.3 The “start” action must be configured such that it cannot inadvertently be left in the “on” position after system shutdown.

EV7.8 Shutdown system sequencing

EV7.8.1 A recommended sequence of operation for the shutdown circuit and related systems is shown in the form of a state diagram in [Figure 41](#).

EV7.8.2 Teams must:

1. Demonstrate that their vehicle operates according to this state diagram,

OR

2. Obtain approval for an alternative state diagram by submitting an electrical rules query on or before the ESF submission deadline, and demonstrate that the vehicle operates according to the approved alternative state diagram.

IMPORTANT NOTE: If during technical inspection, it is found that the shutdown circuit operates differently from the standard or approved alternative state diagram, the car will be considered to have failed inspection.



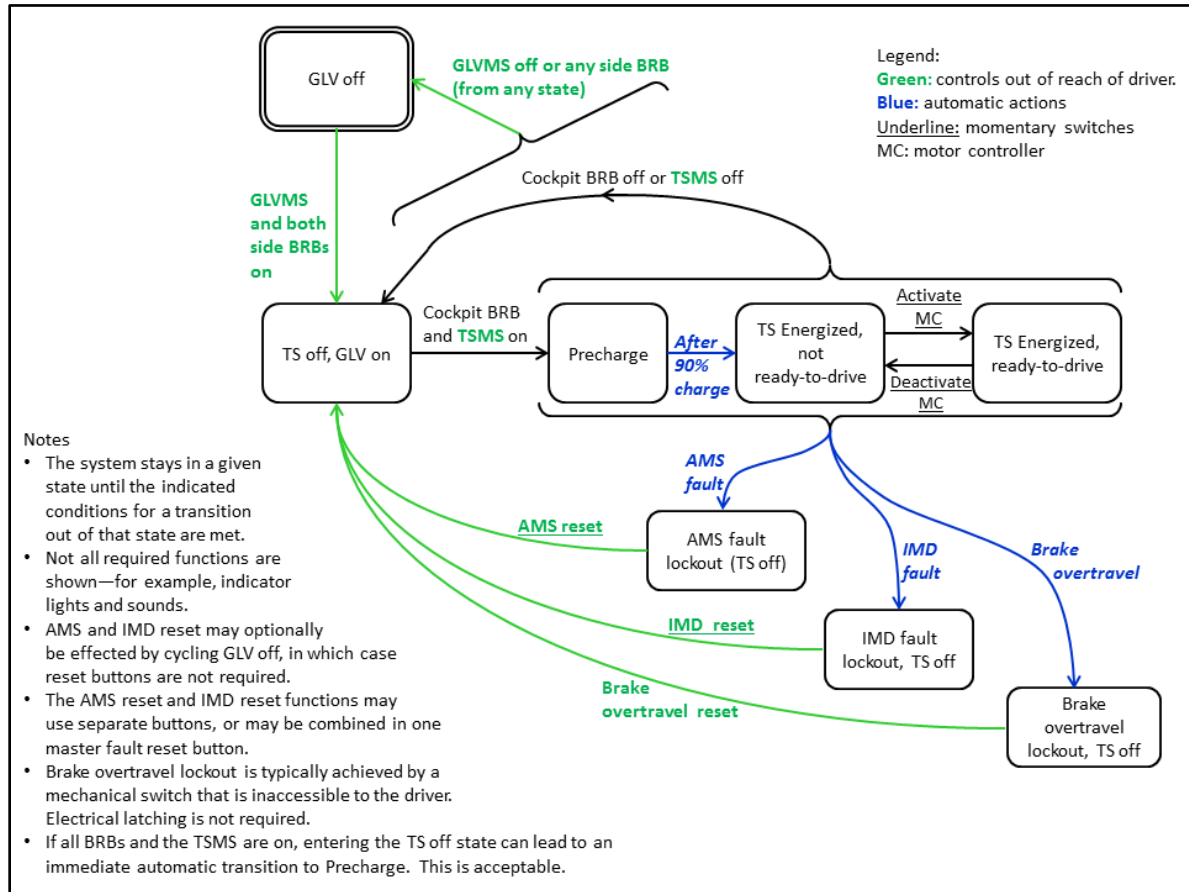


Figure 41 - Example Shutdown State Diagram

EV7.9 Insulation Monitoring Device (IMD)

EV7.9.1 Every car must have an insulation monitoring device installed in the tractive system.

EV7.9.2 The IMD must be designed for Electric Vehicle use and meet the following criteria: robustness to vibration, operating temperature range, availability of a direct output, a self-test function, and must not be powered by the system that is monitored. The IMD must be a stand-alone unit. IMD functionality integrated in an AMS is not acceptable (Bender A-ISOMETER® iso-F1 IR155-3203, IR155-3204, and Iso175C-32-SS are recommended examples).

EV7.9.3 The response value of the IMD needs to be set to no less than 500 ohm/volt, related to the maximum tractive system operation voltage.

EV7.9.4 In case of an insulation failure or an IMD failure, the IMD must shut down all the electrical systems, open the AIRs and shut down the I.C. drive system. (Some GLV systems such as accumulator cooling pumps and fans, may remain energized – See [Figure 37](#))

EV7.9.5 The tractive system must remain disabled until manually reset by a person other than the driver. It must not be possible for the driver to re-activate the tractive system from within the car in case of an IMD-related fault.

EV7.9.6 Latching circuitry added by teams to comply with [EV7.9.5](#) must be implemented using electro-mechanical relays. (See [Appendix G](#) – Example Relay Latch Circuits.)

EV7.9.7 The status of the IMD must be displayed to the driver by a red indicator light in the cockpit. (See [EV9.4](#)). The indicator must show the latched fault and not the instantaneous status of the IMD.

Note: The electrical inspectors will test the IMD by applying a test resistor between tractive system (positive or negative) and GLV system ground. This must deactivate the system. Disconnecting the test resistor may not re-activate the system. i.e. the tractive system must remain inactive until it is manually reset.

EV7.9.8 The IMD high voltage sense connections may be unfused if wiring is less than 30 cm in length, is less than or equal to #16 wire gauge, has an insulation voltage rating of at least 600V, and is “double insulated” (e.g. has additional sleeving on both conductors). If any of these conditions is not met, the IMD HV sense connections must be fused in accordance with [EV3.2.3](#) and [ARTICLE EV6](#).

ARTICLE EV8 GROUNDING

EV8.1 General

- EV8.1.1 Grounding is required for parts of the vehicle which are 100 mm or less from any Tractive System component.
- EV8.1.2 Grounded parts of the vehicle must have a resistance to GLV System Ground less than the values specified below.
1. Electrically conductive parts 300 mOhms (measured with a current of 1 A)
 - (i) Examples: parts made of steel, (anodized) aluminum, any other metal parts.
 2. Parts which may become electrically conductive 5 Ohm
 - (i) Example: carbon fiber parts Carbon fiber parts may need special measures such as using copper mesh or similar to keep the ground resistance below 5 Ohms.
- EV8.1.3 Electrical conductivity of any part may be tested by checking any point which is likely to be conductive. Where no convenient conductive point is available, an area of coating may be removed.
- EV8.1.4 Grounding conductors or straps used for compliance with this section must be a minimum of 16 AWG and be stranded.
- Note:** For GLV system grounding conductor size see [EV4.1.4](#).
- EV8.1.5 **Grounding Tests:** Electrical conductivity of a part may be tested by checking any point on the vehicle which is likely to be conductive, for example the driver's harness attachment bolts. Where no convenient conductive point is available then an area of coating may be removed.
- Tests on track shall be performed with an event provided four terminal ohm meter.
- See: https://en.wikipedia.org/wiki/Four-terminal_sensing
- EV8.1.6 Vehicles that use CANbus as a fundamental element of the safety shutdown system (e.g., AMS communication) will be required to document and demonstrate shutdown equivalent to BRB operation if CAN communication is interrupted. A solution to this requirement may be a separate device with relay output that monitors CAN traffic.

ARTICLE EV9 SYSTEM STATUS INDICATORS

EV9.1 Tractive System Status Indicator (TSSI)

EV9.1.1 The vehicle must have a Tractive System Status Indicator. Refer to the Rules FAQ on the FSAE Online website for additional information

EV9.1.2 The Tractive System Status Indicator must have two different lights:

1. One Green Light
2. One Red Light

EV9.1.3 Each of the TSSI must have:

1. Minimum luminous area of 130 mm²
2. Must be visible in direct sunlight
3. May have one or more of the same elements

EV9.1.4 Mounting location of the Tractive System Status Indicator must be:

1. Near the Main Hoop at the highest point of the vehicle
2. Above the Ready to Move Light
3. In the Rollover Protection Envelope F.1.13
4. No lower than 150 mm from the highest point of the Main Hoop
5. Not able to let the driver's helmet touch it
6. Easily visible from the front of the vehicle

EV9.1.5 The Tractive System Status Indicator must show when the GLV System is energized:

Condition		Green Light	Red Light
a.	No Faults	Always ON	OFF
b.	Fault in one or the two: BMS or IMD	OFF	Flash 2 Hz to 5 Hz, 50% duty cycle

EV9.2 Ready-to-Drive Sound

EV9.2.1 The car must make a characteristic sound, for a minimum of 1 second and a maximum of 3 seconds, when it is ready to drive. (See [EV7.7](#))

Note: The car is ready to drive as soon as the motor(s) will respond to the input of the torque encoder / accelerator pedal.

EV9.2.2 The emitting device must produce a tone between 1000 and 3500 Hz with a minimum loudness of 80dBA measured at any point in a 2m radius around the vehicle from the emitter²⁸.

EV9.2.3 The sound level will be measured with a commercial sound level meter from three points, one in the front, and one from each side of the vehicle.

EV9.2.4 The vehicle must not make any other sounds similar to the Ready-to-Drive sound.

²⁸ Some compliant devices can be found here: <https://www.mspindy.com/>

EV9.3 Electrical Systems OK Lamps (ESOK)

ESOK indicator lamps are required for hybrid and hybrid-in-progress vehicles. They are not required for electric-only vehicles.

The purpose of the ESOK indicators is to ensure that a hybrid vehicle is operating with the electric traction system engaged, and to prevent IC-only operation in dynamic events.

This requirement is in addition to TSSI indicators which are required for all vehicle classes.

- EV9.3.1 Hybrid vehicles must have two ESOK lamps. One mounted on each side of the roll bar in the vicinity of the side-mounted shutdown buttons (EV7.5) that can easily be seen from the sides of the vehicle. The indicators must be Amber, complying with DOT FMVSS 108 for trailer clearance lamps²⁹. See Figure 42. They must be clearly labeled “ESOK”.
- EV9.3.2 The ESOK indicators must be powered from the circuit feeding the AIR (contactor) coils. If there is an electrical system fault, or a “BRB” is pressed, or the TS master switch is opened, the ESOK indicators must extinguish. See Figure 38 for an example of ESOK lamp wiring.
See Figure 38 for an example of ESOK lamp wiring.



Figure 42 – Typical ESOK Lamp

EV9.4 Insulation Monitoring Device (IMD)

- EV9.4.1 The status of the IMD must be shown to the driver by a red indicator light in the cockpit that is easily visible even in bright sunlight. This indicator must light up if the IMD detects an insulation failure or if the IMD detects a failure in its own operation e.g. when it loses reference ground.
- EV9.4.2 The IMD indicator light must be clearly marked with the lettering “IMD” or “GFD” (Ground Fault Detector).

EV9.5 Accumulator Voltage Indicator

- EV9.5.1 Any removable accumulator container must have a prominent indicator, such as an LED, that is visible through a closed container that will illuminate whenever a voltage greater than 60 VDC is present at the vehicle side of the AIRs.
- EV9.5.2 The accumulator voltage indicator must be directly controlled by voltage present at the container connectors using analog electronics. No software control is permitted.

²⁹ https://www.ecfr.gov/cgi-bin/text-idx?node=se49.6.571_1108

EV9.6 Accumulator Monitoring System (AMS)

EV9.6.1 The status of the AMS must be shown to the driver by a red indicator light in the cockpit that is easily visible even in bright sunlight. This indicator must light up if the AMS detects an accumulator failure or if the AMS detects a failure in its own operation.

EV9.6.2 The AMS indicator light must be clearly marked with the lettering “AMS”.

ARTICLE EV10 ELECTRICAL SYSTEM TESTS

Note: During electrical tech inspection, these tests will normally be done in the following order: IMD test, insulation measurement, AMS function then Rain test.

EV10.1 Insulation Monitoring Device (IMD) Test

EV10.1.1 The insulation monitoring device will be tested during Electrical Tech Inspection. This is done by connecting a resistor between the TSMP (See [Figure 43](#)) and several electrically conductive vehicle parts while the tractive system is active, as shown in the example below.

EV10.1.2 The test is passed if the IMD shuts down the tractive system within 30 seconds at a fault resistance of 250 ohm/volt (50% below the response value).

Note: Allowance for the 2 x 10K TSMP resistors can be included in this test.

EV10.1.3 The IMD test may be repeated at any time during the event.

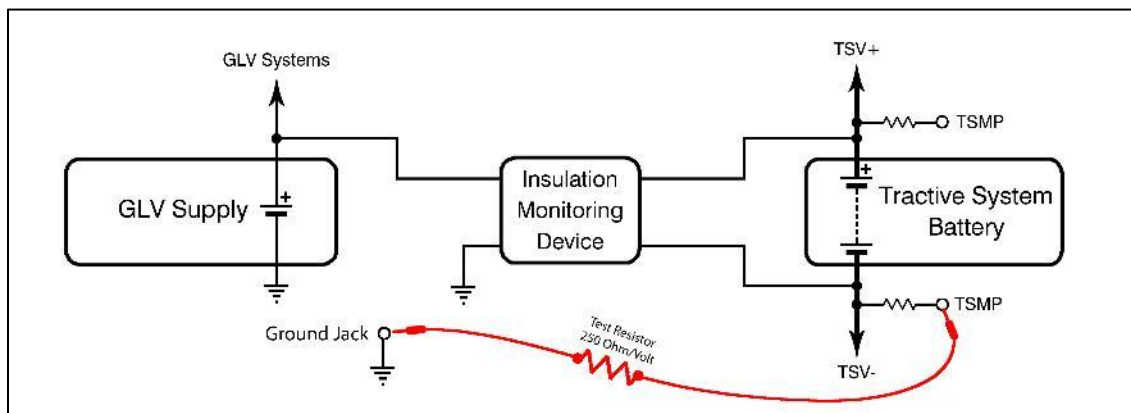


Figure 43 – Insulation Monitoring Device Test

EV10.2 Insulation Measurement Test

EV10.2.1 The technical inspectors may choose to measure the insulation resistance between the tractive system and control system ground. The available measurement voltages are 250 V, 500 V and 750 V DC.

All cars will be measured with the next available voltage level. For example, a 175 V system will be measured with 250 V; a 300 V system will be measured with 500 V, and a 600 V system will be measured with 750 V etc.

EV10.2.2 To pass the insulation measurement test, the measured insulation resistance must be at least 500 ohm/volt related to the maximum nominal tractive system operation voltage.

EV10.3 Tractive System Measuring Points (TSMP)

The TSMPs are used for the IMD and insulation tests, and also for the 5 second discharge rule, [EV2.8.3](#).

They may also be used to ensure the isolation of the tractive system of the vehicle during possible rescue operations after an accident or when work on the vehicle is to be done.

EV10.3.1 Two tractive system voltage measuring points must be installed in an easily accessible³⁰ well marked location. Access must not require the removal of body panels. The TSMP jacks must be marked "Gnd", "TS+" and "TS-".

EV10.3.2 The TSMPs must be protected by a non-conductive housing that can be opened without tools.

EV10.3.3 Shrouded 4 mm banana jacks that accept shrouded (sheathed) banana plugs with non-retractable shrouds must be used for the TSMPs and the system ground measuring point (EV10.3.6).

See Figure 44 for examples of the correct jacks and of jacks that are not permitted because they do not accept the required plugs (also shown).

EV10.3.4 The TSMPs must be connected, via current limiting resistors, to the positive and negative motor controller/inverter supply lines. (See Figure 38) The TSMP must measure motor controller input voltage even if segmenting or TSMS switches are opened.

EV10.3.5 The wiring to each TSMP must be protected with a current limiting resistor sized per Table 15. (Fuses or other forms of overcurrent protection are not permitted). The resistor must be located as close to the voltage source as practical and must have a power rating of:

Maximum TS Voltage	Resistor Value
<= 200 V**	5 kΩ
201 – 400 V	10 kΩ
401 – 600 V	15 kΩ

Table 15-TSMP Resistor Values³¹

The resistor must be located as close to the voltage source as practical and must have a power rating of:

$$2 \left(\frac{TSV_{max}^2}{R} \right)$$

but not less than 5 W.

EV10.3.6 A GLV system ground measuring point must be installed next to the TSMPs. This measuring point must be connected to the GLV system ground, must be a 4 mm shrouded banana jack and marked "GND".

³⁰ It should be possible to insert a connector with one hand while standing next to the car.

³¹ Teams with vehicles with Maximum TS Voltage of 200 V or less may use existing 10 kΩ resistor values upon application to the rules committee



Figure 44

EV10.4 Accumulator Monitoring System (AMS) Test

EV10.4.1 The AMS will be tested in one of two ways:

1. By varying AMS software setpoints so that actual cell voltage is above or below the trip limit.
2. The AMS may be tested by varying simulated cell voltage to the AMS test connector, following open accumulator safety procedures, to make the actual cell voltage above or below the trip limit.

EV10.4.2 The requirement for an AMS test port for commercial accumulator assemblies may be waived, or alternate tests may be substituted, upon application to the rules committee.

EV10.5 Rain test

EV10.5.1 Vehicles that pass the rain test will receive a “Rain Certified” sticker and may be operated in damp or wet conditions. See: [ARTICLE D3](#)

If a vehicle does not pass the rain test, or if the team chooses to forego the rain test, then the vehicle is not rain certified and will not be allowed to operate in damp or wet conditions.

EV10.5.2 During the rain test:

1. The tractive system must be active.
2. It is not allowed to have anyone seated in the car.
3. The vehicle must be on stands such that none of the driven wheels touch the ground.

EV10.5.3 Water will be sprayed at the car from any possible direction for 120 seconds. The water spray will be rain-like. There will be no high-pressure water jet directed at the car.

EV10.5.4 The test is passed if the IMD does not trip while water is sprayed at the car and for 120 seconds after the water spray has stopped. Therefore, the total time of the rain test is 240 seconds, 120 seconds with water-spray and 120 seconds without.

EV10.5.5 Teams must ensure that water cannot accumulate anywhere in the chassis.

ARTICLE EV11 POUCH TYPE LITHIUM-ION CELLS

Important Note: Designing an accumulator system utilizing pouch cells is a substantial engineering undertaking, which may be avoided by using prismatic or cylindrical cells.



EV11.1 Accumulator systems using pouch cells must address: stack arrangement, cell expansion and compression requirements, mechanical constraint, and tab connections.

EV11.2 Teams wishing to use pouch cell accumulator systems must provide details of the design, for review, in their ESF1 and ESF2 submissions.

ARTICLE EV12 HIGH VOLTAGE PROCEDURES & TOOLS

The following rules relate to procedures that must be followed during the Formula Hybrid + Electric competition. It is strongly recommended that these or similar rules be instituted at the team's home institutions.

It is also important that all team members view the Formula Hybrid + Electric electrical safety lecture, which can be found here:

<https://youtu.be/98JBFIbMIQM?si=ovq1hiBMZjnDoBaN>

EV12.1 Working on Tractive Systems or accumulators

- EV12.1.1 Teams must establish a formal lockout/tagout procedure that is documented in the ESF, and that all team members know and follow.
- EV12.1.2 Whenever the accumulator containers are opened the accumulator segments must be separated by using the maintenance plugs. (See [EV2.7](#)).
- EV12.1.3 If the organizers have provided a “Designated Charging Area”, then opening of or working on accumulator containers is only allowed in that charging area (see [EV12.2](#)) and during Electrical Tech Inspection.
- EV12.1.4 Whenever the accumulator is being worked on, only appropriate insulated tools may be used.
- EV12.1.5 Whenever the accumulator is open or being worked on, a “Danger High Voltage” sign (or other warning device provided by the organizers) must be displayed.

Note: Be sure to remove the warning sign or indicator once the hazards are no longer present.

EV12.2 Charging

- EV12.2.1 If the organizers have provided a “Designated Charging Area”, then charging tractive system accumulators is only allowed inside this area.
- EV12.2.2 The chassis or frame of the vehicle must be securely connected to earth ground using a (minimum) 16 AWG green wire during charging.

Note: Earth ground can be the ground terminal of an ac outlet or a positive connection to metallic electrical conduit permanently installed at the competition site.
- EV12.2.3 If the organizers have provided “High Voltage” signs and/or beacons, these must be displayed prominently while charging.
- EV12.2.4 The accumulators may be charged inside the vehicle or outside if fitted with a removable accumulator container.
- EV12.2.5 During charging, the accumulator containers or the car itself (depending on whether the accumulators are charged externally or internally) must have a prominent sign with the following data:
 - 1. Team name
 - 2. RSO Name with cell phone number(s).
- EV12.2.6 Only chargers reviewed and sealed in the garages by tech inspectors are allowed.
- EV12.2.7 All connections of the charger(s) must be isolated and covered, with intact strain relief and no fraying of wires.

- EV12.2.8 No work is allowed on any of the car's systems during charging if the accumulators are charging inside of or connected to the car.
- EV12.2.9 No grinding, drilling, or any other activity that could produce either sparks or conductive debris is allowed in the charging area.
- EV12.2.10 At least one team member who has knowledge of the charging process must stay with the accumulator(s) / car during charging.
- EV12.2.11 Moving accumulator cells and/or stack(s) around at the event site is only allowed inside a completely closed accumulator container.
- EV12.2.12 High Voltage wiring in an offboard charger does not require conduit; however, it must be a UL listed flexible cable that complies with NEC Article 400; jacketed.
- EV12.2.13 The vehicle charging connection must be appropriately fused for the rating of its connector and cabling in accordance with [EV6.1.1](#).
- EV12.2.14 The vehicle charging port must only be energized when the tractive system is energized and the TSSI is flashing, i.e., there must be no voltage present on the charging port when the tractive system is de-energized.
- EV12.2.15 If charging on the vehicle, AMS and IMD systems must be active during charging. The external charging system must shut down if there is an AMS or IMD fault, or if one of the shutdown buttons (See [EV7.5](#)) is pressed. If charging off the vehicle, equivalent AMS and IMD protection must be provided. If charging off the vehicle, the charging cart and any exposed metal must be connected to ac-power ground. 

EV12.3 Accumulator Container Hand Cart

- EV12.3.1 In case removable accumulator containers are used in order to accommodate charging, a hand cart to transport the accumulators must be presented at Electrical Tech Inspection.
- EV12.3.2 The hand cart must have a brake such that it can only be released using a dead man's switch, i.e. the brake is always on except when someone releases it by pushing a handle for example.
- EV12.3.3 The brake must be capable to stop the fully loaded accumulator container hand cart.
- EV12.3.4 The hand cart must be able to carry the load of the accumulator container(s).
- EV12.3.5 The hand cart(s) must be used whenever the accumulator container(s) are transported on the event site.

EV12.4 Required Equipment

Each team must have the following equipment accessible at all times during the event. The equipment must be in good condition, and must be presented during technical inspection. (See also [Appendix F](#))

1. **Multimeter** rated for CAT III use with UL approval. (Must accept shrouded banana leads.)
2. **Multimeter leads** rated for CAT III use with shrouded banana leads at one end and probes at the other end. The probes must have finger guards and no more than 3 mm of exposed metal. (Heat shrink tubing may be used to cover additional exposed metal on probes.)
3. **Multimeter leads** rated for CAT III use with shrouded banana plugs at both ends.

4. **Insulated tools.** (i.e. screwdrivers, wrenches etc. compatible with all fasteners used inside the accumulator housing.)
5. **Face shield** which meets ANSI Z87.1-2003
6. **HV insulating gloves** (tested within the last 14 Months) plus protective outer gloves.
7. **HV insulating blanket(s)** of sufficient size and quantity to cover the vehicle's accumulator(s).
8. **Safety glasses** with side shields (ANSI Z87.1-2003 compliant) for all team members.

Note: All electrical safety items must be rated for at least the maximum tractive system voltage.

ARTICLE EV13 REQUIRED ELECTRICAL DOCUMENTATION

EV13.1 Electrical System Form – Part 1

Part 1 of the ESF requests preliminary design information. This is so that the technical reviewers can identify areas of concern early and provide feedback to the teams.

EV13.2 Electrical System Form – Part 2

Note: Many of the fields in Part 2 ask for the same information that was entered in Part 1. This information must be reentered in Part 2. However, it is not expected that the fields will contain identical information, since many aspects of a design will change as a project evolves.

EV13.2.1 The final ESF must illustrate the interconnection of all electric components including the voltage level, the topology, the wiring in the car and the construction and build of the accumulator(s).

EV13.2.2 Teams must present data pages with rated specifications for all tractive system parts used and show that none of these ratings are exceeded (including wiring components). This includes stress caused by the environment e.g. high temperatures, vibration, etc.

EV13.2.3 A template containing the required structure for the ESF will be made available online.

EV13.2.4 The ESF must be submitted as a Microsoft Word format file.

ARTICLE EV14 - ACRONYMS

AC	Alternating Current
AIR	Accumulator Isolation Relay
AMS	Accumulator Management System
BRB	Big Red Buttons (Emergency shutdown switches)
ESF	Electrical System Form
ESOK	Electrical Systems OK Lamp
GLV	Grounded Low Voltage
GLVMS	Grounded Low Voltage Master Switch
HVD	High Voltage Disconnect
IMD	Insulation Monitoring Device
PCB	Printed Circuit Board
SMD	Segment Maintenance Disconnect
TS	Tractive System
TSMP	Tractive System Measuring Point
TSMS	Tractive System Master Switch
TSV	Tractive System Voltage
TSSI	Tractive System Status Indicator

PART S - STATIC EVENTS

Presentation	150 points
Design	200 points
Total	350 points

Table 15 – Static Event Maximum Scores

ARTICLE S1 TECHNICAL INSPECTION

S1.1 Objective

- S1.1.1 The objective of technical inspection is to determine if the vehicle meets the FH+E rules requirements and restrictions and if, considered as a whole, it satisfies the intent of the Rules.
- S1.1.2 For purposes of interpretation and inspection the violation of the intent of a rule is considered a violation of the rule itself.
- S1.1.3 Technical inspection is a non-scored activity.

S1.2 Inspection & Testing Requirement

- S1.2.1 Each vehicle must pass all parts of technical inspection and testing, and bear the inspection stickers, before it is permitted to participate in any dynamic event or to run on the practice track. The exact procedures and instruments employed for inspection and testing are entirely at the discretion of the Chief Technical Inspector.
- S1.2.2 Technical inspection will examine all items included on the Inspection Form found on the Formula Hybrid + Electric website, all the items on the Required Equipment list ([Appendix F](#)) plus any other items the inspectors may wish to examine to ensure conformance with the Rules.
- S1.2.3 All items on the Inspection Form must be clearly visible to the technical inspectors.
- S1.2.4 Visible access can be provided by removing body panels or by providing removable access panels.
- S1.2.5 Once a vehicle has passed inspection, except as specifically allowed under [T1.2](#) Modification and Repairs, it must remain in the “As-approved” condition throughout the competition and must not be modified.
- S1.2.6 Decisions of the inspectors and the Chief Scrutineer concerning vehicle compliance are final and are not permitted to be appealed.
- S1.2.7 Technical inspection is conducted only to determine if the vehicle complies with the requirements and restrictions of the Formula Hybrid + Electric rules.
- S1.2.8 Technical approval is valid only for the duration of the specific Formula Hybrid + Electric competition during which the inspection is conducted.

S1.3 Inspection Condition

Vehicles must be presented for technical inspection in finished condition, i.e. fully assembled, complete and ready-to-run. Technical inspectors will not inspect any vehicle presented for inspection in an unfinished state.



This requirement will be waived if the vehicle is registered as an HIP ([A2.3](#)) or SEO ([A2.4.1](#)).

Note: Cars may be presented for technical inspection even if final tuning and set-up has not been finished.

S1.4 Inspection Process

Vehicle inspection will consist of five separate parts as follows:

(a) Part 1: Preliminary Electrical Inspection

Vehicles must pass a preliminary electrical safety inspection before they will be permitted to proceed to Mechanical Scrutineering. A sticker will be affixed to the vehicle upon passing the Preliminary Electrical Inspection.

(b) Part 2: Scrutineering - Mechanical

Each vehicle will be inspected to determine if it complies with the mechanical and structural requirements of the rules. This inspection will include examination of the driver's equipment ([ARTICLE T5](#)) a test of the emergency shutdown response time (Rule [T4.9](#)) and a test of the driver egress time (Rule [T4.8](#)).

The vehicle will be weighed, and the weight placed on a sticker affixed to the vehicle for reference during the Design event.

(c) Part 3: Scrutineering – Electrical

Each vehicle will be inspected for compliance with the electrical portions of the rules.

Note: This is an extensive and detailed inspection. Teams that arrive well-prepared can reduce the time spent in electrical inspection considerably.

The electrical inspection will include all the tests listed in [EV9.5.1](#).

Note: In addition to the electrical rules contained in this document, the electrical inspectors will use [SAE Standard J1673 “High Voltage Automotive Wiring Assembly Design”](#) as the definitive reference for sound wiring practices.

Note: Parts 1, 2 and 3 must be passed before a vehicle may apply for Part 4 or Part 5 inspection.

(d) Part 4: Tilt Table Tests

Each vehicle will be tested to insure it satisfies both the 45 degree (45°) fuel and fluid tilt requirement (Rule [T8.5.1](#)) and the 60 degree (60°) stability requirement (Rule [T6.7](#)).

(e) Part 5: Noise, Master Switch, and Brake Tests.

Noise will be tested by the specified method (Rule [IC3.2](#)). If the vehicle passes the noise test then its master switches ([EV7.2](#)) will be tested.

Once the vehicle has passed the noise and master switch tests, its brakes will be tested by the specified method (see Rule [T7.2](#)).

S1.5 Correction and Re-inspection

S1.5.1 If any part of a vehicle does not comply with the Rules, or is otherwise deemed to be a concern, then the team must correct the problem and have the car re-inspected.

S1.5.2 The judges and inspectors have the right to re-inspect any vehicle at any time during the competition and require correction of non-compliance.

S1.6 Inspection Stickers

Inspection stickers issued following the completion of any part of technical inspection must be placed on the upper nose of the vehicle as specified in [T13.4](#) “Technical Inspection Sticker Space”. Inspection stickers are issued contingent on the vehicle remaining in the required condition throughout the competition. Inspection stickers may be removed from vehicles that are not in compliance with the rules or which are required to be re-inspected.

ARTICLE S2 PROJECT MANAGEMENT

S2.1 Project Management Objective

The objective of the Formula Hybrid + Electric Project Management component is to evaluate the team's ability to structure and execute a project management plan that helps the team define and meet its goals.

A well written project management plan will not only aid each team in producing a functional, rules compliant race car on-time, it will make it much easier to create the Change Management Report and Final Presentation components of the Project Management Event.

Several resources are available to guide work in these areas. They can be found in [Appendix C](#).

No team should begin developing its project management plan without FIRST:

- (a) Reviewing "[Application of the Project Management Method to Formula Hybrid + Electric](#)", by Dr. Edward March, former Formula Hybrid + Electric Chief Project Management Judge.
- (b) Viewing in its entirety Dr. March's 12-part video series "[Formula Hybrid + Electric Project Management](#)".
- (c) Reading "Formula Hybrid + Electric Project Management Event Scoring Criteria", found in [Appendix C](#) of the Formula Hybrid + Electric Rules.
- (d) Watching an example of a "[perfect score](#)" [presentation](#), from a previous competition.



S2.2 Project Management Segments

The Formula Hybrid + Electric Project Management component consists of three parts: (1) submission of a written project plan (2) a written change management report, and, (3) an oral final presentation assessing the overall project management experience to be delivered before a review board at the competition.

Segment	Points
Project Plan Report	55
Change Management Report	40
Presentation	55
Total	150

Table 16 - Project Management Scoring

S2.3 Submission 1 -- Project Plan Report

S2.3.1 Each Formula Hybrid + Electric team is required to submit a formal Project Plan Report that reflects its goals and objectives for the upcoming competition, the management structure, the tasks that will be required to accomplish these objectives, and the time schedule over which these tasks will be performed. The topics covered in the Project Plan Report should include:

- (a) **Scope:** What will be accomplished, "SMART" goals and objectives (see [Appendix C](#) for more information), major deliverables and milestones.

- (b) **Operations:** Organization of the project team, Work Breakdown Structure, project timeline, and a budget and funding strategy.
- (c) **Risk Management:** tasks expected to be particularly difficult for the team, but whose completion is essential for achieving team goals and having a functional car ready before shipment to the track; contingency plans to mitigate these risks if problems arise during project execution.
- (d) **Expected Results:** Team goals and objectives quantified into “measure of success”. These attributes are useful for assigning task priorities and allocating resources during project execution. They are also used to determine the extent to which the goals have been achieved.
- (e) **Change Management Process:** The system designed by the team for administering project change and maintaining communication across all team members.

S2.3.2 This Project Plan must consist of at least one (1) page and may not exceed three (3) pages of text. Appendices with supporting information may be attached to the back of the Project Plan.

Note 1: Title pages, appendices and table-of-contents do not count as “pages”.

Note 2: Submittal of the Project Plan is due very soon after registration closes. See the Formula Hybrid + Electric rules and deadlines page at: <https://www.formula-hybrid.org/deadlines> for the exact due date.

Note 3: After the scoring of the Project Plans has been completed and at the discretion of the Project Management judges a one half hour online review session may be made available to each team.

S2.4 Submission 2 – Change Management Report

S2.4.1 Following completion and acceptance of the formal project plan by student team members, the project team begins the execution phase. During this phase, members of the student team and other stakeholders must be updated on progress being made and on issues identified that put the project schedule at risk. The ability of the team to change direction or “pivot” in the face of risks is often a key factor in a team’s successful completion of the project. Changes to the plan are adopted and formally communicated through a change management report. Each team must submit one change management report. See the Formula Hybrid + Electric rules and deadlines page at: <https://www.formula-hybrid.org/deadlines> for the exact due date.

S2.4.2 These reports are not lengthy but are intended to clearly and concisely communicate the documented changes to the project scope and plan to the team that have been approved using the Change Management Process. See Appendix C for details on scoring criteria for this report. The topics covered in the progress report should include:

- A) **Change Management Process:** Detail the Change Management processes and platforms used by the team. At a minimum, the report is expected to cover:
 - a. what triggers the process
 - b. how does information flow through the process
 - c. how are final decisions transmitted to all team members
 - d. team-created process flow diagram

- B) **Implementation of the Change Management Process:** Teams are expected to review in detail how the Change Management Process has been used to modify at least one of their project's main components (Scope or Goals, Operational Plan, Risk Management, and Expected Results). The report is expected to cover an example from start to finish in the process and should also detail the final impact to the project.
- C) **Effectiveness of and Improvements to the Change Management Process:** At the end of the project lifecycle, the team is expected to perform an assessment on the effectiveness of their Change Management Process. Through this assessment, the team should identify at least two areas of opportunity to improve the process and include the plans/details to implement those changes.

S2.4.3 The Change Management Report must not exceed two (2) pages. Appendices may be included with the Report

Note 1: Appendix content does not count as "pages".

Note 2: See the Formula Hybrid + Electric rules and deadlines page at: <https://www.formula-hybrid.org/deadlines> for the due date of the Change Management Report.

S2.5 Submission Formats

S2.5.1 The Project Plan and Change Management Report must be submitted electronically in separate Microsoft Word files (*.doc or .docx file). These documents must each be a single file (text, drawings, and optional content).

S2.5.2 These Report files must be named as follows: **Carnumber_Schoolname_Project Plan.doc(x)** and **Carnumber_Schoolname_ChangeManagement.doc(x)** using the Formula Hybrid + Electric assigned car number and the complete school name, e.g.:

999_University of SAE_Project Plan.doc(x)

999_University of SAE_ChangeManagement.doc(x)

S2.6 Submission Deadlines

The Project Plan and Change Management Report must be submitted by the date and time shown in the Action Deadlines. Submission instructions are in section **A9.2**.

See section **A9.3** for late submission penalties.

S2.7 Presentation

S2.7.1 **Objective:** Teams must convince a review board that the team's project has been carefully planned and effectively and dynamically executed.

S2.7.2 The Project Management presentation will be made on the static events day. Presentation times will be scheduled by the organizers and either posted in advance on the competition website or released during on-site registration (or both).

Note: The presentation schedule set by Formula Hybrid + Electric organizers is final and non-negotiable.

- S2.7.3 Teams that fail to make their presentation during their assigned time period will receive zero (0) points for that section of the event.

Note: Teams are encouraged to arrive fifteen (15) minutes prior to their scheduled presentation time to deal with unexpected technical difficulties.

The scoring of the event is based on the average of the presentation judging forms.

S2.8 Presentation Format

The presentation judges should be regarded as a project management or executive review board.

- S2.8.1 **Evaluation Criteria** - Project Management presentations will be evaluated based on team's accomplishments in project planning, execution, change management, and succession planning. Presentation organization and quality of visual aids as well as the presenters' delivery, timing and the team's response to questions will also be factors in the evaluation.
- S2.8.2 One or more team members will give the presentation to the judges. It is strongly suggested, although not required, that the project manager accompany the presentation team. All team members who will give any part of the presentation, or who will respond to the judge's questions, must be in the podium area when the presentation starts and must be introduced to the judges. Team members who are part of this "presentation group" may answer the judge's questions even if they did not speak during the presentation itself.
- S2.8.3 Presentations are limited to a maximum of twelve (12) minutes. Teams may use handouts, posters, etc. to convey information relevant to their project management case that cannot be contained within a 12-minute presentation.
- S2.8.4 The judges will stop any presentation exceeding twelve minutes. The judges will not interrupt the presentation itself.
- S2.8.5 Feedback on presentations will take place immediately following the eight (8) minute question and answer session. Judges and any team members present may ask questions. The maximum feedback time for each team is eight (8) minutes.
- S2.8.6 Formula Hybrid + Electric may record a team's presentation for publication or educational purposes. Students have the right to opt out of being recorded - however they must notify the chief presentation judge in writing prior to the beginning of their presentation.

S2.9 Data Projection Equipment

Projection equipment is provided by the organizers. However, teams are advised to bring their own computer equipment in the event the organizer's equipment malfunctions or is not compatible with their presentation software.

S2.10 Scoring Formula

The Project Management score is equal to the Project Plan score (max. 55 points), plus the Change Management Report score (max. 40 points), plus the presentation score (max. 55 points). It is then normalized as follows:

$$FINAL PROJECT MANAGEMENT SCORE = 150 * (P_{your}/P_{max})$$

Where:

P_{max} is the highest point score awarded to any team

P_{your} is the point score awarded to your team.

Notes:

1. It is intended that the scores will range from near zero (0) to one hundred and fifty (150) points, providing a good separation range.
2. The Project Management Presentation Captain may at her/his discretion normalize the scores of different presentations judging teams for consistency in scoring.
3. Penalties associated with late submittals (see [A9.3](#) “Late Submission Penalties”) are applied after the scores are normalized up to a maximum of the team’s normalized Project Management Score.

ARTICLE S3 DESIGN EVENT

S3.1 Design Event Objective

The concept of the design event is to evaluate the engineering effort that went into the design of the car (or the substantial modifications to a prior year car), and how the engineering meets the intent of the market. The car that illustrates the best use of engineering to meet the design goals and the best understanding of the design by the team members will win the design event.

Comment: Teams are reminded that Formula Hybrid + Electric is an engineering design competition and that in the Design event, teams are evaluated on their design. Components and systems that are incorporated into the design as finished items are not evaluated as a student designed unit, but are only assessed on the team’s selection and application of that unit. For example, teams that design and fabricate their own shocks are evaluated on the shock design itself as well as the shock’s application within the suspension system. Teams using commercially available shocks are evaluated only on selection and application within the suspension system.

S3.2 Submission Requirements

S3.2.1 Design Report - Judging will start with a Design Review before the event.

The principal document submitted for the Design Review is a Design Report. This report must not exceed ten (10) pages, consisting drawings (see [S3.4](#), “Vehicle Drawings”) and content to be defined by the team (photos, graphs, etc...).

This document should contain a brief description of the vehicle and each category listed in [Appendix D](#) with the majority of the report specifically addressing the engineering, design features, and vehicle concepts new for this year's event. Include a list of different analysis and testing techniques (FEA, dynamometer testing, etc.).

Evidence of this analysis and back-up data should be brought to the competition and be available, on request, for review by the judges. These documents will be used by the judges to sort teams into the appropriate design groups based on the quality of their review.

Comment: Consider your Design Report to be the “resume of your car”.

- S3.2.2 **Design Spec Sheet** - In addition to the above documents, a completed FH+E Design Spec Sheet must also be submitted. The FH+E Design Spec Sheet template can be found on the FH+E website. (Do not alter or re-format the template prior to submission.)

Note: The design judges realize that final design refinements and vehicle development may cause the submitted figures to diverge slightly from those of the completed vehicle. For specifications that are subject to tuning, an anticipated range of values may be appropriate.

The Design Report and the Design Spec Sheet, while related documents, must stand alone and be considered two (2) separate submissions. Two separate file submissions are required.

S3.3 Sustainability Requirement

- S3.3.1 Instead of a Sustainability Report please add a Sustainability section to the Design Report answering **both** of the following prompts:

(a) Quantify, compare, and discuss the green-house gas emissions generated by charging your vehicle and competing in the dynamic events at NHMS and/or the fuel efficiency of your car in each dynamic event. For comparisons, consider a similar style vehicle using ICE power only. How have you improved these quantities compared to previous year vehicles? How might you improve this in your next vehicle? What components or design decisions are the biggest contributors to the emissions or efficiency, positively or negatively?

(b) How does the vehicle's design and construction contribute to the recyclability and re-use of your vehicle, especially when it comes to accumulator storage?

S3.4 Vehicle Drawings

- S3.4.1 The Design report must include all of the following drawings:

- (a) One set of 3 view drawings showing the vehicle from the front, top, and side.
- (b) A schematic of the high voltage wiring showing the wiring between the major components. (See section [EV13.1](#))
- (c) A wiring diagram superimposed on a top view of the vehicle showing the locations of all major high voltage components and the routing of high voltage wiring. The components shown must (at a minimum) include all those listed in the major sections of the ESF (See section [EV13.1](#))

S3.5 Submission Formats

- S3.5.1 The Design Report must be submitted electronically in Adobe Acrobat™ Format files (*.pdf file). These documents must each be a single file (text, drawings, and optional content). These Report files must be named as follows: carnumber_schoolname_Design.pdf using the FH+E assigned car number and the complete school name, e.g.:

999_University of SAE_Design.pdf

- S3.5.2 Design Spec Sheets must be submitted electronically in Microsoft Excel™ Format. The format of the Spec Sheet **MUST NOT** be altered. Similar to the Design Report, the Design Spec Sheet file must be named as follows: carnumber_schoolname_Specs.xls (or .xlsx) using the Formula Hybrid + Electric assigned car number and the complete school name, e.g.

999_University of SAE_Specs.xls (or)

WARNING – Failure to exactly follow the above submission requirements in exclusion from the Design Event. If your files are not submitted in the required format or are not properly named then they cannot be included in the documents provided to the design judges and your team will be excluded from the event.

S3.6 Excess Size Design Reports

If a team submits a Design Report that exceeds ten (10) pages, then only the first ten pages will be read and evaluated by the judges.

Note: If included, cover sheets and tables of contents will NOT count as text pages, but no text information on them will be scored.

S3.7 Submission Deadlines

The Design Report and the Design Spec Sheets must be submitted by the date and time shown in the Action Deadlines. (See [A9.2](#)). You will receive confirmation of receipt via email and/or the event website once report is reviewed for accuracy. Teams should have a printed copy of this reply available at the competition as proof of submission in the event of discrepancy.

S3.8 Penalty for Late Submission or Non-Submission

See section [A9.3](#) for late submission penalties.

S3.9 Penalty for Unsatisfactory Submissions

- S3.9.1 Teams that submit a Design Report or a Design Spec Sheet which is deemed to be unsatisfactory, may not compete in the design event at the discretion of the Lead Design Judge. They may be allowed to compete, but receive a penalty of up to seventy five (75) points. Failure to fully document the changes made for the current year's event to a vehicle used in prior FH+E events, or reuse of any part of a prior year design report, are just two examples of Unsatisfactory Submissions

S3.10 Vehicle Condition

- S3.10.1 With the exception of Static Event Only (See [A2.4.1](#)) or Hybrid In Progress (See [A2.3](#)) cars must be presented for design judging in finished condition, i.e. fully assembled, complete and ready-to-run. The judges will not evaluate any car that is presented at the design event in what they consider to be an unfinished state.



Unfinished cars will be given a fifty (50) point penalty for design. Additional point penalties may be assessed for cars with obvious preparation issues, e.g. notably loose or missing fasteners.

Note: Cars can be presented for design judging without having passed technical inspection, or if final tuning and setup is still in progress.

S3.11 Judging Criteria

The design judges will evaluate the engineering effort based upon the team's Design Report, Spec Sheet, responses to questions and an inspection of the car. The design judges will inspect the car to determine if the design concepts are adequate and appropriate for the application (relative to the objectives set forth in the rules). It is the responsibility of the judges to deduct points on the design judging form, as given in [Appendix D](#) if the team cannot adequately explain the engineering and construction of the car.

S3.12 Judging Sequence

The actual format of the design event may change from year to year as determined by the organizing body. In general, the design event includes a brief overview presentation in front of the physical vehicle followed by detailed Q&A. The presentation may be up to forty (40) minutes with the majority of the time devoted to Q&A. Formula Hybrid + Electric design judging will involve two parts:

- (a) Initial judging of all vehicles
- (b) Final judging ranking the top 2 to 4 vehicles.

S3.13 Scoring Formula

The scoring of the event is based on either the average of the scores from the Design Judging Forms (see Appendix C) or the consensus of the judging team.

$$DESIGN\ SCORE = 200 \frac{P_{your}}{P_{max}}$$

Where: P_{max} is the highest point score awarded to any team in your vehicle category

P_{your} is the point score awarded to your team

Notes:

It is intended that the scores will range from near zero (0) to two hundred (200) to provide good separation.

- The Lead Design Judge may, at his/her discretion, normalize the scores of different judging teams.
- Penalties applied during the Design Event (see [Appendix D](#) “Design Judging Form - Miscellaneous”) are applied before the scores are normalized.
- Penalties associated with late submittals (see [A9.3](#) “Late Submission Penalties”) are applied after the scores are normalized up to a maximum of the teams normalized Design Score.

S3.14 Support Materials

Teams may bring with them to the Design Event any photographs, drawings, plans, charts, example components or other materials that they believe are needed to support the presentation of the vehicle and the discussion of their development process.

PART D - DYNAMIC EVENTS

ARTICLE D1 DYNAMIC EVENTS GENERAL

D1.1 Maximum Scores

Event	
Acceleration	100 Points
Autocross	200 Points
Endurance	350 Points
Total	650 Points

Table 17 - Dynamic Event Maximum Scores

D1.2 Driving Behavior

During the Formula Hybrid + Electric Competition, any driving behavior that, in the opinion of the Event Captain, the Director of Operations or the Clerk of the Course, could result in potential injury to an official, worker, spectator or other driver, will result in a penalty.

Depending on the potential consequences of the behavior, the penalty will range from an admonishment, to disqualification of that driver from all events, to disqualification of the team from that event, to exclusion of the team from the Competition.

D1.3 Safety Procedures

- D1.3.1 Drivers must properly use all required safety equipment at all times while staged for an event, while running the event, and while stopped on track during an event. Required safety equipment includes all drivers gear and all restraint harnesses.
- D1.3.2 In the event it is necessary to stop on track during an event the driver must attempt to position the vehicle in a safe position off of the racing line.
- D1.3.3 Drivers must not exit a vehicle stopped on track during an event until directed to do so by an event official. An exception to this is if there is a fire or a risk of fire due to fuel leakage and/or electrical problems.

D1.4 Vehicle Integrity and Disqualification

- D1.4.1 During the Dynamic Events, the mechanical and electrical integrity of the vehicle must be maintained. Any vehicle condition that could compromise vehicle integrity, e.g. damaged suspension, brakes or steering components, electrical tractive system fault, or any condition that could compromise the track surface, e.g. fluid leaks or dragging bodywork, will be a valid reason for exclusion by the officials.
- D1.4.2 The safety systems monitoring the electrical tractive system must be functional as indicated by an illuminated Safety System OK (ESOK) light to enter or continue in any dynamic event.
- D1.4.3 If vehicle integrity is compromised during the Endurance Event, scoring for that segment will be terminated as of the last completed lap.

ARTICLE D2 WEATHER CONDITIONS

The organizer reserves the right to alter the conduct and scoring of the competition based on weather conditions.

ARTICLE D3 RUNNING IN RAIN

A vehicle may not be operated in damp (**D3.1(b)**) or wet (**D3.1(c)**) conditions unless Rain Certified. (See **EV10.5**)

D3.1 Operating Conditions

The following operating conditions will be recognized at Formula Hybrid + Electric:

- (a) Dry – Overall the track surface is dry.
- (b) Damp – Significant sections of the track surface are damp.
- (c) Wet – The entire track surface is wet and there may be puddles of water.
- (d) Weather Delay/Cancellation – Any situation in which all, or part, of an event is delayed, rescheduled or canceled in response to weather conditions.

D3.2 Decision on Operating Conditions

The operating condition in effect at any time during the competition will be decided by the competition officials.

D3.3 Notification

If the competition officials declare the track(s) to be "Damp" or "Wet",

- (a) This decision will be announced over the public address system, and
- (b) A sign with either "Damp" or "Wet" will be prominently displayed at both the starting line(s) and the start-finish line of the event(s), and the entry gate to the "hot" area.

D3.4 Tire Requirements

The operating conditions will determine the type of tires a car may run as follows:

- (a) Dry – Cars must run their Dry Tires, except as covered in **D3.8**.
- (b) Damp – Cars may run either their Dry Tires or Rain Tires, at each team's option.
- (c) Wet – Cars must run their Rain Tires.

D3.5 Event Rules

All event rules remain in effect.

D3.6 Penalties

All penalties remain in effect.

D3.7 Scoring

No adjustments will be made to teams' times for running in "Damp" or "Wet" conditions. The minimum performance levels to score points may be adjusted if deemed appropriate by the officials.

D3.8 Tire Changing

D3.8.1 During the Acceleration or Autocross Events:

Within the provisions of **D3.4** above, teams may change from Dry Tires to Rain Tires or vice versa at any time during those events at their own discretion.

D3.8.2 During the Endurance Event: Teams may change from Dry to Rain Tires or vice versa at any time while their car is in the staging area inside the "hot" area.

All tire changes after a car has received the "green flag" to start the Endurance Event must take place in the Driver Change Area.

(a) If the track was "Dry" and is declared "Damp":

- (i) Teams may start on either Dry or Rain Tires at their option.
- (ii) Teams that are on the track when it is declared "Damp", may elect, at their option, to pit in the Driver Change Area and change to Rain Tires under the terms spelled out below in "Tire Changes in the Driver Change Area".

(b) If the track is declared "Wet":

- (i) A Red Flag will be shown at the Start/Finish Line and all cars will enter the Driver Change Area.
- (ii) Those cars that are already fitted with "Rain" tires will be allowed restart without delay subject to the discretion of the Event Captain/Clerk of the Course.
- (iii) Those cars without "Rain" tires will be required to fit them under the terms spelled out below in "Tire Changes in the Driver Change Area". They will then be allowed to re-start at the discretion of the Event Captain/Clerk of the Course.

(c) If the track is declared "Dry" after being "Damp" or "Wet":

The teams will NOT be required to change back to "Dry" tires.

(d) Tire Changes at Team's Option:

- (i) Within the provisions of **D3.4** above and **D3.8** below, a team will be permitted to change tires at their option.
- (ii) If a team elects to change from "Dry" to "Rain" tires, the time to make the change will NOT be included in the team's total time.
- (iii) If a team elects to change from "Rain" tires back to "Dry" tires, the time taken to make the change WILL be included in the team's total time for the event, i.e. it will not be subtracted from the total elapsed time. However, a change from "Rain" tires back to "Dry" tires will not be permitted during the driver change.
- (iv) To make such a change, the following procedure must be followed:
 - 1. Team makes the decision
 - 2. Team has tires and equipment ready near Driver Change Area
 - 3. The team informs the Event Captain/Clerk of the Course they wish their car to be brought in for a tire change
 - 4. Officials inform the driver by means of a sign or flag at the checker flag station
 - 5. Driver exits the track and enters the Driver Change Area in the normal manner.

(e) Tire Changes in the Driver Change Area:

- (i) Per Rule **D7.13.2** no more than three people for each team may be present in the Driver Change Area during any tire change, e.g. a driver and two crew or two drivers and one crew member.
- (ii) No other work may be performed on the cars during a tire change.
- (iii) Teams changing from "Dry" to "Rain" tires will be allowed a maximum of ten (10) minutes to make the change.
- (iv) If a team elects to change from "Dry" to "Rain" tires during their scheduled driver change, they may do so, and the total allowed time in the Driver Change Area will be increased without penalty by ten (10) minutes.
- (v) The time spent in the driver change area of less than 10 minutes without driver change will not be counted in the team's total time for the event. Any time in excess of these times will be counted in the team's total time for the event.

ARTICLE D4 DRIVER LIMITATIONS

D4.1 Two Event Limit

D4.1.1 An individual team member may not drive in more than two (2) events.

Note: A minimum of two (2) drivers is required to participate in all of the dynamic events. A minimum of four (4) drivers is required to participate in all possible runs in all of the dynamic events.

D4.1.2 It is the team's option to participate in any event. The team may forfeit any runs in any performance event.

D4.1.3 In order to drive in the endurance event, a driver must have attended the mandatory drivers meeting and walked the endurance track with an official.

D4.1.4 The time and location of the meeting and walk-around will be announced at the event.

ARTICLE D5 ACCELERATION EVENT

D5.1 Acceleration Objective

The acceleration event evaluates the car's acceleration in a straight line on flat pavement.

D5.2 Acceleration Procedure

The cars will accelerate from a standing start over a distance of 75 m (82 yards) on a flat surface. The foremost part of the car will be staged at 0.30 m behind the starting line. A green flag will be used to indicate the approval to begin, however, time starts only after the vehicle crosses the start line. There will be no particular order of the cars in the event. A driver has the option to take a second run immediately after the first.

D5.4 Acceleration Event

D5.4.1 All vehicles may make a total of up to 6 runs. It is permissible for one driver to make all the acceleration runs.

D5.4.2 Hybrid vehicles may use their I.C. engine, Electric motor(s) or any combination of these during any of their acceleration runs.

D5.5 Tire Traction – Limitations

Special agents that increase traction may not be added to the tires or track surface and “burnouts” are not allowed.

D5.6 Acceleration Scoring

The acceleration score is based upon the corrected elapsed time. Elapsed time will be measured from the time the car crosses the starting line until it crosses the finish line.

D5.7 Acceleration Penalties

D5.7.1 Cones Down Or Out (DOO)

A two (2) second penalty will be added for each DOO (including entry and exit gate cones) that occurred on that particular run to give the corrected elapsed time.

D5.7.2 Off Course

An Off Course (OC) will result in a DNF for that run.

D5.8 Did Not Attempt

The organizer will determine the allowable windows for each event and retains the right to adjust for weather or technical delays. Cars that have not run by the end of the event will be scored as a Did Not Finish (DNF).

D5.9 Acceleration Scoring Formula

The equation below is used to determine the scores for the Acceleration Event. The first term represents the “Start Points”, the second term the “Participation Points” and the last term the “Performance Points”.

D5.9.1 A team is awarded “Start Points” for crossing the start line under its own power. A team is awarded “Participation Points” if it completes a minimum of one (1) run. A team is awarded “Performance Points” based on its corrected elapsed time relative to the time of the best team in its class.

$$ACCELERATION\ SCORE = 10 + 15 + \left(75 \times \frac{T_{min}}{T_{your}} \right)$$

Where:

T_{your} is the lowest corrected elapsed time (including penalties) recorded by your team.

T_{min} is the lowest corrected elapsed time (including penalties) recorded by the fastest team in your vehicle category.

Note: A Did Not Start (DNS) will score (0) points for the event

ARTICLE D6 AUTOCROSS EVENT

D6.1 Autocross Objective

The objective of the autocross event is to evaluate the car's maneuverability and handling qualities on a tight course without the hindrance of competing cars. The autocross course will combine the performance features of acceleration, braking, and cornering into one event.

D6.2 Autocross Procedure

- D6.2.1 There will be four (4) Autocross-style heats, with each heat having a different driver. Three (3) timed laps will be run (weather and time permitting) by each driver and the best lap time will stand as the time for that heat.
- D6.2.2 The car will be staged such that the front wheels are 6 m (19.7 feet) behind the starting line. The timer starts only after the car crosses the start line.
- D6.2.3 There will be no particular order of the cars to run each heat, but a driver has the option to take a second run immediately after the first.
- D6.2.4 The organizer will determine the allowable windows for each event and retains the right to adjust for weather or technical delays. Cars that have not run by the end of the event will be scored as a Did Not Start (**DNS**).

D6.3 Autocross Course Specifications & Speeds

- D6.3.1 The following specifications will suggest the maximum speeds that will be encountered on the course. Average speeds should be 40 km/hr (25 mph) to 48 km/hr (30 mph).
 - (a) **Straights:** No longer than 60 m (200 feet) with hairpins at both ends (or) no longer than 45 m (150 feet) with wide turns on the ends.
 - (b) **Constant Turns:** 23 m (75 feet) to 45 m (148 feet) diameter.
 - (c) **Hairpin Turns:** Minimum of 9 m (29.5 feet) outside diameter (of the turn).
 - (d) **Slaloms:** Cones in a straight line with 7.62 m (25 feet) to 12.19 m (40 feet) spacing.
 - (e) **Miscellaneous:** Chicanes, multiple turns, decreasing radius turns, etc. The minimum track width will be 3.5 m (11.5 feet).
- D6.3.2 The length of each run will be approximately 0.805 km (1/2 mile) and the driver will complete a specified number of runs.

D6.4 Autocross Penalties

The cars are judged on elapsed time plus penalties. The following penalties will be added to the elapsed time:

D6.4.1 Cone Down or Out (DOO)

Two (2) seconds per cone, including any after the finish line.

D6.4.2 Off Course

Driver must re-enter the track at or prior to the missed gate or a twenty (20) second penalty will be assessed. Penalties will not be assessed for accident avoidance or other reasons deemed sufficient by the track officials.

If a paved road edged by grass or dirt is being used as the track, e.g. a go kart track, four (4) wheels off the paved surface will count as an "off course". Two (2) wheels off will not incur an immediate penalty; however, consistent driving of this type may be penalized at the discretion of the event officials.

D6.4.3 Missed Slalom

Missing one or more gates of a given slalom will be counted as one "off-course" per occurrence. Each occurrence will incur a twenty (20) second penalty.

D6.4.4 Stalled & Disabled Vehicles

If a car stalls and cannot restart without external assistance, the car will be deemed disabled. Cars deemed disabled will be cleared from the track by the track workers. At the direction of the track officials team members may be instructed to retrieve the vehicle. Vehicle recovery may only be done under the control of the track officials.

D6.5 Corrected Elapsed Time

The elapsed time plus any penalties from that specific run will be used as the corrected elapsed time.

D6.6 Best Run Scored

The time required to complete each run will be recorded and the team's best corrected elapsed time will be used to determine the score.

D6.7 Autocross Scoring Formula

The equation below is used to determine the scores for the Autocross Event. The first term represents the "Start Points", the second term the "Participation Points" and the last term the "Performance Points". A team is awarded "Start Points" for crossing the start line under its own power. A team is awarded "Participation Points" if it completes a minimum of one (1) run. A team is awarded "Performance Points" based on its corrected elapsed time relative to the time of the best team in its vehicle category.

$$AUTOCROSS\ SCORE = 20 + 30 + \left(150 \times \frac{T_{min}}{T_{your}} \right)$$

Where:

T_{min} is the lowest corrected elapsed time (including penalties) recorded by the fastest team in your vehicle category over their four heats.

T_{your} is the lowest corrected elapsed time (including penalties) recorded by your team over the four heats.

Note: A Did Not Start (DNS) in all four heats will score zero (0) points for the event.

ARTICLE D7 ENDURANCE EVENT

D7.1 Right to Change Procedure

The following are general guidelines for conducting the endurance event. The organizers reserve the right to establish procedures specific to the conduct of the event at the site. All such procedures will be made known to the teams through newsletters, or the Formula Hybrid + Electric website, or on the official bulletin board at the event.

D7.2 Endurance Objective

The endurance event is designed to evaluate the vehicle's overall performance, reliability and efficiency. Unlike fuel economy tests that result in vehicles going as slow as possible in order to use the least amount of fuel, Formula Hybrid + Electric rewards the team that can cover a designated distance on a fixed amount of energy in the least amount of time.

D7.3 Endurance General Procedure

- D7.3.1 In general, the team completing the most laps in the shortest time will earn the maximum points available for this event. Formula Hybrid + Electric uses an endurance scoring formula that rewards both speed and distance traveled. (See [D7.18](#))
- D7.3.2 The endurance distance is approximately 44km (27.3 miles) comprised of four (4) 11 km (6.84 mile) segments.
- D7.3.3 Driver changes will be made between each segment.
- D7.3.4 Wheel to wheel racing is prohibited.
- D7.3.5 Passing another vehicle may only be done in an established passing zone or under the control of a course marshal.

D7.4 Endurance Course Specifications & Speeds

Course speeds can be estimated by the following course specifications. Average speed should be 48 km/hr (29.8 mph) to 57 km/hr (35.4 mph) with top speeds of approximately 105 km/hr (65.2 mph). Endurance courses will be configured, where possible, in a manner which maximizes the advantage of regenerative braking.

- (a) **Straights:** No longer than 77.0 m (252.6 feet) with hairpins at both ends (or) no longer than 61.0 m (200.1 feet) with wide turns on the ends. There will be passing zones at several locations.
- (b) **Constant Turns:** 30.0 m (98.4 feet) to 54.0 m (177.2 feet) diameter.
- (c) **Hairpin Turns:** Minimum of 9.0 m (29.5 feet) outside diameter (of the turn).
- (d) **Slaloms:** Cones in a straight line with 9.0 m (29.5 feet) to 15.0 m (49.2 feet) spacing.
- (e) **Minimum Track width:** The minimum track width will be 4.5 m (14.76 feet).
- (f) **Miscellaneous:** The organizers may include chicanes, multiple turns, decreasing radius turns, elevation changes, etc.

D7.5 Energy

- D7.5.1 All vehicles competing in the endurance event must complete the event using only the energy stored on board the vehicle at the start of the event plus any energy reclaimed through regenerative braking during the event. Alternatively, vehicles may compete in the endurance event with accumulator capacities greater than the values in [Table 1](#) if equipped with an FH+E approved energy meter. Any endurance laps that reach or exceed the [Table 1](#) maximum energy value will not be counted towards the endurance score. If energy meter data is missing or appears inaccurate, the vehicle may receive start and performance points as appropriate, but “performance points” may be forfeited at organizers discretion.
- D7.5.2 Prior to the beginning of the endurance event, all competitors may charge their electric accumulators from any approved power source they wish.
- D7.5.3 Once a vehicle has begun the endurance event, recharging accumulators from an off-board source is not permitted.

D7.6 Hybrid Vehicles

- D7.6.1 All Hybrid vehicles will begin the endurance event with the defined amount of energy on board.

- D7.6.2 The amount of energy allotted to each team is determined by the Formula Hybrid + Electric Rules Committee and is listed in [Table 1 – 2024 Energy and Accumulator Limits](#)
- D7.6.3 The fuel allocation for each team is based on the tables in [Appendix A](#), adjusted downward by an amount equal to the stated energy capacity of the vehicle's accumulator(s).
- D7.6.4 There will be no extra points given for fuel remaining at the end of the endurance event.
- D7.7 Fueling - Hybrid Vehicles**
- D7.7.1 Prior to the beginning of the endurance event, the vehicle fuel tank and any downstream fuel accumulators, e.g., carburetor float bowls, will be drained. The allocated amount of fuel will then be added to the tank by the organizers and the filler will be sealed.
- D7.7.2 Teams must arrive at the fueling station with jacks and jack stands appropriate for raising and supporting the vehicle to facilitate draining the fuel tank.
- D7.7.3 Once fueled, the vehicle must proceed directly to the endurance staging area.
- D7.8 Charging - Electric Vehicles**
- Each Electric vehicle will begin the endurance event with whatever energy can be stored in its accumulator(s).
- D7.9 Endurance Run Order**
- Endurance run order will be determined by the team's corrected elapsed time in the autocross. Teams with the best autocross corrected elapsed time will run first. If a team did not score in the autocross event, the run order will then continue based on acceleration event times, followed by any vehicles that may not have completed either previous dynamic event. Endurance run order will be published at least one hour before the endurance event is run.
- D7.10 Entering the Track**
- At the start of the event and after driver changes, vehicles will be directed to enter the track by the starter based on traffic conditions.
- D7.11 Endurance Vehicle Restarting**
- D7.11.1 The vehicle must be capable of restarting without external assistance at all times once the vehicle has begun the event.
- D7.11.2 If a vehicle stops out on the track, it will be allowed one (1) lap by the vehicle that is following it (approximately one (1) minute) to restart.
- D7.11.3 At the end of Driver Change, the vehicle will be allowed two (2) minutes to ensure the electrical tractive system is safe and restart the vehicle drive system(See: [D7.13.8](#)).
- D7.11.4 If restarts are not accomplished within the above times, the vehicle will be deemed disabled and scored as a DNF for the event.
- D7.12 Breakdowns & Stalls**
- D7.12.1 If a vehicle breaks down it will be removed from the course and will not be allowed to re-enter the course.
- D7.12.2 If a vehicle spins, stalls, ingests a cone, etc., it will be allowed to restart and re-enter the course where it went off, but no work may be performed on the vehicle
- D7.12.3 If a vehicle stops on track and cannot be restarted without external assistance, the track workers will push the vehicle clear of the track. At the discretion of event officials, two (2) team members may retrieve the vehicle under direction of the track workers.

D7.13 Endurance Driver Change Procedure

- D7.13.1 There must be a minimum of two (2) drivers for the endurance event, with a maximum of four (4) drivers. One driver may not drive in two consecutive segments.
- D7.13.2 Each driver must attempt to drive an 11 km (6.83 miles) segment, and then be signaled into the driver change area. If a driver elects to come into driver change before the end of their 11km segment, they will not be allowed to make up the laps remaining in that segment.
- D7.13.3 Only three (3) team members, including the driver or drivers, will be allowed in the driver change area. Only the tools necessary to adjust the vehicle to accommodate the different drivers and/or change tires will be carried into this area (no tool chests, electronic test equipment, computers, etc.).

Extra people entering the driver change area will result in a twenty (20) point penalty to the final endurance score for each extra person entering the area.

Note: Teams are permitted to “tag-team” in and out of the driver change area as long as there are no more than three (3) team members present at any one time.

- D7.13.4 The vehicle must come to a complete stop, the IC engine turned off and the TSV shut down. These systems must remain shut down until the new driver is in place. (See [D7.13.8](#))
- D7.13.5 The driver will exit the vehicle and any necessary adjustments will be made to the vehicle to fit the new driver (seat cushions, head restraint, pedal position, etc.). The new driver will then be secured in the vehicle.
- D7.13.6 Three (3) minutes are allowed for the team to change drivers. The time starts when the vehicle comes to a halt in the driver change area and stops when the team indicates that the new driver is in place and that his/her restraints and safety gear are correctly adjusted. Anytime taken over the allowed time will incur a penalty. (See [D7.17.2\(k\)](#))
- D7.13.7 During the driver change, teams are not permitted to do any work on, or make any adjustments to the vehicle with the following exceptions:
 - (a) Changes required to accommodate each driver
 - (b) Tire changing as covered by [D3.8](#) “Tire Changing”,
 - (c) Actuation of the following buttons/switches to assist the driver with re-energizing the electrical tractive system
 - (i) Ground Low Voltage Master Switch
 - (ii) Tractive System Master Switch
 - (iii) Side Mounted BRBs
 - (iv) IMD Reset (Button/Switch must be clearly marked “IMD RESET”)
 - (v) AMS Reset (Button/Switch must be clearly marked “AMS RESET”)
- D7.13.8 Once the new driver is in place and an official has verified the correct adjustment of the driver restraints and safety equipment, a maximum of two (2) minutes are allowed to ensure the electrical tractive system is safe (as indicated by the ESOK indicator), restart the vehicle drive system (IC engine, electrical tractive system, or both) and begin moving out of the driver change area.

The ESOK indicator must be illuminated and verified by the driver change area official prior to the vehicle being released out of the driver change area.

- D7.13.9 The process given in [D7.13.8](#) through [D7.13.8](#) will be repeated for each 11 km (6.83 mile) segment. The vehicle will continue until it completes the total 44 km (27.34 miles) distance or until the endurance event track closing time, at which point the vehicle will be signaled off the course.
- D7.13.10 The driver change area will be placed such that the timing system will see the driver change as an extra-long lap. Unless a driver change takes longer than three (3) minutes, this extra-long lap will not count into a team's final time. If a driver change takes longer than three minutes, the extra time will be added to the team's final time.
- D7.13.11 Once the vehicle has begun the event, electronic adjustment to the vehicle may only be made by the driver using driver-accessible controls.

D7.14 Endurance Lap Timing

- D7.14.1 Each lap of the endurance event will be individually timed either by electronic means, or by hand.
- D7.14.2 Each team is required to time their vehicle during the endurance event as a backup in case of a timing equipment malfunction. An area will be provided where a maximum of two team members can perform this function. All laps, including the extra-long laps must be recorded legibly and turned in to the organizers at the end of the endurance event. Standardized lap timing forms will be provided by the organizers.

D7.15 Exiting the Course

- D7.15.1 Timing will stop when the vehicle crosses the start/finish line.
- D7.15.2 Teams may elect to shut down and coast after crossing the start/finish line, but must fully enter the driver change area before coming to a stop. There will be no "cool down" laps.
- D7.15.3 The speed limit when entering the shut-down area is 15 MPH. Excessive speed will be penalized.

D7.16 Endurance Minimum Speed Requirement

- D7.16.1 A car's allotted number of laps, including driver's changes, must be completed in a maximum of 120 minutes elapsed time from the start of that car's first lap.

Cars that are unable to comply will be flagged off the course and their actual completed laps tallied.
- D7.16.2 If a vehicle's lap time becomes greater than *Max Average Lap Time* (See: [D7.18](#)) it may be declared "out of energy", and flagged off the course. The vehicle will be deemed disabled and scored as a DNF for the event.

Note: Teams should familiarize themselves with the Formula Hybrid + Electric endurance scoring formulas. Attempting to complete additional laps at too low a speed can cost a team points.

D7.17 Endurance Penalties

- D7.17.1 Penalties will not be assessed for accident avoidance or other reason deemed sufficient by the track official.

D7.17.2 The penalties in effect during the endurance event are listed below.

(a) Cone down or out (DOO)

Two (2) seconds per cone. This includes cones before the start line and after the finish line.

(b) Off Course (OC)

For an OC, the driver must re-enter the track at or prior to the missed gate or a twenty (20) second penalty will be assessed.

If a paved surface edged by grass or dirt is being used as the track, e.g. a go kart track, four (4) wheels off the paved surface will count as an "off course". Two (2) wheels off will not incur an immediate penalty. However, consistent driving of this type may be penalized at the discretion of the event officials.

(c) Missed Slalom

Missing one or more gates of a given slalom will incur a twenty (20) second penalty.

(d) Failure to obey a flag

Penalty: 1 minute

(e) Over Driving (After a closed black flag)

Penalty: 1 Minute

(f) Vehicle to Vehicle Contact

Penalty: DISQUALIFIED

(g) Running Out of Order

Penalty: 2 Minutes

(h) Mechanical Black Flag

See **D7.23.3(b)**. No time penalty. The time taken for mechanical or electrical inspection under a "mechanical black flag" is considered officials' time and is not included in the team's total time. However, if the inspection reveals a mechanical or electrical integrity problem the vehicle may be deemed disabled and scored as a DNF for the event.

(i) Reckless or Aggressive Driving

Any reckless or aggressive driving behavior (such as forcing another vehicle off the track, refusal to allow passing, or close driving that would cause the likelihood of vehicle contact) will result in a black flag for that driver.

When a driver receives a black flag signal, he/she must proceed to the penalty box to listen to a reprimand for his/her driving behavior.

The amount of time spent in the penalty box will vary from one (1) to four (4) minutes depending upon the severity of the offense.

If it is impossible to impose a penalty by a stop under a black flag, e.g. not enough laps left, the event officials may add an appropriate time penalty to the team's elapsed time.

(j) Inexperienced Driver

The Event Captain or Clerk of the Course may disqualify a driver if the driver is too slow, too aggressive, or driving in a manner that, in the sole opinion of the event officials,

demonstrates an inability to properly control their vehicle. This will result in a DNF for the event.

(k) Driver Change

Driver changes taking longer than three (3) minutes will be penalized.

D7.18 Endurance Scoring Formula

The scoring for the endurance event will be based upon the total laps completed, the on-track elapsed time for all drivers (less the uncharged extra-long laps for the driver changes, change to wet tires, etc.), plus any penalty time and penalty points assessed against all drivers and team members.

Vehicles scored as a Did Not Finish (DNF) for the event will get credit for all laps completed prior to the DNF.

D7.18.1 The equation below is used to determine the scores for the Endurance Event. The first term represents the “Start Points”, the second term the “Participation Points” and the last term the “Performance Points”.

A team is awarded “Start Points” for crossing the start line under its own power. A team is awarded “Participation Points” if it completes a minimum of one (1) lap. A team is awarded “Performance Points” based on the number of laps it completes relative to the best team in its vehicle category (distance factor) and its corrected average lap time relative to the event standard time and the time of the best team in its vehicle category (speed factor).

ENDURANCE SCORE

$$= 35 + 52.5 + 262.5 \left(\frac{LapSum(n)_{your}}{LapSum(n)_{max}} \right) \frac{\left(\frac{Max\ Average\ Lap\ Time}{T_{your}} \right) - 1}{\left(\frac{Max\ Average\ Lap\ Time}{T_{min}} \right) - 1}$$

Where:

Max Average Lap Time is the event standard time in minutes and is calculated as

$$Max\ Average\ Lap\ Time = \frac{105}{the\ number\ of\ laps\ required\ to\ complete\ 44\ km}$$

T_{min} = the lowest corrected average lap time (including penalties) recorded by the fastest team in your vehicle category over their completed laps.

T_{your} = the corrected average lap time (including penalties) recorded by your team over your completed laps.

$LapSum(n)_{max}$ = The value of *LapSum* corresponding to number of complete laps credited to the team in your vehicle category that covered the greatest distance.

$LapSum(n)_{your}$ = The value of *LapSum* corresponding to the number of complete laps credited to your team

Notes:

- (a) If your team completes all of the required laps, then $LapSum(n)_{your}$ will equal the maximum possible value of $LapSum(n)$. (990 for a 44 lap event).
- (b) If your team does not complete the required number of laps, then $LapSum(n)_{your}$ will be based on the number of laps completed. See [Appendix B](#) for $LapSum(n)$ calculation methodology.
- (c) Negative “performance points” will not be given.
- (d) A Did Not Start (DNS) will score (0) points for the event

D7.18.2 Teams exceeding 120 minutes elapsed clock time since starting their first lap will have their results truncated at the last lap completed within the 120 minute limit.

D7.19 Post Event Engine and Energy Check

The organizer reserves the right to impound any vehicle immediately after the event to check accumulator capacity, engine displacement (method to be determined by the organizer) and restrictor size (if fitted).

D7.20 Endurance Event – Driving

D7.20.1 During the endurance event when multiple vehicles are running on the course it is paramount that the drivers strictly follow all of the rules and driving requirements. Aggressive driving, failing to obey signals, not yielding for passing, etc. will result in a black flag and a discussion in the penalty box with course officials. The amount of time spent in the penalty box is at the discretion of the officials and is included in the run time. Penalty box time serves as a reprimand as well as informing the driver of what he/she did wrong. Drivers should be aware that contact between open wheel racers is especially dangerous because tires touching can throw one vehicle into the air.

D7.20.2 Endurance is a timed event in which drivers compete only against the clock not against other vehicles. Aggressive driving is unnecessary.

D7.21 Endurance Event – Passing

D7.21.1 Passing during the endurance event may only be done in the designated passing zones and under the control of the track officials. Passing zones have two parallel lanes – a slow lane for the vehicles that are being passed and a fast lane for the vehicles that are making a pass. On approaching a passing zone a slower leading vehicle will be blue flagged and must shift into the slow lane and decelerate. The following faster vehicle will continue in the fast lane and make the pass. The vehicle that had been passed may reenter traffic only under the control of the passing zone exit marshal.

The passing lanes, e.g. the slow lanes, may be either to the left or the right of the fast lane depending on the design of the specific course.

D7.21.2 These passing rules do not apply to vehicles that are passing disabled vehicles on the course or vehicles that have spun out and are not moving. When passing a disabled or off-track vehicle it is critical to slow down, drive cautiously and be aware of all the vehicles and track workers in the area.

D7.21.3 Under normal driving conditions when not being passed all vehicles use the fast lane.

D7.22 Endurance Event – Driver’s Course Walk

The endurance course will be available for walk by drivers prior to the event. All endurance drivers should walk the course before the event starts.

D7.23 Flags

D7.23.1 The flag signals convey the commands described below, and must be obeyed immediately and without question.

D7.23.2 There are two kinds of flags for the competition: Command Flags and Informational Flags. Command Flags are just that, flags that send a message to the competitor that the competitor must obey without question. Informational Flags, on the other hand, require no action from the driver, but should be used as added information to help him or her maximize performance. What follows is a brief description of the flags used at the competitions in North America and what each flag means.

Note: Not all of these flags are used at all competitions and some alternate designs are occasionally displayed. Any variations from this list will be explained at the drivers meetings.

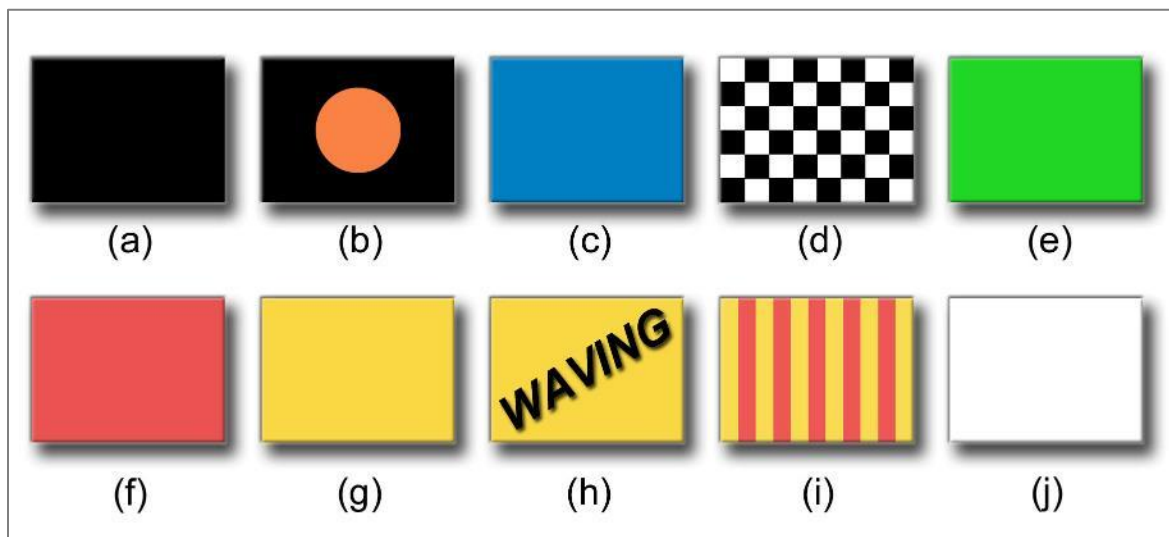


Table 18 - Flags

D7.23.3 Command Flags

- (a) **BLACK FLAG** - Pull into the penalty box for discussion with the Director of Operations or other official concerning an incident. A time penalty may be assessed for such incident.
- (b) **MECHANICAL BLACK FLAG** (Black Flag with Orange Dot) (“Meatball”) - Pull into the penalty box for a mechanical inspection of your vehicle, something has been observed that needs closer inspection.
- (c) **BLUE FLAG** - Pull into the designated passing zone to be passed by a faster competitor or competitors. Obey the course marshal’s hand or flag signals at the end of the passing zone to merge into competition.

- (d) **CHECKER FLAG** - Your segment has been completed. Exit the course at the first opportunity after crossing the finish line.
- (e) **GREEN FLAG** - Your segment has started, enter the course under direction of the starter. **NOTE:** If you are unable to enter the course when directed, await another green flag as the opening in traffic may have closed.
- (f) **RED FLAG** - Come to an immediate safe controlled stop on the course. Pull to the side of the course as much as possible to keep the course open. Follow course marshal's directions.
- (g) **YELLOW FLAG** (Stationary) - Danger, SLOW DOWN, be prepared to take evasive action, something has happened beyond the flag station. NO PASSING unless directed by the course marshals.
- (h) **YELLOW FLAG** (Waved) - Great Danger, SLOW DOWN, evasive action is most likely required, BE PREPARED TO STOP, something has happened beyond the flag station, NO PASSING unless directed by the course marshals.

D7.23.4 Informational Flags

- (a) **RED AND YELLOW STRIPED FLAG** - Something is on the racing surface that should not be there. Be prepared for evasive maneuvers to avoid the situation. (Course marshals may be able to point out what and where it is located, but do not expect it.)
- (b) **WHITE FLAG** - There is a slow moving vehicle on the course that is much slower than you are. Be prepared to approach it at a cautious rate.

ARTICLE D8 RULES OF CONDUCT

D8.1 Competition Objective – A Reminder

The Formula Hybrid + Electric event is a design engineering competition that requires performance demonstration of vehicles and is NOT a race. Engineering ethics will apply. It is recognized that hundreds of hours of labor have gone into fielding an entry into Formula Hybrid + Electric. It is also recognized that this event is an “engineering educational experience” but that it often times becomes confused with a high stakes race. In the heat of competition, emotions peak and disputes arise. Our officials are trained volunteers and maximum human effort will be made to settle problems in an equitable, professional manner.

D8.2 Unsportsmanlike Conduct

In the event of unsportsmanlike conduct, the team will receive a warning from an official. A second violation will result in expulsion of the team from the competition.

D8.3 Official Instructions

Failure of a team member to follow an instruction or command directed specifically to that team or team member will result in a twenty five (25) point penalty.

Note: This penalty can be individually applied to all members of a team.

D8.4 Arguments with Officials

Argument with, or disobedience to, any official may result in the team being eliminated from the competition. All members of the team may be immediately escorted from the grounds.

D8.5 Alcohol and Illegal Material

Alcohol, illegal drugs, weapons or other illegal material are prohibited on the event site during the competition. This rule will be in effect during the entire competition. Any violation of this rule by a team member will cause the expulsion of the entire team. This applies to both team members and faculty advisors. Any use of drugs, or the use of alcohol by an underage individual, will be reported to the local authorities for prosecution.

D8.6 Parties

Disruptive parties either on or off-site must be prevented by the Faculty Advisor.

D8.7 Trash Clean-up

- D8.7.1 Cleanup of trash and debris is the responsibility of the teams. The team's work area should be kept uncluttered. At the end of the day, each team must clean all debris from their area and help with maintaining a clean paddock.
- D8.7.2 Teams are required to remove all of their material and trash when leaving the site at the end of the competition. Teams that abandon furniture, or that leave a paddock that requires special cleaning, will be billed for removal and/or cleanup costs.
- D8.7.3 All liquid hazardous waste (engine oil, fuel, brake fluid, etc.) must be put in well-marked, capped containers and left at the hazardous waste collection building. The track must be notified as soon as the material is deposited by calling the phone number posted on the building. See the map in the registration packet for the building location.

ARTICLE D9 GENERAL RULES

D9.1 Dynamometer Usage

- D9.1.1 If a dynamometer is available, it may be used by any competing team. Vehicles to be dynamometer tested must have passed all parts of technical inspection.
- D9.1.2 Fuel, ignition and drivetrain tuning will be permitted while testing on the dynamometer.

D9.2 Problem Resolution

Any problems that arise during the competition will be resolved through the Formula Hybrid + Electric management team and the decision will be final.

D9.3 Forfeit for Non-Appearance

It is the responsibility of teams to be in the right place at the right time. If a team is not present and ready to compete at the scheduled time they forfeit their attempt at that event. There are no make-ups for missed appearances.

D9.4 Safety Class – Attendance Required

An electrical safety class is required for all team members. The time and location will be provided in the team's registration packet.

D9.5 Drivers Meetings – Attendance Required

All drivers for an event are required to attend the pre-event drivers meeting(s). The driver for an event will be disqualified if he/she does not attend the driver meeting for the event.

D9.6 Personal Vehicles

D9.6.1 Personal cars, motorcycles and trailers must be parked in designated areas only. Only FH+E competition vehicles will be allowed in the track areas.

All vehicles and trailers must be parked behind the white “Fire Lane” lines.

D9.6.2 Some self-powered transport such as bicycles and skate boards are permitted, subject to restrictions posted in the event guide

D9.6.3 The use of self-propelled pit carts, tool boxes, tire carriers or similar motorized devices in any part of the competition site, including the paddocks, is prohibited.

D9.7 Exam Proctoring

D9.7.1 Formula Hybrid + Electric will not provide proctors, exam materials, or exam resources for students attending the competition. The team Advisor, or designated representative ([A6.3.1](#)) attending the competition will be responsible for communicating with college and university officials, administering exams, and ensuring all exam materials are returned to the college and university. Students who need to take exams during the competition may use designated spaces provided by the Formula Hybrid + Electric Officials with at least two weeks advance notice to the Officials.

ARTICLE D10 PIT/PADDOCK/GARAGE RULES

D10.1 Vehicle Movement

D10.1.1 Vehicles may not move under their own power anywhere outside of an officially designated dynamic area.

D10.1.2 When being moved outside the dynamic area:

- (a) The vehicle TSV must be deactivated.
- (b) The vehicle must be pushed at a normal walking pace by means of a “Push Bar” ([D10.2](#)).
- (c) The vehicle’s steering and braking must be functional.
- (d) A team member must be sitting in the cockpit and must be able to operate the steering and braking in a normal manner.
- (e) A team member must be walking beside the car.
- (f) The team has the option to move the car either with
 - (i) all four (4) wheels on the ground **OR**
 - (ii) with the rear wheels supported on a dolly or push bar mounted wheels, provided that the external wheels supporting the rear of the car are non-pivoting such that the vehicle will travel only where the front wheels are steered.

D10.1.3 Cars with wings are required to have two team members walking on either side of the vehicle whenever the vehicle is being pushed.

NOTE: During performance events when the excitement is high, it is particularly important that the car be moved at a slow pace in the pits. The walking rule will be enforced and a point penalty of twenty-five (25) points will be assessed for each violation.

D10.2 Push Bar

Each car must have a removable device that attaches to the rear of the car that allows two (2) people, standing erect behind the vehicle, to push the car around the event site. This device must also be capable of decelerating, i.e. slowing and stopping the forward motion of the vehicle by pulling it rearwards. It must be presented with the car at Technical Inspection.

D10.3 Smoking – Prohibited

Smoking is prohibited in all competition areas.

D10.4 Fueling and Refueling

Officials must conduct all fueling and refueling. The vehicle must be de-energized when refueling, and no other activities (including any mechanical or electrical work) are allowed while refueling.

D10.5 Energized Vehicles in the Paddock or Garage Area

Any time a vehicle is energized such that it is capable of motion (i.e. the TSSI lamp is illuminated), the drive wheels must be removed or properly supported clear of the ground.

D10.6 Engine Running in the Paddock

Engines may be run in the paddock provided:

(a) The car has passed all technical inspections.

AND

(b) The drive wheels are removed or properly supported clear of the ground.

D10.7 Safety Glasses

Safety glasses must be worn at all times while working on a vehicle, and by anyone within 10 ft. (3 meters) of a vehicle that is being worked on.

D10.8 Curfews

A curfew period may be imposed on working in the garages. Be sure to check the event guide for further information

ARTICLE D11 DRIVING RULES**D11.1 Driving Under Power**

D11.1.1 Cars may only be driven under power:

(a) When running in an event.

(b) When on the practice track.

(c) During the brake test.

(d) During any powered vehicle movement specified and authorized by the organizers.

D11.1.2 For all other movements cars must be pushed at a normal walking pace using a push bar.

D11.1.3 Driving a vehicle outside of scheduled events or scheduled practice will result in a two hundred (200) point penalty for the first violation and expulsion of the team for a second violation.

D11.2 Driving Off-Site - Prohibited

Driving off-site is absolutely prohibited. Teams found to have driven their vehicle at an off-site location during the period of the competition will be disqualified from the competition.

D11.3 Practice Track

D11.3.1 A practice track for testing and tuning cars may be available at the discretion of the organizers. The practice area will be controlled and may only be used during the scheduled practice times.

D11.3.2 Practice or testing at any location other than the practice track is absolutely forbidden.

D11.3.3 Cars using the practice track must have passed all parts of the technical inspection.

D11.4 Situational Awareness

Drivers must maintain a high state of situational awareness at all times and be ready to respond to the track conditions and incidents. Flag signals and hand signals from course marshals and officials must be immediately obeyed.

ARTICLE D12 DEFINITIONS

DOO - A cone is “Down or Out”—if the cone has been knocked over or the entire base of the cone lies outside the box marked around the cone in its undisturbed position.

DNF- Did Not Finish

Gate - The path between two cones through which the car must pass. Two cones, one on each side of the course define a gate: Two sequential cones in a slalom define a gate.

Entry Gate -The path marked by cones which establishes the required path the vehicle must take to enter the course.

Exit Gate - The path marked by cones which establishes the required path the vehicle must take to exit the course.

Staging Area - An area prior to the entry to an event for the purpose of gathering those cars that are about to start.

OC - A car is Off Course if it does not pass through a gate in the required direction.

Appendix A

Accumulator Rating & Fuel Equivalency

Each accumulator device will be assigned an energy rating and fuel equivalency based on the following:

Battery capacities are based on nominal (data sheet values). Ultracap capacities are based on the maximum operating voltage (V_{peak}) and the effective capacitance in Farads (N_{parallel} x C / N_{series}).

Batteries:	$Energy (Wh) = V_{nom} \times I_{nom} \times Total\ Number\ of\ Cells \times 80\%$
Capacitors:	$Energy(Wh) = \left(\frac{C(V_{peak}^2 - V_{min}^2)}{2} \right) / 3600$ where V_{min} is assumed to be 10% of V_{peak}.

Table 19 - Accumulator Device Energy Calculations

Liquid Fuels	Wh / Liter ³²
Gasoline (Sunoco ³³ Optima)	2,343

Table 20 - Fuel Energy Equivalencies

Examples:

A battery with an energy rating of 3110 Wh has a fuel equivalency of 1.327 liters.

If using 89 Maxell MC 2600 ultracaps in series (2600 F/89 = 29.2 F, 2.7 x 89 = 240 V), the fuel equivalency is 231.9 Wh resulting in a 99cc reduction of gasoline.

³² Formula Hybrid + Electric assumes a mechanical efficiency of 27%

³³ Full specifications for Sunoco racing fuels may be found at: <https://www.sunocoracefuels.com/>

Appendix B

Determination of *LapSum(n)* Values

The parameter *LapSum(n)* is used in the calculation of the scores for the endurance event. It is a function of the number of laps (*n*) completed by a team during the endurance event. It is calculated by summing the lap numbers from 1 to (*n*), the number of laps completed. This gives increasing weight to each additional lap completed during the endurance event.

For example:

If your team is credited with completing five (5) laps of the endurance event, the value of *LapSum(n)*_{your} used in compute your endurance score would be the following:

$$LapSum(5)_{your} = 1 + 2 + 3 + 4 + 5 = 15$$

Number of Laps Completed (<i>n</i>)	<i>LapSum(n)</i>
0	0
1	1
2	3
3	6
4	10
5	15
6	21
7	28
8	36
9	45
10	55
11	66
12	78
13	91

Number of Laps Completed (<i>n</i>)	<i>LapSum(n)</i>
23	276
24	300
25	325
26	351
27	378
28	406
29	435
30	465
31	496
32	528
33	561
34	595
35	630
36	666

14	105
15	120
16	136
17	153
18	171
19	190
20	210
21	231
22	253

37	703
38	741
39	780
40	820
41	861
42	903
43	946
44	990

Table 21 - Example of LapSum(n) calculation for a 44-lap Endurance event

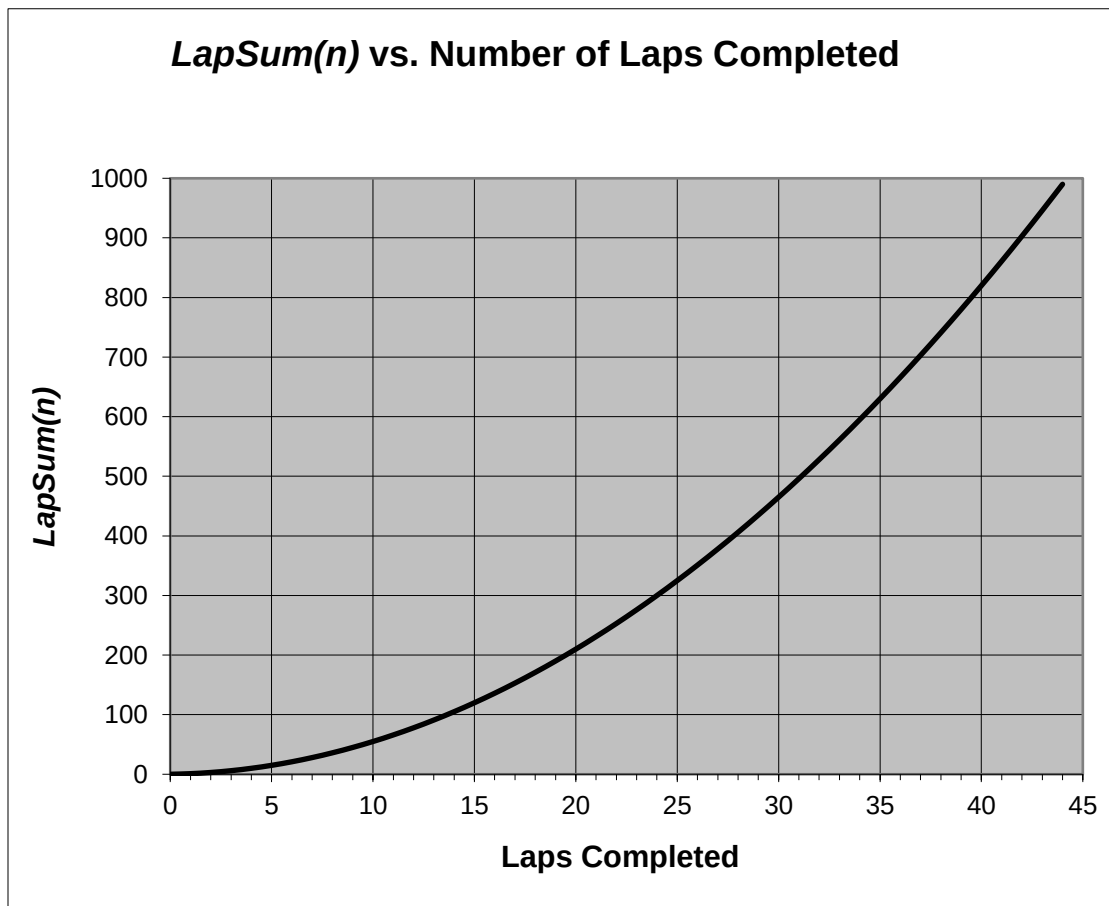


Figure 45 - Plot of *LapSum(n)* calculation for a 44-lap Endurance event

Appendix C

Formula Hybrid + Electric Project Management Event Scoring Criteria

INTRODUCTION

The Formula Hybrid + Electric Project Management Event is comprised of three components:

Project Plan Report

Change Management Report

Final Presentation.

These components cover the entire life cycle of the Formula Hybrid + Electric Project from design of the vehicle through fabrication and performance verification, culminating in Formula Hybrid + Electric competition at the track. The Project Management Event includes both written reports and an oral presentation, providing project team members the opportunity to develop further their communications skills in the context of a challenging automotive engineering team experience.

Design, construction, and performance testing of a hybrid or electric race car are complex activities that require a structured effort to increase the probability of success. The Project Management Event is included in the Formula Hybrid + Electric competition to encourage each team to create this structure specific to their set of circumstances and goals. Comments each team receives from judges relative to their project plan and change management report, offer guidance directed at project execution. Verbal comments made by judges following the presentation component offer suggestions to improve performance in future competitions.

In scoring the Project Management Event judges assess (1) if a well-thought-out project plan has been developed, (2) that the plan was executed effectively while addressing challenges encountered and managing change and (3) the significance of lessons learned by team members from this experience and quality of recommendations proposed to improve future team performance.

Basic Areas Evaluated



Five categories of effort are evaluated across the three components of Formula Hybrid + Electric Project Management, but the scoring criteria differ for each, reflecting the phase of the project life-cycle that is being assessed. The criteria includes: (1) Scope (2) Operations (3) Risk Management (4) Expected Results (5) Change Management. Each is briefly defined below.

1. **Scope:** A brief introduction of the project documenting what will be accomplished: team goals and objectives beyond simply winning the competition, known as “secondary goals”, major deliverables such as critical sub-systems, innovative designs, or new technologies, and milestones for achieving the goals. Overall, this information is called the Statement of Work.

2. **Operations:** How the project team is structured, usually shown with an organizational chart; Work Breakdown Structure, a cascading representation of tasks that will be completed; a project schedule for completing these tasks, usually this is depicted in a Gantt chart that shows the timeline and interdependencies among different activities; also included are the approved budget for the project and plan for obtaining these funds.
3. **Risk Management:** Arriving at the track with a completed, rules compliant race car increases the probability of full participation in all events during the competition. Even though numerous tasks are involved in the design, build, and test of a hybrid race car, there is a smaller subset of tasks that present a high risk to completing the car on schedule. These are high risk tasks because the team may lack knowledge, experience, or sufficient resources necessary for completing them successfully. However, to achieve the team goals, these tasks must be done. Identify the high risk tasks along with contingency plans for advancing the project forward if they become barriers to progress.
4. **Expected Results:** In general, project teams are expected to deliver what is defined in the scope statement, on time according to project schedule, and within the budget constraint. Goals and objectives are more specifically defined by “measures of success”, quantified attributes that give numerical targets for each goal. These “measures” are of value throughout project execution for setting priorities and making resource decisions. At project completion, they are used to determine the extent to which the team’s goals and objectives have been accomplished.
5. **Change Management:** The need for change is a normal occurrence during project execution. Change is good because it refocuses the team when new information is obtained or unexpected challenges are encountered. But if it is not managed correctly change can become a destructive element to the project.

Change Management is a process designed by the team for administering project change and managing uncertainty. The process includes built-in controls to ensure that change is managed in a disciplined way, adequately documented and clearly communicated to all team members.

SCORING GUIDELINES

The guidelines used by judges for scoring the three components of the Project Management Event are given in the following sections: (1) Project Plan Report, (2) Change Management Report, and (3) Project Management Presentation at the competition.

1. Project Plan Report

Each Formula Hybrid + Electric team is required to submit a formal Project Plan that reflects team goals and objectives for the upcoming competition, the management structure and tasks that will be completed to accomplish these objectives, and the time schedule over which these tasks will be performed. In addition, the formal process for managing change must be defined. A maximum of fifty-five (55) points is awarded for the Project Plan.

Quality of the Written Document: The plan should look and read like a professional document. The flow of information is expected to be logical; the content should be clear and concise. The reader should be able to understand the plan that will be executed.

The Project Plan must consist of at least one (1) page and not exceed three (3) pages of text. Appendices may be attached to the Project Plan and do not count as “pages”, but they must be relevant and referenced in the body of the report.

“SMART” Goals

Projects are initiated to achieve predetermined goals, which are identified in the Scope statement. The project plan is a “roadmap” for effectively deploying team resources to accomplish these goals. Overall, the requirements of the Project Plan incorporate the basic principles of “SMART” goals. These goals have the following characteristics:

- **Specific:** Sufficient detail is provided to clearly identify the targeted objectives; goals are specifically stated so that all individual project team members understand what the team must accomplish.
- **Measurable:** The objectives are quantified so that progress toward the goals can be measured; “measures of success” are defined for this purpose.
- **Assignable:** The person or group responsible for achieving the goals is identified; each task, milestone, and deliverable has an owner, someone responsible for seeing that each is completed.
- **Realistic:** The goals can actually be achieved given the resources and time available. Along with realistic goals, a team might define “stretch goals” which are more aggressive objectives that challenge the team. If the “stretch goals” become barriers to progress during project execution, the change management process is used to pull back “stretch goals”, re-focusing the team on more realistic objectives.
- **Time-Related:** Deadlines are set for achieving each goal, milestone, or deliverable. These deadlines are consistent with the overall project schedule and can be extracted from the project timeline.

Characteristics of an Excellent Project Plan Report Submission



- **Scope:** A brief overview is included covering the team’s past performance and recommendations received from previous teams for improvement. Achievable primary and secondary goals for this year’s team are clearly stated. These goals are more than simply winning the competition. Milestones, with due dates, and major deliverables that support accomplishing the goals are listed.
- **Operations:** An organizational chart showing the structure of the team and Work Breakdown Structure showing the cascading linkage of tasks comprising the project are included. The timeframe and interdependencies of each task are shown in a Gantt chart timeline. The project budget is specified and a brief overview of how these funds will be obtained is given.
- **Risk Management:** Careful thought is demonstrated to understand the weakest areas of the project plan. Several “High Risk” tasks are identified that might have a significant impact on a functional car being produced on time. A contingency plan is described to mitigate these risks if they become barriers during project execution.
- **Expected Results:** All teams are expected to complete the project on schedule, within budget, and to deliver a functional, rules complaint race car to the track. But each team has a set of

primary and secondary goals specific to its project plan. Additional depth is given to these goals by quantifying them, defining measurable targets helpful for directing team efforts. At least two “measures of success” are defined that are related to the team’s specific performance goals.

- **Change Management:** A process for administering changes has been carefully thought-out; it is briefly described and shown schematically in terms of: (1) what triggers the process, (2) how information flows through the process, (3) how decision making is handled, and (4) how changes are communicated to the team. A process flow diagram may be included in the Appendix to save space, however is referenced in the body of the report. At a minimum, the diagram shows the “trigger”, information flow, decision making tasks and responsibilities, and communication tasks. The diagram is specific to and created by the team. There are sufficient controls in place to prevent un-managed changes. The team has an effective communication plan in place to keep all team members informed throughout project duration.

Applying the Project Plan Report Scoring Guidelines

The guidelines for awarding points in each Project Plan Report category are given in **Figure 46**. Four performance designations are also specified: Excellent, Good, Marginal, and Deficient. A range of points is suggested for each designation to give reviewers flexibility in evaluating the quality and completeness of the submitted plans.

While the Appendix may be used to refer to diagrams, tables, etc., a brief explanation for each component must be summarized in the body of the report to receive full credit. Any diagram or table (or other appropriate figure type) must be referenced and linked appropriately to the correct content in the Appendix.

Project Plan Report Characteristics	Excellent 10 - 8 points	Good 7 - 4 points	Marginal 3 - 1 points	Deficient 0 points	Project Plan Report Score
Scope	Successfully addresses all of the following elements required for the characteristic: 1. Project Introduction 2. Primary and Secondary Goals 3. Major Deliverables Defined 4. Milestones Established over Project Duration	Contains all of the elements listed under Excellent in which some are unclear / lacking	Missing 1 of the elements listed under Excellent of the characteristic	Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/10 points
Operations	Successfully addresses all of the following elements required for the characteristic: 1. Organizational Chart 2. Work Breakdown Structure 3. Gantt Chart 4. Budget and Funding Plan Note: Diagrams/tables/figures may be included in Appendix but summary of each must be in the body of the report.	Contains all of the elements listed under Excellent in which some are unclear / lacking	Missing 1 of the elements listed under Excellent of the characteristic	Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/10 points
Risk Management	Successfully addresses all of the following elements required for the characteristic: 1. Several high risk tasks identified 2. Explanation of how high risk tasks impact project 3. Risk contingency plans for identified high risk tasks	Contains all of the elements listed under Excellent in which some are unclear / lacking	Missing 1 of the elements listed under Excellent of the characteristic	Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/10 points
Expected Results	Successfully addresses all of the following elements required for the characteristic: 1. Two attributes are given for measuring project success 2. These attributes are quantified goals	Contains all of the elements listed under Excellent in which some are unclear / lacking	Missing 1 of the elements listed under Excellent of the characteristic	Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/10 points
Change Management	Successfully addresses all of the following elements required for the characteristic: 1. What triggers/initiates the process? 2. How does information flow through the process? 3. How are final decisions communicated to the team? 4. How are decisions made and who makes those decisions? 5. Process flow diagram (may be in Appendix but must be referenced in body of report)	Contains all of the elements listed under Excellent in which some are unclear / lacking	Missing 1 of the elements listed under Excellent of the characteristic	Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/10 points
Document Quality	5 points: 1. Understandable 2. Professional appearance of report 3. Acceptable spelling and grammar 4. Compliant with Formula Hybrid + Electric Rules 5. Effective use of tables, diagrams, and appendix	4-3 points Contains all of the elements listed under Excellent in which some are unclear / lacking	2-1 points Missing 1 of the elements listed under Excellent of the characteristic	0 points Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete	___/5 points
Project Plan Report Total:					/55 points

Figure 46 - Scoring Guidelines: Project Plan Component³⁴

³⁴ This score sheet is available on the Formula Hybrid+Electric website in the [Project Management Resources](#) folder.

2. Change Management Report

The purpose of the Change Management Report is to give each judge a sense of the team's ability to deal with issues that will put the project schedule at risk.

Frequently, as the project progresses or as problems arise, changes may be required to vehicle specifications or the plan for completing the project. This is an area that many teams find difficult to manage. Of particular interest to judges are barriers encountered and creative actions taken by the team to overcome these challenges and advance the project forward.

The Change Management Report is a maximum two (2) page summary, clearly communicating the documented changes to the project scope and plan that have been approved using the change management process. Appendices may be included with supporting information; they do not have a page limit but the content is expected to be supportive of the information covered in the main body of the report.

A maximum of forty (40) points is awarded for the Change Management Report. The content of the report is evaluated in three broad areas: (1) Explanation of the Change Management Process (2) Example of the Implementation of the Change Management Process, and (3) Evaluation of the Effectiveness and Areas of Improvement for the Change Management Process. Additionally, a final area evaluated is the quality of the overall report document.

Characteristics of an Excellent Change Management Report Submission



- **Change Management Process:** In the body of the report, the team provides a thorough and clear explanation of their Change Management Process, including the following characteristics:
 - What triggers/initiates the process?
 - How does information flow through the process?
 - How are decisions made, and who makes those decisions?
 - How are final decisions communicated to the team?
 - Process flow diagram (see below)
- A process flow diagram may be included in the Appendix to save space; however, it is referenced in the body of the report. At a minimum, the diagram shows the “trigger”, information flow, decision making tasks and responsibilities, and communication tasks. The diagram is specific to and created by the team.
- **Implementation of the Change Management Process:** In the body of the report, the team provides a single thorough and clear example of the implementation of the Change Management Process. The example provided addresses all stages of the process from start to finish, including the overall impact to the project.
- **Effectiveness of and Improvements to the Change Management Process:** In the body of the report, the team provides their own assessment of the effectiveness of their Change Management Process, primarily based upon the example provided. This includes the effectiveness of each stage of the process (trigger, information flow, decision making, and communication) as well as an overall assessment. Based on their assessment, the team provides two areas of opportunity to improve the Change Management Process, including what stage of the process is impacted and why this change is being made. The team addresses plans for what the team must do to make the

proposed changes, timing of changes, and a definition of how to measure the effectiveness of the changes for the following year.

- **Proper Use of Appendix:** While the Appendix may be used to refer to diagrams, tables, etc., a brief explanation for each component must be summarized in the body of the report to receive full credit. Any diagram or table (or other appropriate figure type) must be referenced and linked appropriately to the correct content in the Appendix.

Applying the Change Management Scoring Guidelines

The guidelines for awarding points in each Change Management Report evaluation category are given in [Figure 47](#). Similar to the Project Plan guidelines, four performance designations are specified: Excellent, Good, Marginal, and Deficient. A range of points is suggested for each designation to give reviewers flexibility in weighing quality and completeness of information for each category.

Change Management Report Characteristics	Excellent	Good	Marginal	Deficient	Change Management Score
Process	15 - 11 points Successfully addresses all of the following elements required for the characteristic: 1. What triggers/initiates the process? 2. How does information flow through the process? 3. How are final decisions communicated to the team? 4. How are decisions made and who makes those decisions? 5. Process flow diagram (may be in Appendix but must be referenced in body of report).	10 - 6 points Contains all of the elements listed under Excellent in which some are unclear / lacking.	5 - 1 points Missing 1 of the elements listed under Excellent of the characteristic.	0 points Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete.	___ / 15 points
Implementation	10 - 8 points Successfully addresses all of the following elements required for the characteristic: 1. A single thorough and clear example of the implementation of the process. 2. The example addresses all of the stages of the process listed under Process (above). 3. Includes overall impact to the project.	7 - 4 points Contains all of the elements listed under Excellent in which some are unclear / lacking.	3 - 1 points Missing 1 of the elements listed under Excellent of the characteristic.	0 points Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete.	___ / 10 points
Effectiveness & Improvements	10 - 8 points Successfully addresses all of the following elements required for the characteristic: 1. Team's own assessment on the effectiveness of their process based on example provided. 2. Two areas of opportunity identified throughout the year for improving the process. 3. Plans provided to address areas of opportunity for improving the process.	7 - 4 points Contains all of the elements listed under Excellent in which some are unclear / lacking.	3 - 1 points Missing 1 of the elements listed under Excellent of the characteristic.	0 points Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete.	___ / 10 points
Document Quality	5 points 1. All characteristics are clearly conveyed and verified with either descriptions, data, tables, diagrams or photographs 2. Professional appearance of report. 3. Acceptable spelling and grammar. 4. Compliant with Formula Hybrid + Electric Rules.	4 - 3 points Contains all of the elements listed under Excellent in which some are unclear / lacking.	2 - 1 points Missing 1 of the elements listed under Excellent of the characteristic.	0 points Missing 2 or more of the elements listed under Excellent and/or section is missing/incomplete.	___ / 5 points
Change Management Report Total:					___ / 40 points

Figure 47: Change Management Report Scoring Guidelines

While the Appendix may be used to refer to diagrams, tables, etc., a brief explanation for each component must be summarized in the body of the report to receive full credit. Any diagram or table (or other appropriate figure type) must be referenced and linked appropriately to the correct content in the Appendix.

3. Project Management Presentation at the Competition

The Presentation component gives teams the opportunity to briefly explain their project plans, assess how well their project management process worked, and identify recommendations for improving their team's project management process in the future. In addition, the Presentation component enables team leaders to enhance their communication skills in presenting a complex topic clearly and concisely.

The Presentation component is the culmination of the project management experience, which included submission by each team of a project plan and change management report. Scoring of each presentation is based on how well project management practices were used by the team in the planning, execution, and change management aspects of the project. Of particular interest are innovative approaches used by each team in dealing with challenges and how lessons learned from the current competition will be used to foster continuous improvement in the design, development, and testing of their cars for future competitions.

Project Management presentations are made on the Static Events Day during the competition. Each presentation is limited to a maximum of twelve (12) minutes. An eight (8) minute question and answer period will follow along with a feedback discussion with judges lasting no longer than five (5) minutes.

This format will give each team an opportunity to critique their project management performance, clarify major points, and have a discussion with judges on areas that can be improved or strengths that can be built upon for next year's competition. Only the team members present who are introduced at the start of the presentation will be allowed to speak and/or respond to questions and comments.

Scoring the Project Management Presentation

In awarding points, each judge must determine if the team demonstrated an understanding of project management, if the described accomplishments are credible, and if the approaches used to manage the project were effective. These judgements are formed after listening to each presentation and probing the teams for clarity and additional details during the questioning period afterward. Presentation quality and communications skills of team members are extremely important for establishing a positive impression.

A chart summarizing the evaluation criteria for the Presentation component is given in **Figure 48 – Evaluation Criteria**. This provides more detailed guidelines used by the judging review team for evaluating each project management presentation.

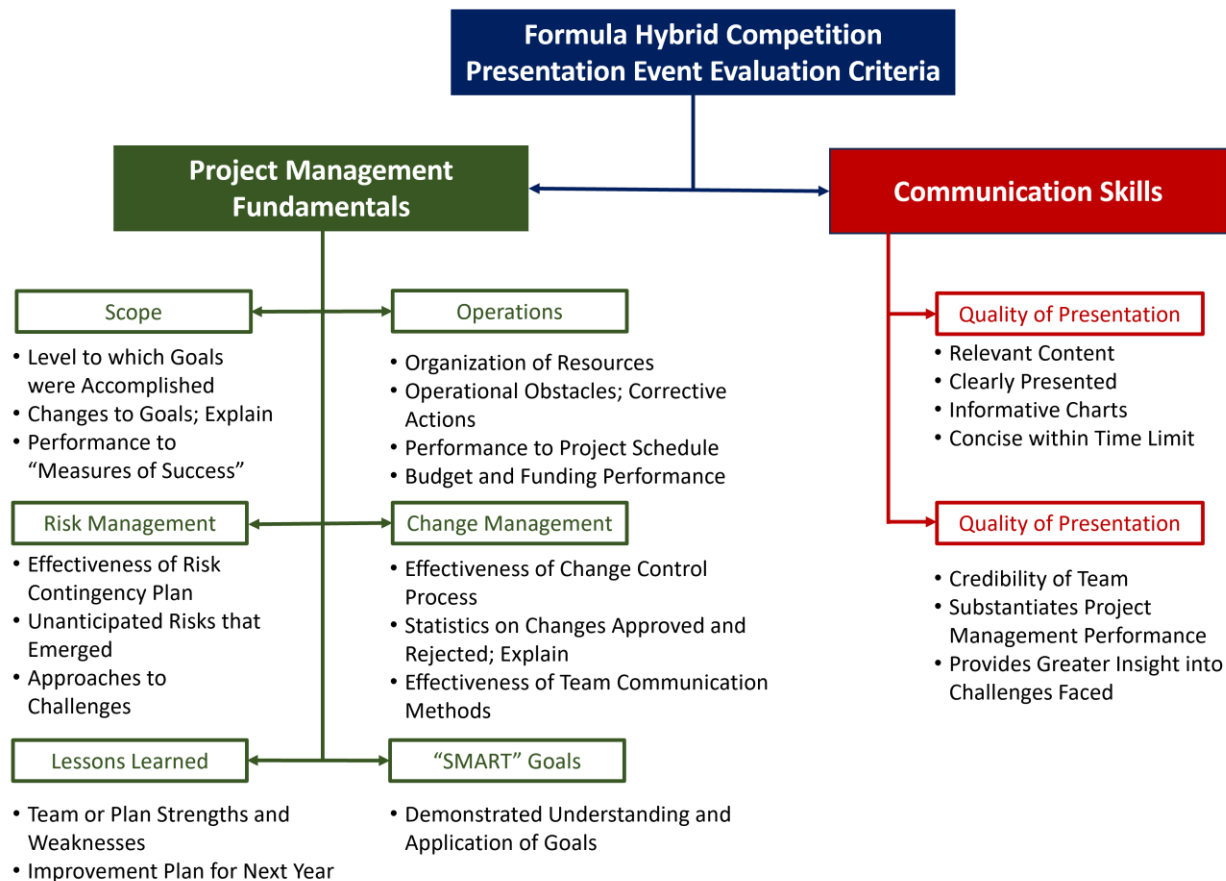


Figure 48 – Evaluation Criteria

Characteristics on an Excellent Project Management Presentation

- Scope:** The primary and secondary goals defined in the project plan are addressed; the degree to which each has been accomplished is explained. Where applicable, the "Measures of Success" are used to support the stated level of accomplishment. If the original goals have been changed, the reasons for modification are explained.
- Operations:** Was this project a success? This conclusion is supported by Team performance against the budget, project schedule, and deliverables produced. The effectiveness of the team's structure is explained. If the operation was disorderly or inefficient during project execution, an overview of corrective actions taken to fix the problem is given.
- Risk Management:** The team's success to correctly anticipate risk in the original project plan and effectiveness of the risk mitigation plan are described. An overview of unanticipated risks that were encountered during project execution and approaches to overcome these barriers are explained.
- Change Management:** A need for change occurs naturally in almost every project; it is anticipated that every Formula Hybrid + Electric team will have experienced some type of change during project execution. Change management strives to align the efforts of all team members working in the dynamic project environment. The effectiveness of the overall change management process in dealing with needed modifications is described. This is supported with



statistics on the number of changes approved and rejected. The effectiveness of the team's communication methods, both written and verbal, is explained. Areas of opportunity for improvement to the change management process are briefly described as well as their status of implementation for next year.

- **Lessons Learned:** When a project is completed, the team should conduct an assessment of overall performance to determine what worked well and what failed. The strengths and weaknesses of the team are described. This evaluation is used to propose recommendations for improving the performance of next year's team. Also, a plan for conveying this information to the new leadership team is described. A leadership succession plan is briefly described.
- **"SMART" Goals:** Team's demonstrated understanding and application of "SMART" goals.
- **Communications Skills:** The team establishes credibility by demonstrating that it has taken the time necessary to carefully plan and create the presentation. The presentation is well organized, content is relevant to the purpose of the presentation. Charts have a professional appearance and are informative. Without the speaker rushing through the material, a large amount of information is conveyed within the allowed time limit. Tables, diagrams, and graphs are used effectively.

Team members demonstrate mutual accountability with shared responses to questions. Answers are conveyed in a manner that instills confidence in the team's ability to plan and execute a complex project like Formula Hybrid + Electric.

Car # _____ Group _____ Judge # _____		
Presentation Topics (30 Points)		Presentation Topics Total Points:
Scope / Expected Results (7 Points)	Operations (5 Points)	Risk Management (4 Points)
☐ Project Introduction	☐ Organizational Chart	☐ Risk Contingency Plan
☐ Primary Goal(s) Defined	☐ Organizational Chart Effectiveness	☐ Risk Contingency Plan Effectiveness
☐ Primary Goal(s) Measure of Success	☐ Budget	☐ Unanticipated Risk(s) Example(s)
☐ Primary Goal(s) Status of Completion	☐ Project Timeline/Schedule	☐ Approach to Overcome Unanticipated Risk(s)
☐ Secondary Goal(s) Defined	☐ Operations Corrective Actions	
☐ Secondary Goal(s) Measure of Success		Overall "SMART" Goals (5 Points)
☐ Secondary Goal(s) Status of Completion	Lessons Learned (5 Points)	☐ Specific: Objectives detailed & goals clearly stated
Change Management (4 Points)	☐ Team Strengths	☐ Measurable: Quantifiable objectives to track & measure success
☐ Change Management Process Explanation	☐ Team Weaknesses	☐ Assignable: Clear ownership for each task, milestone, deliverable, etc.
☐ Change Management Process Assessment	☐ Improvement Plan for Next Year	☐ Realistic: Achievable goals set with possible stretch goals
☐ Team Communication Process & Effectiveness	☐ Leadership Succession Plan	☐ Time-Related: Achievable deadlines for goals, milestones, deliverable
☐ Change Management/Communication Process Areas of Improvement	☐ Succession Plan Status	
Quality of Presentation (25 Points)		Quality of Presentation Total Points:
Use of Time (6 Points)		Visual Quality (4 Points)
☐ Team Discussed All 5 Presentation Topics within Time Limit. Clearly demonstrated the team has practiced the presentation and used time efficiently. (deduct for multiple sections not addressed until Q&A or not addressed at all)		☐ Presentation visuals were clearly defined, easy to read, informative, and well crafted. (deduct for each missing visual quality aspect)
Communications (5 Points)		Q&A Response Team Accountability (3 Points)
☐ Presenter(s) spoke in clear voice, at reasonable speed, with minimal breaks, and maintained eye contact with judges. All team members present were introduced. (deduct for each missing communications aspect)		☐ In Q&A, team displayed mutual accountability, balanced responsibility, and collective authority. (deduct for each missing response aspect)
Presentation Professionalism (4 Points)		Q&A Response Quality (3 Points)
☐ Presenter(s) showed enthusiasm, conveyed confidence in the material, clearly defined messages being delivered, and displayed professionalism throughout presentation. (deduct for each missing presentation aspect)		☐ In Q&A, the team displayed confidence, clarity, and understanding in their answers. (deduct for each missing response aspect)
Comments:		Overall Total Presentation Total: _____/55

Figure 49 - Project Management Scoring Sheet

Appendix D

Design Judging Form³⁵

Car #: _____ Group: _____ Judge # _____

Design Presentation (140 Points)			
Chassis / Cockpit (20 Points) - What are the requirements for the chassis design? Are load paths direct and short? Are components sized properly for the loads? Were weight distribution and C.G. height optimized? - Is the vehicle designed to accommodate & function with a wide variety of body sizes? Are controls and instruments easy to use? Are electrical systems well isolated? Does the design consider occupant safety beyond the requirements?	_____	Suspension (10 Points) What were the requirements for suspension design? How were kinematics, lateral load transfer, adjustability, etc., addressed? How was vehicle handling developed?	_____
Brakes, Steering, and Wheel Assembly (10 Points) How was the brake system designed? How was the steering system designed?	_____	Powertrain Architecture (50 Points) - Was the balance between the IC engine, electric drive, and energy storage well thought out? What were the resulting requirements? Were different architectures considered, and why was the current one selected? How does the system architecture relate to scoring points in the FH competition? - Are the accumulator, power electronics, and electrical machine well matched? What were the requirements for the power electronics and electrical machine, and why were the components chosen? How well do the power electronics and electrical machine meet the requirements? What are the design requirements for the high-voltage wiring harness, fuse block, and connectors? - What were the requirements for the IC engine? Was the engine modified (optimized) for the hybrid application?	_____
Energy Storage (20 Points) Did the team utilize a commercial battery pack or design their own? Did they use batteries or capacitors (or perhaps a combination)? Was it sized specifically for the competition? How is SOC measured or derived? Accumulator devices can be expensive – was cost a factor in the design?	_____	Electrical and Controls (15 Points) What are the requirements for the electronics and controls system, and what determined these requirements? Did the students design the electronic systems? Is there closed-loop control of the engine? Data acquisition?	_____
Aerodynamics (10 Points) What requirements determined the aerodynamics package on the vehicle? How were they constructed? What calculations/simulations were done? What advantages does the design give during dynamic events?	_____	Aesthetics and Craftsmanship (5 Points) Fit and finish, use of appropriate materials, professional quality fabrication (e.g., wiring routed, loomed, and labeled; quality of fabrication, welding, machine work), detail work completed. Does the vehicle look attractive? Does it have a high-performance appearance?	_____
Design Presentation Total Points: _____ / 140			
Design Report (60 Points)			
Chassis / Cockpit (8 Points)	_____	Suspension (5 Points)	_____
Brakes, Steering, and Wheel Assembly (5 Points)	_____	Powertrain Architecture (15 Points)	_____
Energy Storage (8 Points)	_____	Electrical and Controls (5 Points)	_____
Aerodynamics (4 Points)	_____	Report only: Sustainability (10 Points) How well thought out was the sustainability discussion? How relevant was it to the team, competition, or industry? Was the discussion supported by research or calculations?	_____
Design Report Total Points: _____ / 60			
Penalties: _____			
Design Total: _____ / 200			
Comments:			

³⁵ This form is for informational use only – the actual form used may differ. Check the Formula Hybrid + Electric website prior to the competition for the latest Design judging form.

Appendix E

Wire Current Capacity (DC)

Wire Gauge Copper AWG	Conductor Area mm ²	Max. Continuous Fuse Rating (A)	Standard Metric Wire Size mm ²	Max. Continuous Fuse Rating (A)
24	0.20	5	0.50	10
22	0.33	7	0.75	12.5
20	0.52	10	1.0	15
18	0.82	14	1.5	20
16	1.31	20	2.5	30
14	2.08	28	4.0	40
12	3.31	40	6.0	60
10	5.26	55	10	90
8	8.37	80	16	130
6	13.3	105	25	150
4	21.2	140	35	200
3	26.7	165	50	250
2	33.6	190	70	300
1	42.4	220	95	375
0	53.5	260	120	425
2/0	67.4	300	150	500
3/0	85.0	350	185	550
4/0	107	405	240	650
250 MCM	127	455	300	800
300 MCM	152	505		
350 MCM	177	570		
400 MCM	203	615		
500 MCM	253	700		

Table 22 – Wire Current Capacity (single conductor in air)

Reference: US National Electrical Code Table 400.5(A)(2), 90C Column D1 (Copper wire only)

Appendix F

Required Equipment

□ Fire Extinguishers

Minimum Requirements

Each team must have at least two (2) 2.3 kg (5 lb.) dry chemical (Min. 3-A:40-B:C) Fire extinguishers

Extinguishers of larger capacity (higher numerical ratings) are acceptable.

All extinguishers must be equipped with a manufacturer-installed pressure/charge gauge that shows “FULL”.

Special Requirements

Teams must identify any fire hazards specific to their vehicle’s components and if fire extinguisher/fire extinguisher material other than those required in section **T15.1** are needed to suppress such fires, then at least two (2) additional extinguishers/material (at least 5 lb. or equivalent) of the required type must be procured and accompany the car at all times. As recommendations vary, teams are advised to consult the rules committee before purchasing expensive extinguishers that may not be necessary.

□ Chemical Spill Absorbent

Teams must have chemical spill absorbent at hand, appropriate to their specific risks. This material must be presented at technical inspection.

□ Insulated Gloves

Insulated gloves, rated for at least the voltage in the TS system, with protective over-gloves. Electrical gloves require testing by a qualified company. The testing is valid for 14 months after the date of the test. All gloves must have the test date printed on them.

□ Safety Glasses

Safety glasses must be worn as specified in section **D10.7**

□ Additional

Any special safety equipment required for dealing with accumulator mishaps, for example correct gloves recommended for handling any electrolyte material in the accumulator.

Appendix G

Example Relay Latch Circuits

The diagrams below are examples of relay-based latch circuits that can be used to latch the momentary output of the Bender IMD (either high-true or low-true) such that it will comply with Formula Hybrid + Electric rule [EV7.1](#). Circuits such as these can also be used to latch AMS faults in accordance with [EV2.1.3](#). It is highly recommended that IMD and AMS latching circuits be separate and independent to aid fault identification during the event.

Note: It is important to confirm by checking the data sheets, that the output pin of the IMD can power the relay directly. If not, an amplification device will be required³⁶

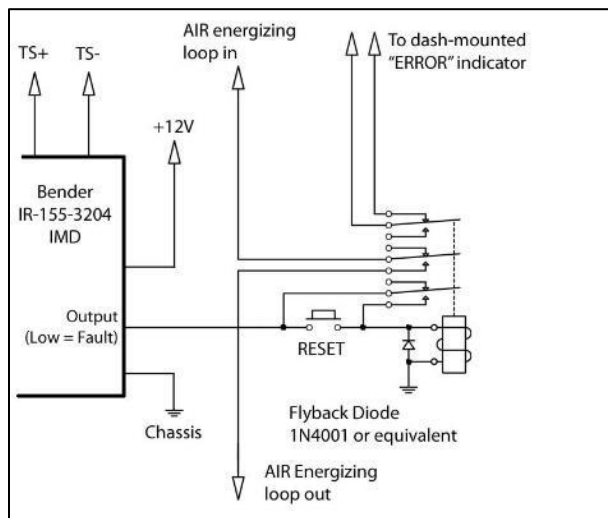


Figure 50 - Latching for Active-High output

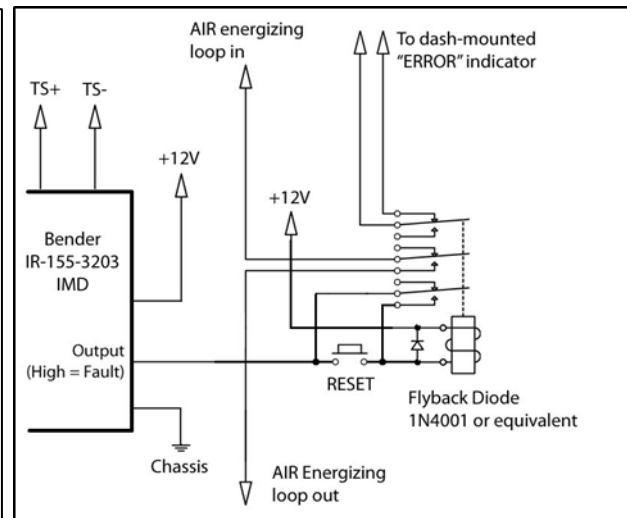


Figure 51 - Latching for Active-Low output

³⁶ An example relay is the Omron LY3-DC12:

<http://www.ia.omron.com/product/item/6403/>

Appendix H

Firewall Equivalency Test

To demonstrate equivalence to the aluminum sheet specified in rule **T4.5.2**, teams should submit a video, showing a torch test of their proposed firewall material.

Camera angle, etc. should be similar to the video found here:

<https://www.youtube.com/watch?v=Qw6kzG97ZtY>

A propane plumber's torch should be held at a distance from the test piece such that the hottest part of the flame (the tip of the inner cone) is just touching the test piece.

The video must show two sequential tests and be contiguous and unedited (except for trimming the irrelevant leading and trailing portions).

The first part of the video should show the torch applied to a piece of Aluminum of the thickness called for in **T4.5.2**, and held long enough to burn through the aluminum. The torch should then be moved directly to a similarly sized test piece of the proposed material without changing any settings, and held for at least as long as the burn-through time for the Aluminum.

There must be no penetration of the test piece. The equivalent firewall construction must have similar mechanical strength to the aluminum barrier called for in **T4.5.2**. This can be demonstrated by equivalent resistance to deformation or puncturing.

Appendix I Virtual Racing Challenge

The Virtual Racing Challenge for 2026 is TBD

END